

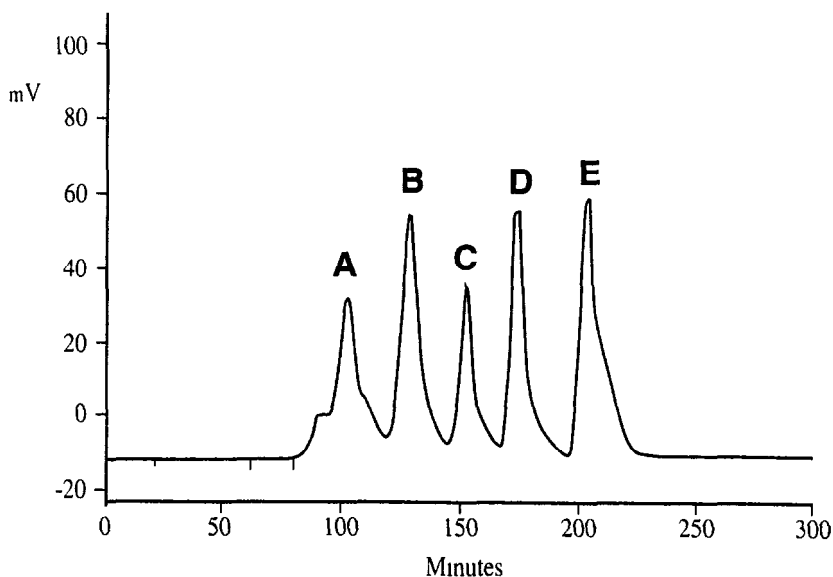


Fig. 4. Size exclusion of five mol-wt markers on Superdex 200 (Pharmacia) column (1.6×75 cm). Thyroglobin, 670,000 (A); γ globulin, 158,000 (B), ovalbumin, 44,000 (C), myoglobin, 17,000 (D), and vitamin B₁₂, 1350 (D).

agent, which combines good solubilizing activity with prevention of endotoxin formation. It may, however, lead to chemical breakdown of certain matrices.

4. Notes

1. If the protein elutes before the void volume ($V_e < V_o$), this suggests channelling through the column owing to improper packing or operation of the column. If the protein elutes after the total volume ($V_e > V_t$), then some interaction must have occurred between the matrix and the protein of interest.
2. Size exclusion is particularly suited for the resolution of protein aggregates from monomers. Aggregates are often formed as a result of the purification procedures used. Size exclusion is often incorporated as a final polishing step to remove aggregates and act as a buffer exchange mechanism into the final solution.
3. Desalting gels are used to remove low-mol-wt material rapidly, such as chemical reagents from proteins and for buffer exchange. Since the molecules to be separated are generally very small, typically < 1 kDa, gels are generally used with an exclusion limit of approx 2–5 kDa. The protein appears in the void volume (V_o) and the reagents and buffer salts are

retained. Because of the distinct mol-wt differences, the columns are shorter than other size-exclusion columns and operated at higher flow rates (e.g., 30 cm/h).

4. In some instances, the length of the column required to obtain a satisfactory separation exceeds that which can be packed into a commercially available column (>1 m). In these cases, columns can be packed in series. The tubing connecting the columns should be as narrow and as short as possible to avoid zone spreading.
5. The majority of protein separations performed using size exclusion are carried out in the presence of aqueous phase buffers. Size exclusion of proteins in organic phases (sometimes called gel permeation) is not normally undertaken, but is sometimes used for membrane protein separations. The agarose- and dextran-based matrices are not suitable for separations with organic solvents. Synthetic polymers, and to a certain extent, silica are suitable for separations in organic phases. For separation in acidified organic solvents, polyacrylamide matrices are particularly suitable. The separation of the proteins may be influenced by the denatured state of the protein in the organic phase. The equipment must be compatible with the solvent system, e.g., glass or Teflon™.
6. Interactions with the matrix are commonly seen when charged proteins are being resolved. Protein-matrix interactions can be minimized by use of ionic strength in excess of 0.1 M. If low ionic strength is necessary, then the risk of interaction can be reduced by manipulating the charge on the protein via the pH of the buffer. This is best achieved by keeping the mobile phase above or below the pI of the protein as appropriate. Protein-matrix interaction is a common cause of protein loss during size exclusion. This may be owing to complete retention on the column or retardation sufficient for the material to elute in an extremely broad dilute fraction, thereby evading detection above the baseline of the buffer system. Another important consideration is when enzymes are detected off line by activity assay. The active enzyme may resolve into inactive subunits. In such cases, a review of the mobile phase is advisable.
7. In addition to determination of the number of theoretical plates (N) described, the performance of the column can be assessed qualitatively by the shape of the eluted peaks (Fig. 5). The theoretically ideal peak is sharp and triangular with an axis of symmetry around the apex. Deviations from this are seen in practice. Some peak shapes are diagnostic of particular problems that lead to broadening and poor resolution. If the down slope of the peak is significantly shallow, it is possible that the concentration of the load was too high or the material has disturbed the equilibrium between the mobile and stationary phases. If the down slope tends to be symmetrical initially,

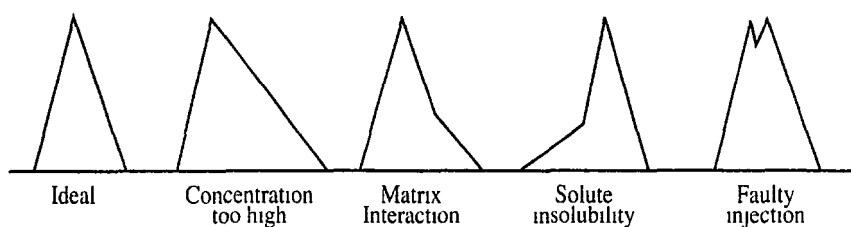


Fig. 5. Diagnosis of column performance by consideration of peak shape.

but then becomes shallow, it is common to assume that there is a poorly resolved component. However it may be suspected that interaction with the matrix is taking place. A shallow up slope may represent insolubility of the loaded material. A valley between two closely eluting peaks may suggest poor resolution, but can also be the result of a faulty sample injection.

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CHAPTER 26

Fast Protein Liquid Chromatography (FPLC) Methods

David Sheehan

1. Introduction

High-resolution protein separation techniques critically depend on the availability of column packings of small average particle size. This gives a minimum of peak broadening on the column owing to the direct relationship between the theoretical plate height parameter, H , and particle size (lowest values of H give highest resolution). High-performance liquid chromatography (HPLC) procedures exploit column packings with average diameters of as little as 5–40 μm . However, these are used in high-pressure systems (up to 400 bar) often with organic solvents and are generally limited to rather low sample loadings (1). To provide a more biocompatible high-resolution separation of biopolymers, including (although not exclusive to) proteins, Pharmacia LKB (Uppsala, Sweden) developed fast protein liquid chromatography (FPLC) in 1982 (2).

FPLC provides a full range of chromatography modes, such as ion exchange, chromatofocusing (3), gel filtration, hydrophobic interaction (4), and reverse phase (5), based on particles with average diameter sizes in the same range as those used for HPLC separations. These columns can accommodate much higher protein loadings than conventional HPLC, however, and use a wide range of aqueous, biocompatible buffer systems. Although almost 1000 reports of the use of FPLC have appeared in the literature in the decade since its introduction, two of the most popular FPLC modes are ion exchange and gel filtration. The objective of this chapter is to introduce the reader to these two FPLC chromatography

From *Methods in Molecular Biology*, Vol 59 *Protein Purification Protocols*
Edited by S Doonan Humana Press Inc, Totowa, NJ

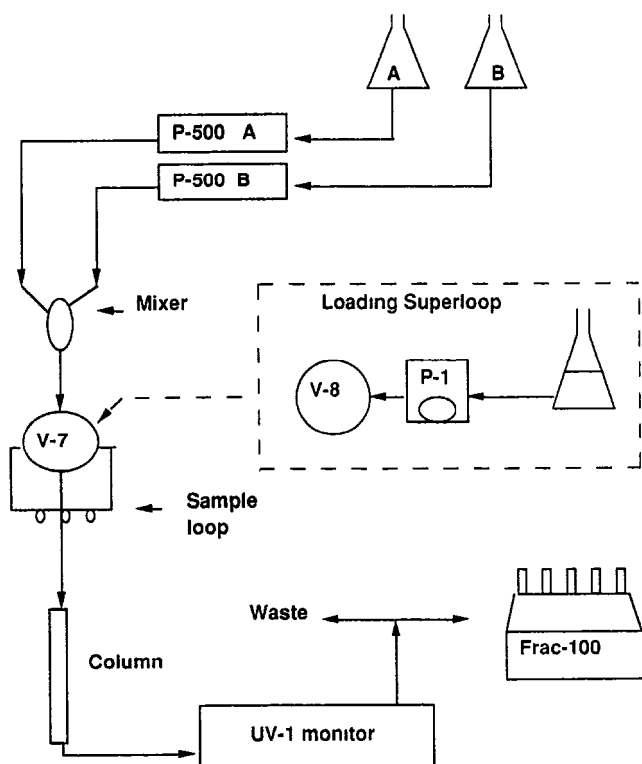


Fig. 1. Format used for FPLC chromatography. Information on other accessories and automation may be obtained from Pharmacia LKB. Superloop may be loaded using the arrangement shown within dashed lines.

modes as used in a typical protein chemistry laboratory. References 3–5 give examples of the use of the other main FPLC chromatography modes, which are not described in detail here.

2. Materials

1. The FPLC system consists of a program controller (LCC 500 Plus), two P-500 pumps (one each for buffers A and B), a mixer, prefilter, seven-port M-7 valve, assorted sample loops (0.025–10 mL), column, UV-1 UV monitor (fitted with an HR-10 flow cell and 280-nm filter), and a Frac-100 fraction collector. Associated equipment includes a superloop attached to an eight-port M-8 valve and a P-1 peristaltic pump for the introduction of samples up to 10-mL volume. All of this equipment is available from Pharmacia LKB. The most commonly used configuration for ion-exchange FPLC is shown in Fig. 1

2. Mono Q HR5/5 anion-exchange, Mono S HR5/5 cation exchange, HR10/10 rapid desalting, and Superose 12 HR 10/30 gel-filtration columns are obtained from Pharmacia LKB, and are stored in 20% ethanol. Sephadex G-25 is also obtained from Pharmacia LKB. A glass column (45 × 3 cm in diameter) is required. Filters (0.22 μ m) are from Millipore.
3. All reagents for buffer (*see* Notes 3 and 5) preparation are of Analar grade. Buffers should be prepared in HPLC or milli-Q-grade water, and require filtering and degassing immediately before use.

Buffer A: 10 mM Tris-HCl, pH 7.0.

Buffer B: 10 mM Tris-HCl, pH 7.0, 1M NaCl.

Buffer C: 50 mM Tris-HCl, pH 7.0, 100 mM KCl.

3. Methods

3.1. Sample Preparation

Generally, FPLC chromatography is performed on material that has already been subjected to some preliminary chromatography steps (e.g., for ion exchange using cellulose matrices, *see* Chapter 14; for affinity chromatography, *see* Chapter 16). For ion-exchange chromatography, the sample must then be desalted into buffer A (*see* Note 2). This may be accomplished by passing it through a column of Sephadex G-25 (45 × 3 cm in diameter, total bed volume 318 mL).

1. New Sephadex G-25 resin is de-fined by aspiration before packing as a slurry in water into glass column with a no. 1 glass sinter.
2. Column is equilibrated in three to four column volumes of buffer A. Equilibration is complete when the pH and conductivity of eluate are the same as buffer A at the same temperature. The sample (50 mL) is loaded under gravity flow. Buffer A is passed through the column. Protein is monitored by measuring A_{280} , and collected as a single large peak.
3. Conductivity and pH of desalted protein should be the same as those of buffer A at the same temperature.

Sephadex G-25 gel filtration is performed at 4°C in a cold room or cabinet (*see* Note 6). Sample is centrifuged (bench centrifuge, 3–5 min) before application to column.

3.2. Ion-Exchange FPLC

The FPLC system is operated usually at room temperature, taking care to hold samples on ice and to return eluted protein fractions to ice as soon as possible (*see* Note 6).

1. With no column in system (this may be replaced with some tubing), prime P-500 pumps A and B with filtered (0.22- μ m filter) and degassed buffers A and B, respectively.

- 2 Set pressure limits on both P-500 pumps well below the maximum for column in use (*see* Notes 4 and 11)
3. Equilibrate Mono Q column (1-mL vol) with 5 vol buffer A and 10 vol buffer B followed by 5 vol buffer A.
4. Wash loading loop with buffer A.
5. Load 0.5–10 mL sample (approx 1 mg/mL), and wash with buffer A. Collect wash through, and assay for protein of interest.
6. If protein has not bound, immediately replace Mono Q with Mono S, and repeat procedure described in steps 3–5 (generally, at pH 7.0, proteins that do not bind to one resin will bind to the other). If protein still does not bind, follow procedure given in Section 3.3.
5. Mono Q (or Mono S) column is developed with a 0–100% gradient of buffer B. Elution of protein is determined by A_{280} while protein of interest is assayed
- 6 Regenerate column by washing with 10 vol buffer B, followed by 5 vol buffer A. Column is now ready to receive another sample.

3.3. Scouting FPLC Methods

The aforementioned procedure describes a basic FPLC ion-exchange chromatography experiment. If the protein of interest has not previously been purified using FPLC, a new method of purification may be needed. Usually, 25–500 μ L loadings are used to scout a new procedure. The general approach used is as follows.

- 1 By varying % B at different time-points, create gradients of varying degrees of shallowness (down to and including isocratic). Knowing % B at which the protein of interest elutes, separation from other proteins can be improved by varying the gradient program around this value (6)
2. Having identified column (either Mono Q or Mono S) to which protein of interest binds, carry out chromatography at different pH values using various buffer systems (Table 1; *see* Notes 3 and 5). By carrying out rechromatography of a single sample at a number of different pH values on a Mono Q or Mono S column, this can often separate two proteins that may elute close to each other at a single pH.
3. Having identified a suitable buffer and gradient program for optimum separation of our protein of interest, the 500- μ L sample loading loop may be replaced with the superloop. This allows loading of up to 10 mL (20 mg protein) sample on the column. Fractions may be collected on the Frac-100 for further analysis.
4. Final adjustments to the gradient may be required to allow for the effect of scaling up the loading.

Table 1
Some Buffers Suitable for Ion-Exchange FPLC

pH range	Buffer ^a	Column
3.8–4.3	50 mM Na formate-formic acid	Mono S
4.5–5.2	20 mM Na acetate-acetic acid	Mono S
5.5–6.0	20 mM Histidine-HCl	Mono Q
6.0–7.6	50 mM Na phosphate	Mono S/Mono Q
7.6–7.8	50 mM HEPES	Mono S
7.0–8.0	10 mM Tris-HCl	Mono Q
8.0–9.0	50 mM Tris-HCl	Mono Q
9.0–9.5	20 mM Ethanolamine-HCl	Mono Q

^aThese may be used as buffer A (see Section 3.3.)

3.4. Gel Filtration FPLC

Gel filtration is often used as a final “polishing” step to remove minor contaminants from partially purified protein (see Note 9). It is also especially useful in determining native molecular weights for oligomeric proteins. Run times for samples may be as little as 30 min. Superose 12 has a fractionation range of 1–300 kDa. An identical procedure may be used for Superose 6 (fractionation range: 5–5000 kDa).

1. With no column in system, prime both pumps A and B with buffer C.
2. Set pressure limits on both P-500 pumps to the same value (that for the column in use; see Notes 4 and 11).
3. Connect Superose 12, and operate system at a flow rate of 0.5 mL/min for 90 min to equilibrate column.
4. Wash sample loading loop with buffer C.
5. Load 200 μ L sample (maximum) at approx 2 mg/mL.
6. Monitor A_{280} , collect peaks with Frac-100 fraction collector. Note elution volume of peak.

4. Notes

1. Prime pumps with water to remove 20% ethanol in which the system is stored. This avoids any risk of NaCl precipitation on contact of buffer B with 20% ethanol.
2. If the protein of interest is stable, then desalting may also be achieved by dialysis against three changes of 1 L buffer A (minimum of 6 h/change). Centrifuge sample after dialysis to remove any precipitated material. If the protein is particularly unstable then the rapid desalting HR 10/10 column may be included in the system before the ion-exchange column as shown in Fig. 1. This allows for desalting of sample in as little as 4 min.

3. The purpose of using a fairly neutral pH to begin with is to allow assessment of both the anion- and cation-exchange columns at the same pH. It is much quicker to swap columns than to reprime the entire system with new buffers at different pH values. The procedure outlined in Section 3.2. takes < 2 h to complete.
4. Flow rates of 1 mL/min are routinely achievable with newly purchased ion-exchange columns, but even with rigorous cleaning, back-pressure in the system soon rises appreciably after a number of uses. It is often better to accept a comparatively slow flow rate of 0.5 mL/min and modest back-pressures, since this seems to give better chromatography and to extend column life-times.
5. Although it is possible to carry out ion-exchange FPLC across a wide pH range, extremes of pH should be avoided if possible unless it is known that the protein of interest is stable at these pH values
6. Operation of the FPLC system in a cold room at 4°C may be essential for purification of some particularly labile proteins. However, permanent location in the cold may lead to deterioration of the system. A convenient approach to this problem is to mount the system on a trolley with the components electrically connected to an extension lead on the trolley. This may be wheeled into the cold room, the extension lead connected to a power point, and the system operated once it has cooled to the temperature of the room. After use, the system may be easily wheeled out of the cold room again.
7. It is sometimes desired to operate the system (gel filtration or ion exchange) in the presence of urea. It is not advisable to use high concentrations (e.g., 6–8M) of urea though, because urea may easily precipitate, clogging valves and leading to excessive wear on seals and gaskets. Urea is rarely used at concentrations higher than 6M, and special care has to be taken if the system is being operated at 4°C. It is essential when chromatography in the presence of urea is complete that the entire system is washed extensively with water before storing it in 20% ethanol. Otherwise, urea will precipitate on contact with the ethanol solution. Also, it is important not to allow air (e.g., air bubbles) to come into contact with buffers containing urea, since this will lead to crystal formation. Once formed, crystals are very difficult to remove.
8. It is sometimes advisable to rechromatograph the peak of interest. This is achieved by rapidly desalting the peak into buffer A and reapplying it to the column. This often produces a considerable improvement in purification, since contaminants “move” to one side of the chromatogram away from the peak of interest.
9. Although gel filtration is often used toward the end of a purification procedure, in the particular case of immunoglobulin purification from serum, it may be used with advantage at an early stage of the purification (7).

10. Calibration of Superose columns should be performed whenever a different chromatography buffer is used. A quick way to achieve this is to load standard proteins with widely separated molecular masses (e.g., cytochrome *c* and ovalbumin) in pairs (i.e., as a single sample).
11. Decrease in flow rate usually improves the resolution of gel-filtration chromatography.
12. When FPLC is complete, wash the system with water. Fill pumps with 20% filtered, degassed ethanol, wash loading loop with 20% ethanol, and place arm of Frac-100 in center of carousel (to minimize risk of damage to optical sensor).

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CHAPTER 27

Reversed-Phase Chromatography of Proteins

Bill Neville

1. Introduction

Reversed-phase HPLC (RP-HPLC) is now well established as a technique for isolation, analysis, and structural elucidation of peptides and proteins (1,2). Its use in protein isolation and purification may have reached a peak owing to recent developments in high efficiency ion-exchange and hydrophobic interaction supports, which are now capable of equivalent levels of resolution to RP-HPLC without concomitant risk of denaturation and loss of biological activity. Nevertheless, there are many applications in which denaturation may be unimportant and high concentrations of organic modifier can be tolerated (e.g., purity analyses, structural studies, and micropreparative purification prior to microsequencing). It is a widely used tool in Biotechnology for process monitoring, purity studies, and stability determinations.

1.1. Principle of Separation

Application of a solvent gradient is generally superior to isocratic elution because separation is then achievable within a reasonable time frame and peak broadening of later-eluting peaks is reduced and thus sensitivity increased. In gradient elution RP-HPLC, proteins are retained essentially according to their hydrophobic character. The retention mechanism can be considered either as adsorption of the solute at the hydrophobic stationary surface or as a partition between the mobile and stationary phases (3).

From *Methods in Molecular Biology, Vol. 59 Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

In the first case, retention is related to total interfacial surface of the RP packing and is expressed by the adsorption coefficient K_A . Retention is based on a hydrophobic association between the solute and the hydrophobic ligands of the surface (4). By increasing solvent strength of the mobile phase, attractive forces are weakened and the solute is eluted. The process can be regarded as being entropically driven and endothermic, i.e., both ΔS and ΔH are positive (5). Through the relation between the solute capacity factor k' and the mobile and stationary phase properties, solvophobic theory permits prediction of the effect of the organic modifier of the mobile phase, the ionic strength, the ion-pairing reagent, the type of ligand of the RP packing, and other variables on the chromatographic retention of proteins.

The second case assumes a partitioning of the solute between the mobile and stationary phases, the latter being regarded as a hydrophobic bulk phase. This system resembles an *n*-octanol/water two-phase system where the selectivity is expressed by the partition coefficient P of the solute. Both retention principles have their pros and cons (3). In essence, the effective process is largely dependent on the molecular organization of the bonded *n*-alkyl chains of the RP packing in the solvated state and the size and conformation of the solute.

Several methods have been developed for correlating peptide structures with RP-HPLC retention. One method sums empirically derived retention coefficients, representing the hydrophobic contribution of each amino acid residue (6). The assumption behind this approach is that each amino acid residue contacts the adsorbent surface and that the total hydrophobicity of the solute determines RP-HPLC retention. This approach works well for peptides of 20 amino acids or less (7), but generally is not accurate for predicting retention of larger proteins. This is caused by the peculiarities, especially of polar basic moieties, leading to either less retention or irreversible adsorption by residual matrix silanols. Studies with peptides containing amphiphilic helices indicate that secondary structure plays a large role in determining the surface of a polypeptide, which is exposed to the hydrophobic adsorbent (8). Other studies have shown that proteins are less sensitive to changes in the HPLC support or bonded phase than are corresponding small molecules (9). These results imply that protein retention in RP-HPLC is governed mainly by the protein surface and not by the support surface chemistry. Although some proteins show a loss of biological activity during RP-HPLC, others

retain full biological activity. Thus, the hydrophobic forces necessary for RP binding compete with those required to maintain the protein's secondary and tertiary structures. Through steric, hydrophobic, and ionic constraints, these structural features define a chromatographic surface or "footprint" of the protein which in turn defines its RP binding. This concept of multipoint attachment of a protein to adsorbent surface is consistent with a relatively large chromatographic footprint for a protein compared with that of a small molecule.

1.2. RP-HPLC Columns

Optimization of chromatographic separation is determined by an appropriate choice of the column matrix, the mobile phase, and the temperature of separation. RP-HPLC has rapidly become the most widely used tool for the separation, purification, and analysis of peptides. For proteins, RP-HPLC has suffered from problems associated with denaturation, including loss of activity, poor recoveries, wide and misshapen peaks, and ghost peaks (10,11). Some of these problems may be column related (12) and insurmountable, although others can be overcome by selective optimization of extra-column variables, such as sample pretreatment and mobile phase and hardware considerations (13). As with small-molecule HPLC, there are a large number of RP-HPLC columns for use with peptides and proteins. In trying to select the best column for a new application or for broad applicability, one should consider such column variables as support, bonded phase, pore size, particle size, and column dimensions. These factors are dealt with in the following.

1.2.1. Packing Support

Most commercial RP-HPLC columns for peptides and proteins are silica based, as a consequence of its use for the separation of small molecules over many years. Silica offers good mechanical stability and allows a wide range of selectivities by virtue of the bonding of various phases. Although it is well known that silica-based columns are not stable at basic pH, recent reports suggest that in the acidic mobile-phases typically used, the bonded phase may also be slowly dissolved from the base silica (14,15). Degradation can thus affect not only reproducibility, stability, and lifetime of a column, but also recovery and, in addition, modify selectivity as the silica surface becomes uncovered. An added problem could be contamination of recovered products with silica and bonded-phase material. Because of the limitations associated with silica-based

columns, polymer-based columns have gained increased popularity for peptide and protein separations by RP-HPLC. Polymer-based columns, such as divinylbenzene-crosslinked polystyrene are usually stable over the pH range 1.0–14.0, making them more widely applicable with the added advantage that they are easier to clean up after use. Although most polymer-based columns are still inferior to their silica-based counterparts in terms of mechanical strength, selectivity, and efficiency (*see* Note 1), newer materials, such as PLRP-S (Polymer Laboratories, Church Stretton, Shropshire, UK) show superior performance in complex protein separations (13). Recent studies have shown that the use of polymer-based columns at high pH (>8.0) can offer unique selectivities and thus complement separations with classical acidic mobile phases (16,17). These advantages, coupled with superior stability, longer column life, and better reproducibility make polymer-based columns a good first choice for complex protein separations.

1.2.2. Bonded Phases

A wide variety of bonded phases are available for the separation of peptides and proteins by RP-HPLC, but the most common are *n*-alkyl bonded phases, namely C₄, C₈, and C₁₈. In general, C₄ and C₈ phases are preferable for more hydrophobic samples and the C₁₈ phase for hydrophilic samples. In general, significant differences may be observed in the separations achieved on nominally the same column but from different sources (12). For proteins, a C₈ column is generally a good compromise. If the proteins of interest are too strongly retained a C₄ column could be substituted.

1.2.3. Particle Size

As with small molecules, theory predicts that column performance should increase with decreasing particle size. This is generally the case for most commercially available RP-HPLC columns used for protein separations. Columns with particle sizes of 5–10 μm show little difference in performance. Columns with particle sizes of 2–3 μm are now available and can give much greater resolution (*see* Note 2).

1.2.4. Pore Size

The standard pore size for separation of small molecules is 80 Å but for the separation of proteins a pore size of 300 Å has become accepted as the norm. For proteins >100 kDa, pore sizes of >1000 Å may be even

better (12). For complex samples, if all other column variables are equal, it would be better to use a larger pore size to ensure that restricted diffusion or exclusion from the pores is not encountered by very large proteins.

1.2.5. Column Dimensions

The two factors to consider when selecting a column size are the efficiency and sample loading capacity. In the case of proteins, column length contributes little to efficiency. In addition, longer columns may have an adverse effect on protein recovery. When considering gradient rather than isocratic elution, resolution of a mixture of proteins depends directly on the particle diameter of the packing, but less on the column length and flow rate for gradients of equal time. However peak height, which will determine the detection limit, depends inversely on the particle diameter, flow rate and, to a lesser extent, gradient time. An increase in column length would have little effect on resolution since by the time a protein band reaches the extra column length, it would be overtaken by the gradient and the organic modifier concentration would ensure that the protein spent little time in the adsorbed state. The effect of additional column length is thus largely passive and may even lead to increased band broadening. Long columns might be useful in isocratic protein separations but if extra capacity were needed, a short fat column might be a better choice than a long thin column. Having selected the column length, choice of column diameter should be based on required sample capacity. Columns of 1-mm internal diameter (id) (microbore) to 2.1-mm id (narrowbore) are best suited to submicrogram levels of material thus minimizing sample loss and also increasing sensitivity of detection. For analytical applications in the microgram to low milligram range, columns of 4.6-mm id (analytical) are best.

1.3. Mobile Phase

The mobile-phase composition is the most readily changed variable in an RP-HPLC separation. In normal usage the mobile phase consists of a mixture of water, a miscible organic solvent, and dissolved buffers and salts. A buffer, such as a low concentration of a strong acid or salt, is essential for chromatography. If a protein is adsorbed in the absence of a salt or acid, no increase in the proportion of organic component will elute it. In addition, other components, either solvents or solutes, may be added in order to affect the separation. This may be to accelerate or delay elution, improve peak shape, or adjust the elution position of some compo-

nents with respect to others thus affecting selectivity. Other important variables are the pH of the mobile phase and the concentration and type of ions present. The ionic strength is a factor in limiting "mixed-mode" retention involving silica hydroxyls, but the choice of ions can also affect the solubility and stability of the protein in the mobile phase and, through ion pairing, the distribution of the protein between stationary and mobile phases. The net effect depends on the combination of column packing and mobile phase.

Numerous studies have examined the effect of mobile phase on the retention and selectivity of proteins in RP-HPLC. Systematic studies by several researchers over the past 10 yr have resulted in the publication of a number of fairly standardized sets of elution conditions. The behavior can be altered by the variation of a number of mobile-phase parameters. Altering the nature of the organic component of the mobile phase can alter column behavior but not usually in a predictable manner. The most successful way to alter column behavior is to manipulate the ionic component of the solvent system. By altering the nature and strength of the ion-pairing reagent and the pH of the mobile-phase, it is possible to exploit both the acidic and basic character of proteins in a systematic and predictable way. Without prior knowledge of the primary structure of a peptide or protein, one can be certain of some hydrophobic character by virtue of the content of the lipophilic amino acids leucine, iso-leucine, methionine, phenylalanine, tyrosine, tryptophan, and valine. It will also have hydrophilic character largely because of the presence of aspartic acid, glutamic acid, arginine, and lysine. The ratio between hydrophobic and hydrophilic amino acids determines the initial behavior observed for any protein in the most common RP-HPLC solvent system in use today, namely, aqueous acetonitrile containing 0.1% trifluoroacetic acid (TFA). To some extent hydrophilic character of solutes is suppressed at pH 2.0. All carboxylic acid groups are protonated under these conditions and their contribution to hydrophilic character is reduced. This in turn will improve peak shape. Basic amino acids (e.g., arginine, lysine, and histidine) will remain as cations which in turn will lead to a marked reduction in silanophilic solute-matrix interactions.

1.3.1. Organic Component of Mobile Phase

The most popular organic solvents are acetonitrile, 1-propanol, and 2-propanol. Propanols are much more viscous than acetonitrile, giving

higher column backpressures (*see* Note 3) but these are not significant when low flow rates and short columns are used. The acetonitrile concentration needed to elute a protein is considerably higher than the equivalent 2-propanol concentration, perhaps half as high again, and thus proteins are likely to be seriously denatured. There is also evidence that acetonitrile is intrinsically a more powerful denaturant than the alcohols. Recoveries are often higher in propanol-containing mobile phases, especially for the more "difficult" model proteins, such as ovalbumin. Acetonitrile and 2-propanol can be frozen in dry-ice/alcohol mixtures and lyophilized. All solvents should be of the highest quality available (*see* Note 4).

1.3.2. Minor Mobile-Phase Additives

The most popular minor components of the mobile-phase are the perfluoroalkanoic acids (e.g., TFA and heptafluorobutyric acid). These appear to solubilize proteins in organic solvents. TFA is used in preference to other fully dissociated acids because it is completely volatile and will not corrode stainless steel. TFA also acts as a weak hydrophobic ion-pairing reagent. Most RP columns employed for RP-HPLC are silica-based and apart from the suppression of silanol ionization under acidic conditions (thereby suppressing undesirable ionic interactions with basic residues), silica-based columns are more stable at low pH. As mentioned previously (Section 1.3.), an acid pH is preferred although many polypeptides will lose their tertiary structure at pH 2.0–3.0 (*see* Note 5).

Other popular additives are pyridinium acetate buffers and buffers based on phosphoric acid, particularly triethylammonium phosphate (TEAP). TEAP is useful in circumstances where tailing is suspected because of nonspecific interactions with surface silanols. Proteins are eluted at a higher percentage of organic component with an equivalent concentration of phosphoric acid. Sulphate, phosphate, perchlorate, and chloride are all transparent in the far UV, while acetate, formate, and fluoroalkanoic acids must be used at concentrations <20 mM if wavelengths below 220 nm are to be monitored. Pyridine-containing eluents cannot be monitored by UV. Ion-pairing reagents, such as heptane and octane sulphonic acids, sodium dodecyl sulphate, or alkyl ammonium salts may be added to the mobile phase to selectively increase the retention of proteins carrying larger charges of opposite sign. Other additives might be needed depending on the nature of the protein being analyzed (*see* Note 6)

1.3.3. Gradient Elution

The theory of gradient elution is now well established and a good understanding exists concerning the effects of gradient steepness on separation. Geng and Regnier (18) developed the stoichiometry factor, Z , which represents the number of solvent molecules displaced during the binding of the protein to the column packing. Thus, the value of Z is a measure of the size of the effective chromatographic footprint of that protein.

For RP-HPLC, Eq. (1) has been derived which shows that Z is proportional to the term S :

$$Z = 2.3\phi S \quad (1)$$

where ϕ is the volume fraction of the organic in the mobile-phase (19). Under defined gradient conditions, every protein will have an S value described by Eq. (2):

$$\log k' = \log k_w - S\phi \quad (2)$$

where k' is the retention of the solute (capacity factor) and k_w is the value of k' when water is the mobile phase ($\phi = 0$). Values of k_w and S are characteristic of each solute in a sample. Although empirically derived, Eq. (2) is a good approximation of RP-HPLC retention in the gradient mode. For a series of related proteins of nearly identical composition, the hydrophobic contribution to RP retention will be similar and thus observed changes in Z (and S) should reflect changes in the structure of the protein which contacts the adsorbent surface. Examination of Eq. (2) reveals why small changes in the organic composition of RP eluents result in large changes in protein retention as opposed to small molecule separations. Proteins will have a much larger chromatographic footprint (Z and S) than small molecules. Thus, changes in ϕ (% organic) will be amplified by large values of S , resulting in even larger changes in retention ($\log k'$). Many examples of the applicability of Eq. (2) for peptide or protein samples have been reported. Values of S and k_w for each solute can be obtained from two experimental gradient separations of the sample. Retention times in gradient elution then can be predicted as a function of gradient conditions. Two components that elute adjacent to one another in a chromatogram will often show significant changes in band spacing when isocratic solvent strength or gradient steepness is varied. The resolution of such band pairs can usually be accomplished when values of S for the two components differ by 5% or more.

1.4. Temperature

Protein samples will often contain many individual components. A complete separation of such samples poses a real challenge because statistical considerations suggest that one or more peak pairs usually will be resolved poorly. To overcome this dilemma, a systematic variation of separation selectivity would be required. This approach has been widely used for the separation of typical small-molecule samples. In the case of peptide and protein samples, however, the control of selectivity has received little attention. When a change in selectivity is desired, the usual approach is a change of column or mobile phase. The use of elevated temperatures for RP-HPLC of protein samples has been advocated primarily as a means of increasing column efficiency or shortening run time. For samples of this type, a few studies have shown that a change in column temperature can also affect separation selectivity. A combination of low pH and higher temperature can result in a very short life for commonly used alkyl-silica columns which in turn limits the application of temperature optimization. This problem has recently been overcome by the development of "sterically protected" RP packings that are extremely stable at low pH (<2.0) and high temperature (>90°C). The combined use of temperature and gradient steepness would provide an efficient procedure for the control of peak spacing and optimization of separation. At the same time, this approach to selectivity control is more convenient than alternatives, such as change of column or mobile phases because temperature and gradient steepness can be varied via the HPLC system controller.

2. Materials

To illustrate the general points in Section 1., the separation of a mixture of standard proteins is described.

2.1. Equipment

Analysis of proteins by RP-HPLC is suited to most commercially available equipment. Systems require a column packed with a suitable matrix giving adequate retention and resolution, a detector to monitor eluted proteins, a chart recorder or data system to view detector response, and a fraction collector for recovery of eluted proteins, if further analysis is intended. If high sensitivity is a requirement, then narrowbore or microbore chromatography should be considered and specialized equipment may be necessary (*see* Note 7). Proteins are usually monitored by

their absorbance in the UV either at 280 nm, which is suitable for proteins with aromatic amino acids, or at 206 nm, which is characteristic of the peptide bond (*see* Note 8). Most modern chromatographic systems now have computers to control operating parameters and to collect and store data. Microfraction collectors are also available with low internal diameter tubing to reduce the volume to a minimum between detector and collection tube in order to more nearly reflect the real time elution situation.

2.2. Column

A wide variety of suitable C₈ or C₁₈ columns are commercially available for protein analysis. The price of modern columns is such that little advantage is gained by self-packing a column (*see* Note 9). The column must be able to withstand moderate pressures (250 bar) and be resistant to the mobile phase (pHs typically 1.0–8.0 for silica columns). A guard column (containing the same or a very similar packing material) should be used as a matter of course to extend analytical column lifetime.

2.3. Injector

For most purposes a conventional HPLC manual injection valve (e.g., Valco or Rheodyne) with a fixed volume loop (e.g., 10 μ L) is adequate. For microbore analysis, injectors with small internal loops may be better. Many modern chromatographs are now fitted with auto-samplers whereby injection volume can be varied (e.g., 0.1–200 μ L) via operating computer.

2.4. Eluents

Use Analar-grade or even Aristar-grade reagents and distilled/deionized water for all solutions (*see* Note 10). All eluents should be filtered through a 0.2- μ m filter and degassed under vacuum before use.

1. Eluent A: 0.1% TFA (v/v). To 1 L of water add 1 mL of TFA.
2. Eluent B: acetonitrile:water (70:30) + 0.085% TFA (v/v). To 700 mL of acetonitrile add 300 mL of water and 0.85 mL of TFA (*see* Note 8).

Eluents should be degassed by a slow stream of helium while on the chromatograph, if at all possible, to counteract the effect of dissolved air. If a ternary system is available, acetonitrile can be programmed into a gradient to fully elute hydrophobic materials which otherwise might be retained by the column packing.

2.5. Detector

By far the most common detector used in chromatography is the variable wavelength UV detector, operated at 210–220 or 280 nm. Most pro-

teins have an absorption maximum at about 280 nm because of the presence of aromatic amino acids, which falls to a minimum at 254 nm, a popular wavelength for fixed-wavelength HPLC detectors. Highest sensitivities are obtained by monitoring the strong absorption bands which peak below 220 nm owing to the "peptide bond" itself (*see* Note 8).

Proteins are usually eluted as fairly broad peaks (*see* Fig. 1, a protein test mixture of insulin, cytochrome *c*, lactalbumin, carbonic anhydrase, and ovalbumin) and thus place fairly modest demands on the dimensions of a normal flowcell (typically 8 μ L with pathlength 1 mm). An alternative to UV detection is to monitor intrinsic fluorescence. The usual excitation wavelength is 280 nm with emission monitored at either 320 (for tyrosine) or 340 nm (for tryptophan). The main advantage is a several-fold increase in sensitivity and an improvement in selectivity of detection, particularly in distinguishing protein peaks from "ghost" peaks originating from eluent impurities.

2.6. Protein Sample

A mixture of test proteins, namely insulin (from bovine pancreas), cytochrome *c* (from horse heart), α -lactalbumin (from bovine milk), carbonic anhydrase (from bovine erythrocytes), and ovalbumin (from hen egg whites) can be used as a column test. Two of the proteins, insulin and cytochrome *c*, are difficult to resolve. A partial to complete separation should be achievable when a column with a high enough number of theoretical plates is used. Of the five standard proteins, ovalbumin is not only the most hydrophobic, but also the most difficult to recover. The utility of the column from a recovery perspective will be illustrated by the size of this peak.

3. Method

1. Set up appropriate pump, temperature, injection, detector, and integration methods, as required by chromatograph being used.
2. Install appropriate column (C_8 or C_{18} bonded silica, 300 Å, 5 μ m, 10–25-cm length, 2.1 or 4.6 mm id).
3. Equilibrate the column by pumping eluent A at 200 μ L/min (narrowbore, 2.1 mm) or 1 mL/min (normalbore, 4.6 mm) for several column volumes. Monitor detector baseline (214 nm) until flat and then zero the detector. Inject 5 μ L of HPLC quality water and run a blank gradient to identify any system peaks (*see* Note 10) and to balance baseline ($\pm$ TFA), if necessary (*see* Note 8).
4. Dissolve protein mixture in HPLC quality water at a concentration of approx 2 μ g/ μ L to produce a stock solution. The stock solution should be

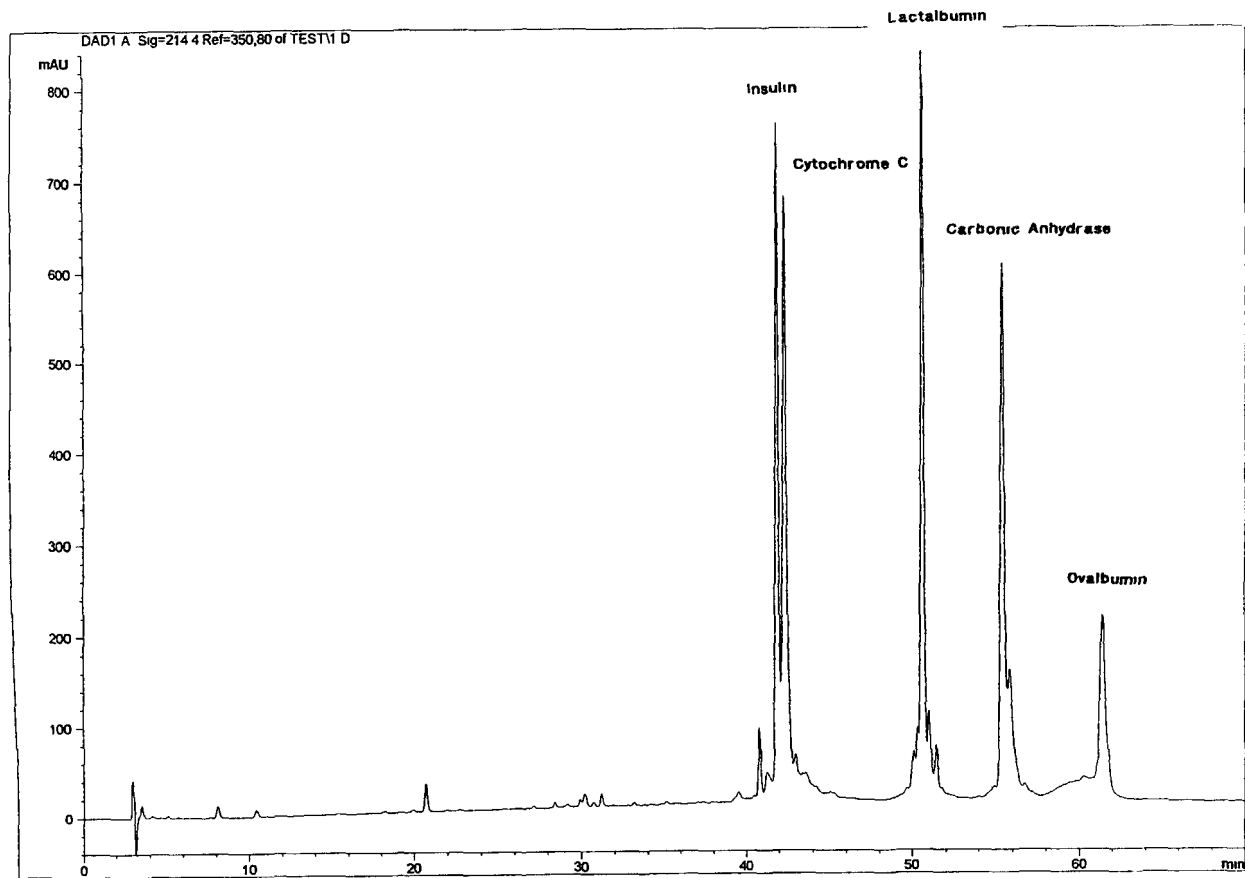


Fig. 1. RP-HPLC of a protein test mixture. The conditions used for chromatography are described in Section 3.

diluted further with eluent A (1:1) to produce a working solution of 1 $\mu\text{g}/\mu\text{L}$ of each protein.

5. Apply the sample (5 μL) at initial gradient conditions (see Note 11).
6. Start gradient to elute proteins.

The elution profile obtained is shown in Fig. 1.

4. Notes

1. Polymer columns unfortunately suffer from a relatively low pressure resistance and, in general, pressure should be kept below 2000 psi. Furthermore, the physicochemical properties are governed by a solvent-dependent swelling and shrinking of the organic matrix which is associated with concomitant changes in pore diameters, leading to changes in mass transfer characteristics of the column. This effect may be encountered during gradient elution starting with an aqueous phase (by which only little or no wetting of the column matrix takes place) and termination with a pure organic solvent (e.g., methanol, acetonitrile, 2-propanol). Pore diameter will continuously change and often may be the cause of poor chromatographic performance. In contrast, silica gel as the starting material for subsequent alkyl silylation offers the advantage that a great number of different alkyl silica substituents can be bound to the silica surface.
2. Any gain in column efficiency can be outweighed quickly by increased column backpressure, susceptibility to plugging, and shorter column lifetime.
3. The pressure of an acetonitrile-containing mobile phase decreases approximately linearly with increasing percentage of organic modifier, whereas aqueous mixtures of methanol, ethanol, 1-propanol, and 2-propanol exhibit a more or less convex pressure dependence on the percentage of organic modifier. Maximum values are reached on average between 40 and 70% and thus a maximum in viscosity. To reduce unacceptably high column pressures, separations may have to be carried out at higher temperatures.
4. Water needs to be free of UV-absorbing organic impurities, which can be adsorbed onto the column during equilibration and the early part of a gradient and which may be eluted later as "ghost" peaks. Water should be freshly distilled and deionized and stored out of contact with plastic thus avoiding dissolution of plasticizers.
5. This may not necessarily be fatal for resultant activity but if the polypeptide consists of subunits stabilized by noncovalent forces, the individual chains will almost certainly be separated during the chromatography.
6. Metal ions, such as calcium, may be included to enhance the stability of a protein and low concentrations of nonionic detergents may improve the

behavior of hydrophobic membrane proteins. Finally, guanidinium hydrochloride or urea, at moderate concentrations, may be added to elute very hydrophobic membrane proteins.

7. In order to obtain good performance from microcolumns, it is very important to have an HPLC system with very low dead volume so that extra-column peak broadening does not destroy the resolution achieved by the column.

Consideration should be given to the minimization of dead volume at all points in the chromatographic system. If there is a large dead volume between the point at which the gradient is mixed and the injector, then lag will be introduced into the gradient. Also the gradient system must be capable of delivering the gradient accurately and reproducibly at low flow rates. A suitable mixing device (volume <200 μ L) may need to be added to the system to enable a reproducible gradient to be delivered at very low flow rates

8. During a gradient there is usually an associated change in background absorption. This may be caused directly by the increase in the concentration of organic component or by the effect of this concentration change on other mobile-phase components. In addition, refractive index changes may be detected in the flowcell. Usually not much can be done about flowcell design but careful choice of wavelength and pH can minimize the change in absorption spectrum of mobile-phase components. For TFA and acetonitrile, the optimum wavelength is 214 nm. Alternatively, the concentration of other components in the mobile phase can be adjusted to compensate or a small amount of a UV absorbing solute can be added to one of the eluent reservoirs. For example, in a TFA/water/acetonitrile system monitored at 214 nm, the rise in background can be balanced by using 0.1% (v/v) aqueous TFA as eluent A and 70% aqueous acetonitrile, containing 0.085% (v/v) TFA as eluent B.
9. It is generally accepted that Vydac columns are the "industry standard," against which other manufacturers' columns are compared. Packing columns is based on know-how. Packing being commercially important, know-how is not readily imparted. Packing LC columns is still not fully understood. Commercial columns are usually supplied with a test and a guarantee to replace the column if performance does not match that specified. Although packed columns cannot be expected to last indefinitely, a usual lifetime of several months and several hundred injections is a reasonable objective.
10. HPLC quality water is commercially available from a number of suppliers if access to high-quality distilled or distilled/deionized water is not possible. A definition of HPLC quality is when 40 mL water is pumped onto the column and eluted with a linear water/acetonitrile gradient at 5%/min, at a flow rate of 2 mL/min, and absorbance of largest eluted peak, monitored at 254 nm, is <0.002 AU.

11. Proper gradient conditions are critical in optimizing separations of proteins by RP-HPLC. The gradient time, range, and shape are all important and must be optimized for a given sample. A typical starting gradient would be a 2%/min linear increase in acetonitrile over perhaps 70 min. A washing step of 100% eluent B then should be included before returning to initial conditions, allowing sufficient equilibration before the next injection.

Column re-equilibration time is an extremely important variable. The time must be sufficient to allow the initial mobile phase to equilibrate in the pores of the column packing (at least 3 column volumes) and should be exactly repeatable if precise retention times are required. A linear gradient of the above type (0.5–5%/min increase in acetonitrile) should allow resolution of the proteins in the test mixture. For increased resolution of certain of the proteins, a number of steps may have to be included in the gradient profile.

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CHAPTER 28

Extraction of Membrane Proteins

Kay Ohlendieck

1. Introduction

The fluid mosaic model of biomembrane structure, proposed in 1972 by Singer and Nicolson (1), defines two classes of membrane proteins that are associated, to varying degrees, with the phospholipid bilayer. In addition to peripheral and integral membrane proteins, a third class of membrane proteins is represented by lipid-anchored proteins. As well as these protein-membrane interactions, certain components of the membrane cytoskeleton and the extracellular matrix are also directly or indirectly associated via binding proteins or receptor molecules with membranes. Treatment of biological membranes with salt solutions or change in pH usually dissociates peripheral proteins, since these extrinsic membrane proteins interact with the membrane surface mostly via electrostatic and hydrogen bonds. Integral membrane proteins that possess hydrophobic surfaces are more strongly associated with the bilayer, and these intrinsic proteins extend across or are partially inserted into the lipid bilayer. Extraction of integral membrane proteins is commonly accomplished by solubilizing the protein containing membrane fraction using a variety of detergents (2) (Fig. 1).

1.1. Peripheral Membrane Proteins

Since peripheral membrane proteins are extracted by relatively mild treatments, care must be taken not to partially release these membrane proteins accidentally during subcellular fractionation procedures prior to purification. Isolation of a soluble form of an extrinsic membrane protein is usually achieved by a single technique or combinations of mild

From *Methods in Molecular Biology*, Vol. 59 *Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

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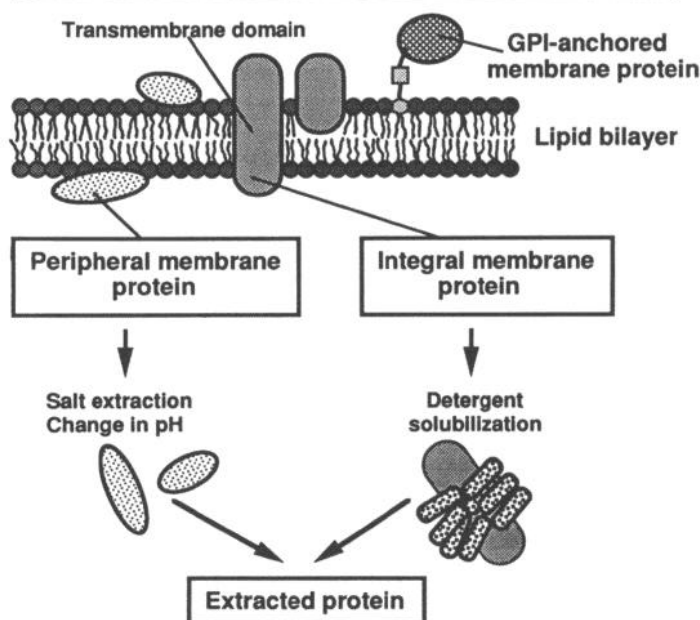


Fig. 1. Diagram of the different kinds of membrane proteins and the most commonly employed methods to extract them from biological membranes.

extraction procedures, depending on the particular properties of the protein under investigation and the starting material, i.e., a particular sub-cellular fraction enriched in the peripheral protein (3–5). Most extraction procedures aim at the disordering of the structure of water, the disruption of weak electrostatic interactions and hydrogen bonds, and occasionally weak hydrophobic interactions in order to break the interactions between the extrinsic proteins and the membrane. Following extraction for 10–30 min at 40°C, the remaining membrane bilayer and its associated integral proteins are separated by centrifugation (30–60 min at 100,000g), and the released peripheral membrane proteins are recovered in the supernatant. The following list summarizes commonly used procedures for the extraction of peripheral membrane proteins:

1. Treatment with alkaline buffers (pH 8.0–12.0).
2. Treatment with acidic buffers (pH 3.0–5.0).
3. Use of metal chelators (10 mM EGTA or EDTA).
4. Treatment with high-ionic-strength (1M NaCl or KCl).

5. Treatment with denaturing agents (i.e., urea).
6. Treatment with organic solvents (i.e., butanol).
7. Sonication of membrane fractions.

Usually extraction procedures employing high-ionic-strength NaCl or KCl, alkaline, or acidic buffers and metal chelators result in a relatively distinct separation between solubilized peripheral proteins and membrane-associated integral membrane proteins. However, the nonspecific association of soluble proteins with membrane fractions and/or the entrapment of cytosolic proteins in membrane vesicles during subcellular fractionation, as well as the partial release of integral membrane proteins even under mild conditions of extraction and the existence of certain proteins in both integral and soluble isoforms, makes the interpretation of these kinds of differential extraction procedures sometimes difficult (3–5).

1.2. Integral Membrane Proteins

Extraction of integral membrane proteins is most conveniently achieved by the use of detergents. Detergents are amphipathic molecules that contain both hydrophobic and hydrophilic moieties, and the preferred form of detergent aggregation in water is the formation of micelles. Detergent micelles are characterized by a unique critical micelle concentration (CMC), and below the specific CMC value, individual detergent molecules predominate in solution. CMC values differ quite significantly between individual classes of detergents, i.e., the CMC for octylglucoside is 25 mM, whereas Triton X-100 exhibits a CMC of 0.3 mM (2,6,7). With respect to micelle size, certain detergents form relatively large micelles, i.e., Triton X-100 forms micelles with approx 150 detergent molecules/micelle and a molecular mass of 90–95 kDa (2). A detailed listing of detergents used in the biological sciences and a description of their individual properties can be found in a review on detergent structure by Neugebauer (6). Ideally, the detergent of choice should not only be available in pure form and be relatively inexpensive for its usage in large-scale preparation, but it should also sufficiently solubilize the membrane protein under investigation without irreversibly denaturing it. Furthermore, easy removal of excess detergent from the solubilized protein fraction is an additional criterion for the choice of detergent, which will be discussed in Chapter 29. For comprehensive information on the availability, purity, unit prices, and sources of commonly used detergents for the solubilization of biological membranes, *see* the review article by Jones et al. (7).

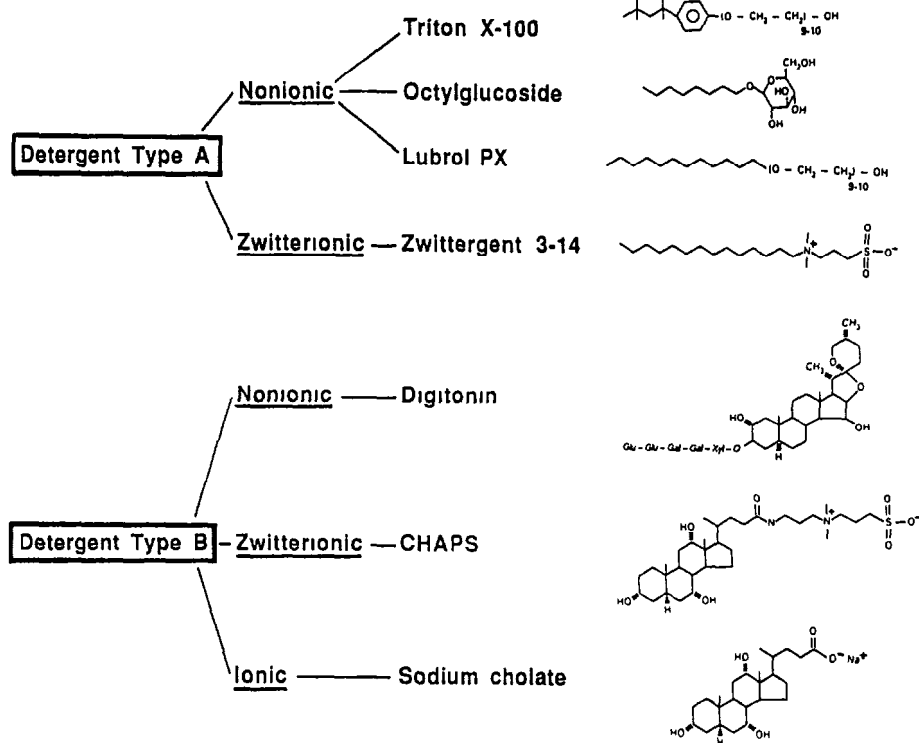


Fig. 2. Commonly used detergents employed in the solubilization of biological membranes.

A good initial selection of suitable detergents for pilot experiments on the extraction of novel integral membrane proteins with little knowledge of their physical and chemical properties is summarized in Fig. 2. As extensively reviewed by Helenius and Simons (2), detergents may be classified by their overall chemical structure as Type A or Type B detergents, and further subdivided according to their electric charge as non-ionic, ionic, or zwitterionic detergents. Although Type A detergents exhibit flexible hydrophobic tails and hydrophilic head groups, Type B detergents are more rigid and are cholesterol-based structures with amphiphilic properties (Fig. 2). Commonly used Type A detergents are the nonionic components Triton X-100, octylglucoside, and Lubrol PX, as well as the zwitterionic detergent sulfobetaine 14 (Zwittergent 3-14). The nonionic detergent digitonin, the zwitterionic component CHAPS,

and the ionic bile salt sodium cholate are typical Type B detergents extensively used in biological research (2,6,7). Many other detergents exist that are closely related to the described Type A and Type B detergents, and are listed in recent review articles on membrane solubilization (6,7). The ionic detergent sodium dodecyl sulfate, although widely used in analytical and preparative polyacrylamide gel electrophoresis techniques (8), is not usually employed in the initial extraction of membrane proteins and therefore is not further discussed in this chapter.

1.3. Experimental Design

Although the design of an extraction experiment may partially be based on the physical and chemical properties of the specific detergent used and the biological characteristics of the particular integral membrane protein and subcellular fraction under investigation, often a series of small-scale trial and error experiments is a good start to determining conditions for optimum solubilization. In pilot experiments, aliquots of the particular biological membrane are incubated with different concentrations of a variety of commonly used detergents. Initially, standard conditions are used with respect to buffer composition, salt solutions, temperature, and incubation time. In order to optimize the extraction process, these conditions may be varied. Most importantly, the denaturing properties of individual detergents, compatibility of detergents with divalent cations, the pH dependency of detergent solubility, spectral properties of detergents, and electrostatic and temperature effects on detergent behavior should be taken into account when choosing the appropriate buffer conditions and the ratio of detergent to protein (6). Effective solubilization is usually achieved in a well-buffered, physiological pH environment and the addition of 0.15M NaCl to the extraction mixture for optimum ionic strength. Membrane preparations are used at a protein concentration of 1–5 mg/mL and are solubilized by detergent concentrations of 0.1–5% (v/v). By convention, retention of a membrane protein in the supernatant following centrifugation for 60 min at 100,000g after solubilization defines this protein operationally as soluble (6). However, in large-scale preparations, shorter spins and lower g-forces might be sufficient to extract a high yield of a particular membrane protein. To determine if the integral protein of interest was sufficiently solubilized by a particular concentration of a detergent, a relatively simple and fast assay procedure should be chosen. Enzyme assays or ligand

binding assays, as well as cell biological test systems are certainly very convenient to evaluate the efficiency of the extraction protocol, and might also be useful in determining the potential loss of biological activity during the solubilization procedure. If a specific labeled probe or antibody is available to the integral membrane protein to be solubilized, overlay assays with immobilized protein fractions and immunoblot analysis using minigel systems can be performed reasonably quickly and with large numbers of samples.

To illustrate the use of commonly employed methods and reagents for the extraction of peripheral and integral membrane proteins, this chapter covers the initial isolation of a typical extrinsic protein, calsequestrin, and that of a typical intrinsic protein, Ca^{2+} -ATPase. Both proteins are major constituents of the sarcoplasmic reticulum from skeletal muscle. The Ca^{2+} -ATPase is localized to the longitudinal tubules and is responsible for the resequestration of calcium into the lumen of the sarcoplasmic reticulum following muscle contraction (9,10). Calsequestrin represents a high-capacity, intermediate affinity calcium-binding protein localized to the luminal site of the junctional sarcoplasmic reticulum and increases the calcium holding capacity of the terminal cisternae in muscle (11,12). These properties make both proteins key components of calcium homeostasis and excitation-contraction coupling in muscle (13).

2. Materials

2.1. Preparation of Sarcoplasmic Reticulum from Rabbit Skeletal Muscle

1. Dissection kit: good quality scissors for fine mincing of muscle tissue.
2. Ice bucket or tray, sturdy glass plate (size of the ice-containing bucket or tray).
3. Skeletal muscle from a New Zealand white rabbit (100 g).
4. Waring blender.
5. Buffer A (homogenization buffer): 10% (w/v) sucrose, 20 mM histidine, pH 7.0, 0.5 mM EDTA, 0.23 mM PMSF, 0.83 mM benzamidine (see Note 1).
6. Refrigerated high-speed centrifuge.
7. Cheese cloth.
8. Ultracentrifuge with fixed-angle rotor.
9. Small glass homogenizer (for resuspension of pellets).
10. Buffer B: 0.6M KCl, 30 mM histidine, pH 7.0, 0.23 mM PMSF, 0.83 mM benzamidine.
11. Buffer C (storage buffer): 10% (w/v) sucrose, 30 mM histidine, pH 7.4, 0.23 mM PMSF, 0.83 mM benzamidine.

2.2. Extraction of the Extrinsic Calcium-Binding Protein Calsequestrin from Sarcoplasmic Reticulum

- 1 Buffer D (extraction buffer): 100 mM sodium carbonate, pH 11.4.
2. Ultracentrifuge and fixed-angle rotor.
3. Magnetic stirrer.
4. Concentrated HCl and a pH meter.
5. NaCl.
6. Morpholinopropanesulfonic acid (MOPS).
7. Dithiothreitol (DTT).

2.3. Assay of Calcium Binding to Calsequestrin

1. Standard dialysis tubing (with a mol-wt cutoff of approx 10,000).
2. Buffer E (to pretreat dialysis membrane): 50 mM EDTA, 100 mM sodium carbonate.
3. Heat plate and large glass beaker (to boil dialysis tubing).
4. Buffer F (calcium-binding buffer): 5 mM Tris-HCl, pH 7.5, 0.1 mM $^{45}\text{CaCl}_2$ (40,000 cpm/mL).
5. Scintillation fluid.
6. Scintillation spectrometer.

2.4. Extraction of the Intrinsic Enzyme Ca^{2+} -ATPase from Sarcoplasmic Reticulum

1. Buffer G. 10% (w/v) sucrose, 50 mM sodium phosphate, pH 7.4, 1M KCl, 0.23 mM PMSF, 0.83 mM benzamidine.
2. Ultracentrifuge and fixed-angle rotor.
3. Magnetic stirrer.
4. 10% (w/v) Sodium deoxycholate.

2.5. Assay of Ca^{2+} -ATPase Activity

1. Stock solutions for assay buffer: 100 mM histidine, pH 7.4, 1M KCl, 50 mM MgCl_2 , 1 mM CaCl_2 , 10 mM EGTA, 25 mM ATP.
2. Stock solutions for phosphate determination:
 - a. Ammonium molybdate: 28.6 g of ammonium molybdate are dissolved in 500 mL of 6M HCl.
 - b. Polyvinyl alcohol: 11.6 g of polyvinyl alcohol are dissolved in 500 mL of boiling water and are allowed to cool.
 - c. Malachite green: 0.81 g of malachite green are dissolved in 1 L of distilled water.
 - d. Mixed reagent: 2 vol of malachite green are mixed with 1 vol each of polyvinyl alcohol and ammonium molybdate, and 2 vol of distilled water. Allow the reagent to stand at room temperature until it turns a golden-yellow color, which takes approx 30 min.

- 3 Potassium hydrogen phosphate standard (10–100 nmol P_i /mL)
- 4 Spectrophotometer.

3. Methods

Isolation of membrane vesicles derived from sarcoplasmic reticulum of skeletal muscle were based on the method of Eletr and Inesi (14). Ca^{2+} -ATPase activity and calcium binding was assayed according to Meissner et al. (15) and MacLennan (16), respectively. Inorganic phosphate was determined by the method of Chan et al. (17), and measurement of protein concentration was performed according to Hartree (18) using bovine serum albumin as a standard. Extraction procedures for the extrinsic membrane protein calsequestrin and the intrinsic membrane protein Ca^{2+} -ATPase were based on the protocols of Cala and Jones (19) and Warren et al. (20), respectively. All preparative steps are performed at 4°C unless otherwise stated.

3.1. Preparation of Sarcoplasmic Reticulum from Rabbit Skeletal Muscle

1. Mince 100 g of trimmed white rabbit skeletal muscle (*see* Note 2) with fine scissors on a sturdy glass plate positioned on top of a tray of crushed ice. Disperse the tissue pieces in 300 mL of ice-cold buffer A.
2. Homogenize the muscle tissue in a Waring blender at full speed four times for 15 s.
3. Centrifuge the homogenate for 20 min at 15,000g, filter the supernatant through three layers of washed cheesecloth, and recentrifuge at 40,000g for 90 min.
4. Resuspend the pellet in 50 mL of buffer B, incubate the suspension for 40 min on ice, and then centrifuge at 15,000g for 20 min. Recentrifuge the supernatant at 40,000g for 90 min, and resuspend the pellet in buffer C. At this stage, the crude sarcoplasmic reticulum preparation may be quick-frozen in liquid nitrogen, and stored at –70°C until future usage (*see* Note 3).

3.2. Extraction of the Extrinsic Calcium-Binding Protein Calsequestrin from Sarcoplasmic Reticulum

1. Pellet freshly thawed sarcoplasmic reticulum, prepared as described in Section 3.1., at 100,000g for 30 min, and resuspend in ice-cold buffer D to a protein concentration of 1.5 mg/mL.
2. Incubate the suspension on ice for 30 min, and then centrifuge at 100,000g for 30 min yielding a pellet and the carbonate-extracted supernatant fraction containing calsequestrin.
3. Stir the supernatant magnetically, and make the suspension to 50 mM MOPS, 0.5M NaCl, and 1 mM DTT by the addition of solid reagents, and adjust the pH to 7.0 by the dropwise addition of concentrated HCl (*see* Note 4).

Table 1
Calcium Binding to Calsequestrin Extracted from Sarcoplasmic Reticulum

Step	Total protein, mg	Specific Ca^{2+} binding, nmol Ca^{2+} /mg protein
Sarcoplasmic reticulum	80	175
Carbonate solubilized supernatant	27	280
Purified calsequestrin	—	713

3.3. Assay of Calcium Binding to Calsequestrin

1. Boil a sufficiently long piece of thin dialysis tubing for 10 min in buffer E, followed by boiling for 10 min in distilled water, and then rinse the tubing extensively in distilled water (*see* Note 5).
2. Place a 0.4-mg protein sample in 1-mL vol and dialyze against 100 mL of buffer F for 24 h in a cold room at 4°C. Use all the necessary precautions when handling the radioactive isotope (*see* Note 6).
3. Dissolve 0.1-mL samples from both inside and outside of the dialysis bag each in 10 mL of scintillation fluid, and count the specific radioactivity in a scintillation spectrometer.
4. Assuming that the dialysis reached an equilibrium, calculate Ca^{2+} binding/mg of protein from the increased radioactivity within the dialysis bag (Table 1).

3.4. Extraction of the Intrinsic Enzyme Ca^{2+} -ATPase from Sarcoplasmic Reticulum

1. Centrifuge freshly thawed sarcoplasmic reticulum, prepared as described in Section 3.1., at 100,000g for 30 min, and resuspend the pellet in ice-cold buffer G.
2. Add sodium deoxycholate from a 10% (w/v) detergent stock solution slowly under magnetic stirring and on ice to the membrane suspension at a ratio of 0.4 mg detergent/mg protein.
3. Centrifuge the resulting mixture at 100,000g for 60 min, and collect the clear supernatant containing the solubilized Ca^{2+} -ATPase (*see* Note 7).

3.5. Assay of Ca^{2+} -ATPase Activity

1. Mix in a microcuvet 10 μL of protein sample with 390 μL of distilled water and with 100 μL each of the assay stock solutions, including histidine, KCl, MgCl_2 , as well as calcium (to measure total ATPase activity) or EGTA (to measure basal ATPase activity).
2. Initiate the enzyme reaction by the addition of 200 μL of ATP stock solution.

Table 2
Enzyme Activity of Ca^{2+} -ATPase Extracted from Sarcoplasmic Reticulum

Sample	Protein, mg	Ca^{2+} -ATPase activity, $\mu\text{mol P}_i/\text{mg protein/min}$
Sarcoplasmic reticulum	52	2.1
Detergent solubilized supernatant	29	3.9
Purified Ca^{2+} -ATPase	—	8.1

3. After an incubation time of 5 min, terminate the reaction by the addition of 2 mL of mixed reagent (malachite green-molybdate-polyvinyl alcohol reagent) (*see* Note 8).
4. Immediately after the addition of the mixed reagent, measure color development at 630 nm against a mixed reagent blank.
5. Compare your measurements with a potassium dihydrogen phosphate standard graph.
6. Calculate the Ca^{2+} -ATPase activity by subtracting the basal ATPase activity (in the presence of EGTA) from the total ATPase activity (in the presence of calcium) (*see* Note 9) (Table 2).

4. Notes

1. No general composition of a protease inhibitor cocktail suitable for all homogenization or extraction procedures can be given. To prevent proteolytic degradation of calsequestrin and the Ca^{2+} -ATPase during isolation from rabbit skeletal muscle, the addition of EDTA, PMSF, and benzamide to the isolation buffer appears to be sufficient. Other widely used protease inhibitors include antipain, aprotinin, bestatin, chymostatin, E-64, leupeptin, pepabloc, pepstatin, and phosphoramidon. Some companies, i.e., Boehringer-Mannheim (Mannheim, Germany), sell protein inhibitor sets for small-scale pilot experiments in order to determine the minimum concentration and composition of a protease inhibitor cocktail sufficient for preventing proteolytic degradation in a novel purification protocol.
2. Muscle tissue should be quickly removed from the freshly killed animal and trimmed of fat tissue. Avoid contaminating the isolation buffer and blender with rabbit hair. Fine mincing of muscle tissue should preferentially be performed in a cold room at 4°C with the muscle positioned on top of an ice-cold glass plate.
3. To purify the sarcoplasmic reticulum further, the crude preparation of Section 3.1. may be centrifuged through a 26–40% sucrose gradient for 150 min at 100,000g. The upper half of the sucrose density gradient contains the purified sarcoplasmic reticulum vesicles with considerably lower

amounts of contaminating mitochondrial membranes and vesicles derived from the transverse tubular membrane system.

4. Solubilized calsequestrin, extracted by alkaline treatment as described in Section 3.2., can be very simply purified to homogeneity using Phenyl-Sepharose chromatography (19). Bound calsequestrin is eluted from the column by 10 mM CaCl_2 , and this results in a homogeneous protein preparation.
5. Alternatively, gentler washing of dialysis tubing may be performed at 60°C by incubation for 2 h in buffer E, followed by extensive washing in distilled water. To avoid time-consuming washing of dialysis tubing in order to remove impurities and heavy metals, more expensive brands of tubing can be purchased that have been prewashed and sterilized (Spectrum, Houston, TX). However, irrespective of the brand of tubing, handle dialysis tubing at all times with gloves to avoid contamination of protein samples with proteases and other impurities from your hands.
6. When using $^{45}\text{CaCl}_2$ solutions, handle the radioactive isotope with extreme caution. Work exclusively in an area of the cold room designated for radioactive work, and confine radioactive experiments to a minimum space. Wear at all times double plastic gloves and protective clothing when handling radioactive solutions, and clearly mark all radioactive glassware and equipment. When removing labeled sample from the inside of the tubing following equilibrium dialysis, ask someone to help you in order to avoid spillage of radioactive solution when transferring it to the scintillation cocktail.
7. Solubilized Ca^{2+} -ATPase may be purified to homogeneity by centrifuging the solubilized preparation (Section 3.4.) through a linear 20–60% sucrose density gradient at 100,000g for 24 h (20). The peak fractions with the highest specific Ca^{2+} -ATPase activity are usually found in the lower part of the sucrose gradient and are greatly depleted of excess detergent.
8. An alternative way to determine Ca^{2+} -ATPase activity is a coupled enzyme assay. Instead of measuring the production of inorganic phosphate, the increase of ADP is measured by coupling the ATPase reaction to the widely employed pyruvate kinase/lactate dehydrogenase system. The production of NAD can conveniently be measured at 340 nm.
9. Contaminating mitochondrial ATPase activity, possibly present in crude membrane preparations, can be inhibited by the addition of 5 mM NaN_3 . The presence of inorganic phosphate in the protein sample prior to initiation of the enzyme reaction or nonenzymatic hydrolysis of ATP can be accounted for by control assays using sample but no ATP, or just ATP with no sample.

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CHAPTER 29

Removal of Detergent from Protein Fractions

Kay Ohlendieck

1. Introduction

The solubilization of biological membranes by detergents plays an important role in the identification, characterization, and extraction of integral membrane proteins. The most commonly used detergents in the biological sciences are described in Chapter 28, and several review articles exist on solubilization procedures (1–3). Since the initial extraction of intrinsic membrane proteins usually involves high detergent concentrations, excess detergent has to be removed or exchanged for another type of detergent at later stages of preparative or analytical procedures involving the solubilized protein fraction. The efficiency of certain chromatographic procedures is often significantly improved by lowering the overall detergent concentration or by exchanging one type of detergent for another. Lectin chromatography, a powerful technique to affinity-purify subsets of glycoproteins (*see* Chapter 18), appears to be especially sensitive to high concentrations of a variety of detergents (4). Furthermore, since high concentrations and/or certain types of detergent interfere with many physical and chemical analyses and detergents also exhibit undesirable side effects on highly sensitive cell biological assays, detergent removal is in many cases of central importance for retaining the biological activity of isolated membrane proteins. Another important area of detergent removal is reconstitution studies with hydrophobic membrane proteins that exhibit vectorial transport (3).

From *Methods in Molecular Biology*, Vol 59 *Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

Table 1
Techniques Used for the Removal
or Exchange of Detergents from Protein Fractions^a

Equilibrium dialysis
Batch or column chromatography
Hydrophobic adsorption
Ion-exchange chromatography
Gel filtration
Affinity chromatography
Lectin chromatography
Sucrose density gradient centrifugation
Protein precipitation procedures
Phase partitioning
Electroelution techniques

^aSuitability of individual techniques for detergent removal depends on the CMC, aggregation number, and HLB of the individual detergent (*see* Section 1)

This chapter summarizes some of the techniques employed in removing or exchanging detergents used in the solubilization of biological membranes. For a more in-depth discussion, see recent reviews on procedures of detergent removal (5–8). Technical bulletins from companies selling commercially available resins for detergent removal, i.e., Biobeads SM-2 (Bio-Rad, Hercules, CA) or Extracti-Gel D (Pierce, Rockford, IL), usually contain a list of references relevant to conditions recommended for detergent removal. It is well worth studying these general guidelines and recommendations prior to attempting to remove or exchange detergent from a precious, novel protein sample. Detergents are amphipathic molecules, and their preferred form of aggregation in water is the formation of micelles whose size and molecular weight may vary considerably between different types of detergent (1–3). Thus, the most important properties of a detergent with respect to removal is its unique critical micelle concentration (CMC), its hydrophile-lipophile balance (HLB), and its micellar molecular weight determined by the aggregation number of detergent molecules (7).

The most commonly employed techniques for detergent removal from protein fractions are summarized in Table 1, and are based on physical and chemical differences between protein–detergent complexes and detergent micelles. Suitability of individual techniques depends on the unique properties of the detergent used, and the choice of procedure is furthermore strongly dependent on the concentration range of the protein

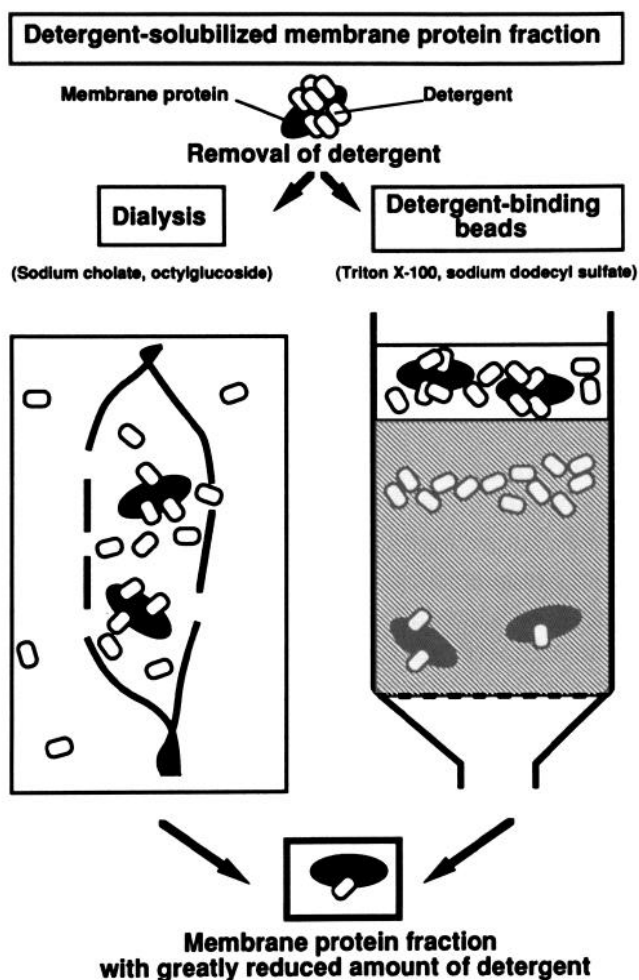


Fig. 1. Commonly used methods to remove detergent from protein fractions.

fraction to be depleted of detergent. Probably the simplest and least work-intensive method to remove detergent from protein fractions is dialysis (Fig. 1). Using large external volumes and frequent changes of dialysis medium, ionic detergents with a relatively high CMC can quite successfully be removed. However, it can take considerable time to reach an equilibrium, and this can lead to undesirable side effects, such as protein degradation. An excellent example of detergent removal, routinely performed by dialysis, is reconstitution experiments (3). Alternatively,

incorporation of hydrophobic membrane proteins into lipid vesicles can be achieved by gel-filtration or dilution procedures.

A variety of chromatographic techniques are available to remove or exchange detergents. These are more work-intensive than dialysis, but take less overall time and thereby tend to keep protein degradation during detergent removal to a minimum. Hydrophobic adsorption chromatography is certainly a very convenient way to exchange different classes of detergent (Fig. 1). The exchange of alkyl detergents, i.e., octyl glucoside and dodecyl sulfate for Triton X-100 type detergents using Phenyl-Sepharose (Pharmacia), was reported by Robinson et al. (9). Reasonably inexpensive matrices for the specific binding of detergents are commercially available from Bio-Rad (Bio-Beads SM-2) and Pierce (Extracti-Gel D). Both adsorbents, however, require a high enough protein concentration to avoid losses in recovering protein samples during detergent removal. The preincubation of columns with bulk carrier proteins in order to saturate nonspecific protein binding sites might at least partially solve this problem. Alternatively, protein-detergent complexes could be bound to an affinity matrix and eluted following washing and/or exchange with a different detergent. Affinity matrices with immobilized ligands for receptor binding or a variety of lectin columns highly specific for binding to subsets of glycosylated membrane proteins are suitable for these kinds of procedures. However, high detergent concentrations often adversely affect ligand-protein interactions, and dilution prior to application to the affinity column is beneficial. Another way to bind charged membrane proteins and remove excess detergent by extensive washing is ion-exchange chromatography. Elution of the bound protein fraction can be achieved by increasing the ionic strength or addition of an ionic detergent (5,7). Furthermore, if the difference in size between detergent micelles and protein-detergent complexes is large enough, gel-filtration chromatography can be employed to exchange detergents in a protein fraction (5,7). The use of sucrose gradient centrifugation in the removal of excess detergent was demonstrated by Warren et al. (10), who could successfully separate excess deoxycholate from an integral membrane protein by this method. Precipitation of solubilized membrane proteins from aqueous solutions can be achieved by treatment with polyethylene glycol, and phase partitioning can also be exploited to precipitate hydrophobic integral proteins (5,7). Finally, electroelution is a widely used method to recover protein samples separated by sodium dodecyl sulfate-

polyacrylamide gel electrophoresis (*see* Chapter 33), and many reasonably priced electroelution units are now commercially available.

To illustrate the practical aspects involved in the removal of detergent from protein fractions, equilibrium dialysis and detergent adsorption chromatography are described in more detail in Section 3. For a typical protocol of detergent-exchange chromatography, see the article by Robinson et al. (9). Since the unique properties of individual classes of integral membrane proteins and their interaction with a variety of ionic, nonionic, and zwitterionic detergents cannot adequately be predicted, the procedures described are only general outlines and do not describe the removal of detergent from a specific solubilized membrane protein. Refer to quoted research papers and review articles for details on specific requirements with respect to individual membrane proteins.

2. Materials

2.1. Dialysis

1. Dialysis tubing with a mol-wt cutoff of approx 10,000.
2. Wash buffer: 100 mM NaHCO₃, 50 mM EDTA.
3. Hot plate and large glass beaker.
4. Reliable, leak-proof plastic clamps for closing dialysis tubing.
5. Dialysis buffer: 20 mM Tris-HCl, pH 7.4, 0.15M NaCl.
6. Large beaker (4–6 L).
7. Small plastic funnel
8. Magnetic stirrer and suitable large stir bar.

2.2. Detergent-Adsorption Chromatography

1. Small columns (1–5 mL bed volume).
2. Detergent adsorption matrix, i.e., macroporous Bio-Beads SM-2 (Bio-Rad), Extracti-Gel D (Pierce), or any other commercially available detergent-adsorption matrix.
3. Blocking buffer: 0.1% (w/v) bovine serum albumin in 50 mM Tris-HCl, pH 7.4, 0.15M NaCl.
4. Washing buffer: 50 mM Tris-HCl, pH 7.4, 0.15M NaCl.
5. Small peristaltic pump and suitable tubing.

3. Methods

All procedures are performed in a cold room at 4°C, unless stated otherwise.

3.1. Dialysis

1. Take a sufficiently long piece of standard dialysis tubing, and boil it for 10 min in washing buffer, followed by boiling for 10 min in distilled water and

extensive washing in distilled water. Alternatively, use prewashed dialysis tubing (*see* Note 5, Chapter 28).

2. Transfer the solubilized membrane protein fraction with the aid of a small funnel into the dialysis tubing, which is securely closed at the lower end (*see* Note 1)
3. Close the dialysis tubing after removal of any air bubbles possibly introduced during transfer of the detergent-containing suspension, and allow for a small increase in volume during equilibrium dialysis. Generally, dialysis tubing is very sturdy, and leakage is not a problem as long as the tubing is tightly closed.
4. Dialysis should be performed with large external volumes (4–6 L) and adequate stirring, as well as frequent exchanges of the external solution (*see* Note 2).
5. At the end of the dialysis, wash the outside of the tubing, and carefully remove the dialyzed protein fraction (*see* Note 3).

3.2. Detergent-Adsorption Chromatography

1. The protein fraction to be treated with respect to detergent removal or detergent exchange should have a relatively high protein concentration (*see* Note 4) and be of large enough molecular weight to avoid entrapment in the pores of the affinity matrix (*see* Note 5).
2. Wash the detergent-removing column matrix first with distilled water, then equilibrate it thoroughly with blocking buffer (*see* Note 6), and wash with two column volumes of washing buffer.
3. Apply the protein sample, preferentially dissolved in the washing buffer or another suitable buffer for optimum detergent binding, to the equilibrated column.
4. Collect 0.5–1 mL fractions, and combine the protein peak fractions. The protein concentration can conveniently be measured by microprotein assays to avoid substantial losses owing to assaying. Peak fractions may then be concentrated by ultrafiltration prior to subsequent analytical or preparative procedures.

4. Notes

1. Dialysis tubing should be carefully closed by tight knotting and preferentially secured by special leak-proof plastic clamps (Spectrum, Los Angeles, CA).
2. Since reaching equilibrium in dialysis procedures can be quite time consuming, the placement of a suitable matrix to bind detergent outside the dialysis tubing might accelerate the process. Although nonionic detergent might be trapped by a hydrophobic adsorption matrix, ionic detergents might be removed by the use of an ion-exchange matrix.

3. Following equilibrium dialysis, avoid losing dialyzed samples when removing them from the tubing. Dialysis bags can be expanded following extensive dialysis, and should be carefully opened surrounded by a larger, clean glass beaker to avoid any accidental spillage.
4. Generally, a very important requirement for avoiding substantial losses of protein during procedures of detergent removal is a high enough protein concentration of the starting material. Especially, chromatographic techniques and precipitation procedures might result in a severe loss of proteins from very dilute solutions. Ideally, protein concentrations >1 mg/mL should be used to avoid these problems. Thus, concentrating a protein fraction prior to detergent removal by a suitable type of ultrafiltration is recommended, although these techniques are usually also not without danger of losing protein samples owing to nonspecific binding to membrane filters. These kinds of problems have to be worked out with every new class of solubilized membrane protein and no general strategy with respect to optimizing overall protein recovery can be given.
5. Using hydrophobic adsorption chromatography to remove or exchange detergent, the molecular weight of the protein to be recovered should be large enough to avoid entrapment of smaller peptides within the pores of the support matrix. The chromatographic matrix of commercially available columns exhibits exclusion limits between 2 and 10 kDa.
6. If dilute starting material cannot be concentrated, the detergent-exchange columns should be pretreated with solutions of bulk protein to avoid substantial protein losses. Bovine serum albumin is usually very useful in blocking nonspecific binding sites for protein on affinity matrices. This precaution should significantly lower the loss of dilute protein samples during detergent removal or exchange.

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CHAPTER 30

Purification of Membrane Proteins

Kay Ohlendieck

1. Introduction

In contrast to soluble proteins, the isolation of a peripheral or integral membrane protein requires extraction of the biological membrane containing the protein of interest prior to purification. The most commonly used extraction procedures to isolate membrane proteins are described in Chapter 28. Following initial solubilization in high concentrations of detergent (1), intrinsic proteins can then be purified to homogeneity using a variety of biochemical techniques (*see* Chapters 14–24) in the presence of relatively low concentrations of detergent. Solubilized membrane proteins from diverse sources and with different properties have successfully been purified by a combination of standard techniques, such as density gradient centrifugation, ion-exchange chromatography, gel filtration, lectin chromatography, and different forms of affinity chromatography methods. Usually separation techniques based on biological differences, such as affinity chromatography using specific ligands, probes, or antibodies highly specific for a membrane protein, result in a higher yield and purity of the isolated protein than chromatographical methods based on physical differences, such as size and charge of the membrane protein (2,3). An especially powerful separation technique for the initial purification of membrane-associated glycoproteins is lectin affinity chromatography. As described in more detail in Chapter 18, this method explores the highly specific interaction between N- and O-linked oligosaccharide chains on glycosylated proteins with immobilized lectins, which usually results in a remarkable enrichment of integral glycoproteins (4).

From *Methods in Molecular Biology, Vol 59 Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

Although the ultimate experiments to be carried out with an isolated membrane protein may influence the overall strategy of purification, it is desirable that the molecules of interest are obtained in their native, biologically active form in high yield and purity. However, biochemical, biophysical, or physiological experiments with homogeneous membrane proteins might require a gentler purification scheme than, for example, the preparation of only partially purified antigen samples for the production of monoclonal antibodies. Once a suitable source, such as a subcellular membrane fraction enriched in the membrane protein of interest, is found (*see* Chapters 6 and 7), small-scale pilot experiments should be performed to test the effects of various detergents on the biological activity of the membrane protein to be purified (*see* Chapter 28). With respect to the choice of detergent, a reagent with a high critical micelle concentration (i.e., cholates) should be tried first, since these detergents can be more easily removed following purification than detergents with a lower critical micelle concentration (*see* Chapter 29).

To illustrate the basic strategy for purifying an integral membrane protein and to monitor its purification and analyze the purity of the final product, this chapter addresses the purification of the sea urchin egg receptor for sperm. This highly glycosylated, integral surface protein represents a novel class of cell recognition molecules and exists in its native configuration as a disulfide-bonded homo-tetrameric complex (5,6). Each sea urchin egg contains approx 1.25×10^6 receptor molecules whose subunits exhibit an apparent mol wt of 350 kDa (7). Based on the previous biochemical characterization of proteolytic receptor fragments (8) and information on the primary structure deduced from its cDNA sequence (9), a strategy to purify the sperm receptor to homogeneity was worked out. The purification protocol, typical for the isolation of integral membrane proteins, includes cell homogenization, subcellular fractionation, solubilization of membrane fractions, lectin affinity chromatography, ion-exchange chromatography, dialysis, ultrafiltration, and lyophilization (7). Since this membrane protein does not exhibit enzyme activity, purification was monitored using immunoblot analysis with an antibody raised to a recombinant protein representing the extracellular sperm binding domain of the receptor molecules. The purity of the final product was analyzed by gradient SDS-PAGE in combination with sensitive silver staining (5,7).

Thus, the techniques described in Section 3. and elaborated on in Section 4. give a good overview of how to approach the purification of a

novel protein based on limited information on its physical and biological properties. The flowchart in Fig. 1 summarizes the different steps in the purification of the sea urchin sperm receptor typical for the isolation of an integral membrane protein. Other excellent examples of detailed descriptions of the methods employed in the purification of membrane proteins, i.e., rat liver 5'nucleotidase and rabbit skeletal muscle Ca^{2+} -ATPase appeared previously in this series (10,11)

2. Materials

2.1. Isolation of Crude Egg Surface Membrane Complex

1. Sea urchin eggs (50 mL of settled, dejellied, and washed eggs from *Strongylocentrotus purpuratus*) (see Note 1).
2. Small plastic beakers (for collection of gametes).
3. 6°C Water bath for storage of sturdy 2-L glass beakers.
4. 120- μm Nitex nylon membrane (Tekton, Inc., Elmsford, NY).
5. 0.5M KCl
6. Buffer A (artificial sea water): 0.48 M NaCl, 10 mM KCl, 27 mM MgCl_2 , 29 mM MgSO_4 , 11 mM CaCl_2 , 2 mM NaHCO_3 , pH 7.5 (see Note 2)
7. Buffer B (homogenization buffer): 0.5M NaCl, 10 mM KCl, 25 mM NaHCO_3 , 63 mM NaOH, 25 mM ethylene glycol-bis(β -aminoethyl ether) *N,N,N',N'*-tetracetic acid (EGTA), 63 mM NaOH, pH 8.0; supplemented with 1 mM of each of aprotinin, soybean trypsin inhibitor, antipain, leupeptin, benzamidine, and phenylmethanesulfonyl fluoride (Sigma, St. Louis, MO) (see Note 3).
8. Buffer C: 20 mM Tris-HCl, 0.15M NaCl, pH 7.4, supplemented with the same protease inhibitor cocktail as described for buffer B.
9. Hand-operated Potter-Elvehjem homogenizer (25-mL vol).
10. Low-speed bench centrifuge with 4×200 mL rotor.
11. Standard bright-field microscope.
12. Water aspirator.

2.2 Solubilization of Sperm Receptor Complex

1. 4% (w/v) Octylglucoside (Boehringer-Mannheim, Indianapolis, IN) in buffer C
2. Ultracentrifuge with 30-mL fixed-angle rotor and appropriate tubes.

2.3. Lectin Affinity Chromatography

1. Wheat germ agglutinin (WGA)-agarose (EY Labs, San Mateo, CA) matrix with a bed volume of approx 10 mL (see Note 4).
2. Wash buffer: 0.1% (w/v) octylglucoside in buffer C

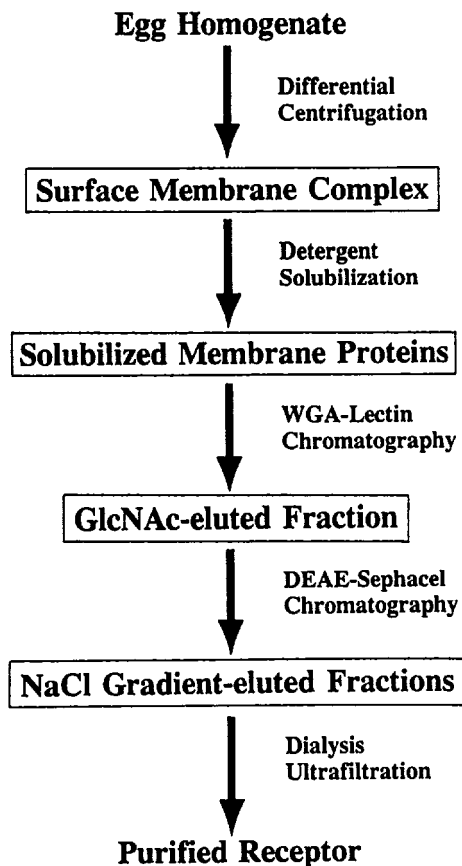
RECEPTOR PURIFICATION PROTOCOL

Fig. 1. Flowchart of the receptor purification protocol. The sea urchin egg receptor for sperm is purified to homogeneity from *S purpuratus* egg homogenates. Following isolation of a crude surface membrane complex by subcellular fractionation, membranes are solubilized, and a subset of glycoproteins purified by lectin affinity chromatography. The eluted glycoprotein fraction is then bound at low ionic strength to an ion-exchange matrix, and eluted with a linear salt gradient. The peak fractions containing homogeneous receptor are dialyzed, concentrated and lyophilized, resulting in a preparation of purified sperm receptor.

3. Elution buffer: 0.5M *N*-acetylglucosamine, 0.1% (w/v) octylglucoside in buffer C.
4. End-over-end mixer

2.4. Ion-Exchange Chromatography

1. DEAE-Sephacel column (Pharmacia, Uppsala, Sweden) with a bed volume of approx 10 mL.
2. Wash buffer: 0.1% (w/v) octylglucoside in buffer C.
3. Elution buffer A: 0.5M NaCl, 0.1% (w/v) octylglucoside, 20 mM Tris-HCl, pH 7.4, supplemented with the protease inhibitor cocktail as described for buffer B.
4. Elution buffer B: 4M NaCl, 0.1% (w/v) octylglucoside, 20 mM Tris-HCl, pH 7.4, supplemented with the protease inhibitor cocktail as described for buffer B.
5. 30-mL linear gradient maker, small magnetic pellet, and magnetic stirrer.
6. Small peristaltic pump and suitable elastic tubing.
7. Fraction collector (optional).

2.5. Dialysis, Ultrafiltration, and Lyophilization

1. Standard dialysis tubing (with a mol wt cutoff of approx 10,000).
2. Ultrafiltration filter, i.e., pre-equilibrated Amicon PM-30 membrane filter.
3. Ultrafiltration apparatus, i.e., 50-mL vol Amicon concentrator.
4. Lyophilizer.

3. Methods

Perform all manipulations of eggs at 6°C and subsequent homogenization and chromatographical purification steps in a cold room at 0–4°C.

3.1. Isolation of Crude Egg Surface Membrane Complex

1. To release mature gametes, treat 10–20 adult sea urchins with intra-coelomic 0.5M KCl injection, and collect eggs in small plastic beakers of artificial sea water. After microscopical examination of individual batches of eggs, combine batches of morphological integrity, and wash them three times by settling in 2 L of artificial sea water.
2. Dejelley washed eggs by 10 slow passages through a 120- μ m Nitex membrane, and then wash twice in filtered artificial sea water and once in 2 L of buffer B (*see* Note 5).
3. Resuspend the dejellied, washed, and settled eggs in 100 mL of ice-cold buffer B, supplemented with the described protease inhibitor cocktail, and homogenize by approx 10 gentle strokes using a hand-operated glass homogenizer and Teflon™ pestle. When breakage of eggs is adequate, the majority of eggs can be seen to release cytoplasmic organelles, such as yolk platelets, when viewed under a light microscope (*see* Note 6).
4. After 10-fold dilution, centrifuge the ghost membranes for 2 min at 1000g, carefully remove the yellowish supernatant with the help of a water aspirator, and combine the white membrane pellets.

- 5 Wash the pellet twice in buffer B, and finally resuspend it in 10 mL of buffer C (*see* Note 7).
6. If the egg surface complex cannot be processed immediately, it may be quick-frozen in liquid nitrogen, and stored at -80°C until usage.

3.2 Solubilization of Sperm Receptor Complex

1. Solubilize the egg surface complex membranes at a final protein concentration of 1 mg/mL by adding an equal volume of 4% (w/v) octylglucoside in buffer C, and incubate this mixture for 60 min on ice.
2. Remove insoluble material by centrifugation at 105,000g for 60 min, and decant the supernatant containing the solubilized membrane protein fraction.
3. Dilute the supernatant 1:4 (v/v) with buffer C.

3.3. Lectin Affinity Chromatography (see Note 8)

1. Equilibrate the WGA matrix by settling five times in 20 vol of buffer C.
2. Add the diluted, solubilized egg surface membrane complex suspension to the settled lectin matrix, and incubate gently for 3 h or overnight in a cold room on an end-over-end mixer.
3. Allow the lectin matrix to settle on ice and carefully remove the supernatant. Wash the matrix three times with 10 vol of wash buffer.
4. Add 1.5 vol of elution buffer to the lectin matrix and incubate the mixture for 20 min in a cold room on a mixer.
5. Allow the matrix to settle on ice, carefully recover the supernatant containing the eluted glycoprotein fraction, and proceed immediately to Section 3.4. For regenerating the WGA matrix, *see* Note 9.

3.4. Ion-Exchange Chromatography

1. Apply the glycoprotein fraction to 10 mL of DEAE-Sephacel ion-exchange resin, which has been pre-equilibrated with wash buffer. Gently shake this mixture for 2 h in a cold room.
2. Allow the ion-exchanger matrix to settle, remove the supernatant, and wash the resin three times with 10 vol of wash buffer.
3. Pour the resin into a clean glass column, and elute bound protein with a linear NaCl gradient of increasing ionic strength. The gradient is formed in a small gradient maker under adequate stirring using elution buffers A and B.
4. Collect 3-mL fractions, and store them on ice prior to analysis.
5. Analyze the fractions by immunoblot and/or lectin blot analysis. The sea urchin egg receptor for sperm exhibits a characteristic brownish color in silver-stained SDS polyacrylamide gels, indicative of highly glycosylated proteins. In addition, the receptor protein band of approx 350 kDa is strongly labeled by peroxidase-conjugated WGA lectin (*see* Fig. 2).

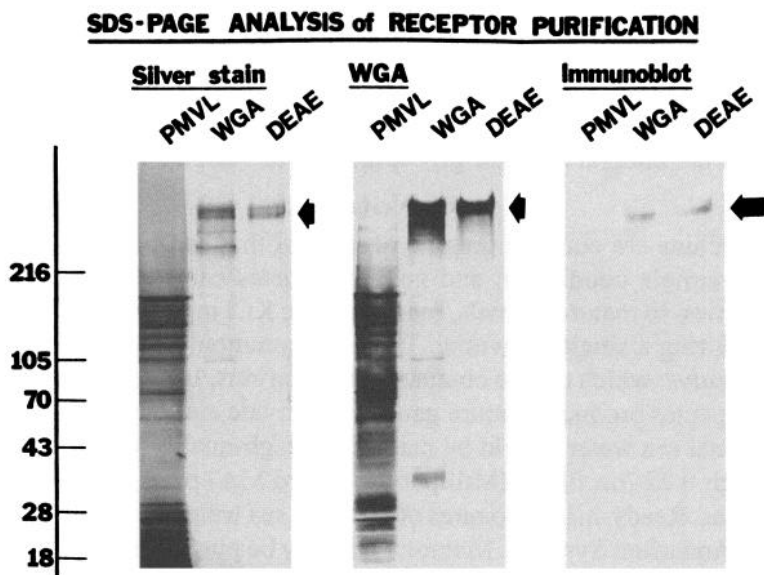


Fig. 2. Analysis of sperm receptor purification. Shown is the SDS-PAGE analysis of the different steps in the purification of the receptor using subcellular fractionation of the crude egg surface (PMVL; plasma membrane/vitelline layer), followed by lectin affinity chromatography (WGA) and finally ion-exchange chromatography (DEAE-Sephacel). The silver-stained and the WGA lectin-blotted 350-kDa protein band corresponds to the receptor band visualized by immunoblotting. The double protein band in silver staining might reflect a partial proteolytic degradation of the high-mol-wt receptor molecule or the existence of two isoforms with slightly different molecular weights. This phenomena is currently under investigation. Mol-wt standards ($M_r \times 10^{-3}$) are indicated on the left.

3.5. Dialysis, Ultrafiltration, and Lyophilization

1. Combine the peak fractions containing homogeneous sperm receptor, and dialyze them overnight against distilled water. For the preparation of dialysis tubing, see Chapter 11 or 29.
2. Concentrate the dialyzed receptor preparation using ultrafiltration with a PM-30 membrane. Following removal of the concentrated protein solution from the ultrafiltration chamber, rinse the filter twice with 1 mL of buffer to elute some of the protein possibly sticking to the membrane.
3. Quick-freeze the concentrated receptor preparation in liquid nitrogen and lyophilize the sample. Resuspend the freeze-dried receptor preparation in a small volume of buffer suitable for the subsequent analytical procedures,

i.e., filtered artificial sea water for fertilization inhibition assays or sample buffer for SDS-PAGE.

4. Figure 2 shows a silver-stained SDS-PAGE and corresponding immunoblot and lectin blot of the different preparative steps involved in the isolation of the integral receptor glycoprotein (see Note 10).

4. Notes

1. Sea urchins are common marine organisms that can be held in captivity under simple conditions, and mature gametes can be obtained in large quantities. In mature animals, intracoelomic KCl injection may release 10^7 eggs during a single spawning. The most commonly used sea urchin is *S. purpuratus*, which can be obtained from Marinus, Inc. (Long Beach, CA). This species produces mature gametes from late autumn to early summer.
2. Artificial sea water should be made from high-quality water, and filtered through 0.22- μ m filters (Millipore, Bedford MA) prior to use on isolated gametes. Ready-made mixtures of artificial sea water (i.e., "Instant Ocean" from Aquarium Systems, Mentor, OH) may be purchased from a pet shop.
3. The composition of the protease inhibitor cocktail used during the purification of the sea urchin egg receptor for sperm may not be suitable for other purification protocols. Since certain protease inhibitors are relatively expensive, small-scale pilot experiments should be performed to determine the lowest concentration of protease inhibitors necessary to prevent protein degradation during preparative steps (see Note 1, Chapter 28).
4. Immobilized WGA is an expensive affinity matrix and can alternatively be prepared quite simply from crude wheat germ purchased in a health store. Following extraction, WGA can be purified to homogeneity by affinity chromatography using *N*-acetylglucosamine agarose (Sigma, St. Louis, MO) (12). Purification of the lectin can be monitored by SDS-PAGE analysis combined with immunoblot analysis using a polyclonal antibody to WGA commercially available from Sigma. Purified WGA can then be immobilized by standard procedures using cyanogen bromide-activated agarose (Pharmacia).
5. If suitable nylon membranes are not available, dejellying of sea urchin eggs can also be achieved by acid treatment. A 10% solution of eggs is treated for 2 min at pH 5.5 with the addition of 0.1M HCl under gentle stirring with a plastic spatular. The solution is then neutralized by the addition of 2M Tris, pH 8.0, and washed three times in filtered artificial sea water.
6. It is important to monitor visually the generation of ghost membranes following homogenization. Figure 2 in ref. 5 illustrates the typical appearance of ghost membranes. Depending on the type of hand-operated homogenizer used, the breakage of the majority of eggs (>95%) can vary from batch to batch. On the one hand, the breakage of sea urchin eggs

should be adequate to release the cytoplasmic content, but on the other hand, the surface membrane should not be greatly fragmented, since this leads to loss in yield of suitable ghost membranes.

7. If a small yellow pellet is seen at the bottom of the final membrane complex pellet, this is owing to eggs that were inadequately broken. Avoid resuspending this part of the otherwise whitish egg surface membrane complex.
8. Alternative to the described batch method, binding and elution of the glycoprotein fraction may be performed by column chromatography. However, elution of the bound glycoprotein fraction using a linear gradient of *N*-acetylglucosamine does not result in a major separation of WGA binding proteins. Since the batch method is technically simpler, this procedure is used for routine purification of the sperm receptor. However, the disadvantage of batch methods is a possible gradual loss of matrix when removing the eluted supernatant fraction. This can be a problem with expensive affinity resins, and in that case, column chromatography would be advantageous.
9. Regeneration of a lectin matrix is usually achieved by salt washes. Treat the WGA matrix, following elution with *N*-acetylglucosamine, with three 1.5M NaCl washes. Then re-equilibrate the WGA matrix with incubation buffer or phosphate-buffered saline complemented with 5 mM NaN₃ for storage at 4°C.
10. For a description of the standard techniques used in the analysis of the sperm receptor, *see* ref. 7. Gel electrophoretic procedures combined with analytical blotting techniques are ideal for monitoring the purification of a membrane protein, since they are not only comparatively inexpensive and simple methods, but also relatively fast and reliable. Silver staining of proteins separated by gel electrophoresis is a very useful method to estimate the purity of the final preparation, since it is highly sensitive in the detection of contaminating proteins.

Acknowledgments

The author thanks W. J. Lennarz, State University of New York, in whose laboratory experiments purifying the sea urchin sperm receptor were worked out. This research was funded by the National Institutes of Health (HD 18590-W. J. Lennarz).

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CHAPTER 31

Lyophilization of Proteins

Ciarán Ó Fágáin

1. Introduction

Lyophilization, or freeze-drying, is a method for the preservation of labile materials in a dehydrated form. It can be particularly suitable for high-value biomolecules, such as proteins. The process involves the removal of bulk water from a frozen protein solution by sublimation under vacuum with gentle heating (primary drying). This is followed by controlled heating to more elevated temperatures for removal of the remaining "bound" water from the protein preparation (secondary drying). Residual moisture levels are often $<1\%$ (1). If the freeze-drying operation is carried out correctly (*see* Section 3.), the protein will preserve all or most of its initial biological activity in the dry state. This dry state offers many advantages for long-term storage of the protein in question.

An isolated protein in an aqueous system can suffer various adverse reactions over time owing to physical, chemical and biological factors. Typical physical phenomena are aggregation (2) and precipitation. There are many deleterious chemical reactions involving the side chains of amino acid residues, notably asparagine, aspartic acid, and cysteine/cystine (2,3), or the glycation of lysine residues with reducing sugars (the Maillard reaction) (4,5). Biological deterioration can result from loss of an essential cofactor or from the action of proteolytic enzymes, either endogenous or arising from microbial contamination. Water participates directly in many of the chemical reactions and in proteolysis. In any case, it provides a medium for molecular movement and interactions. For these reasons, removal of water effectively prevents deterioration of the pro-

From Methods in Molecular Biology, Vol 59 Protein Purification Protocols
Edited by S. Doonan Humana Press Inc., Totowa, NJ

tein. The freeze-dried preparation will be much less bulky than the original solution and can conveniently be stored in a laboratory freezer or refrigerator (or perhaps even at room temperature). When one wishes to use the protein preparation, one can rehydrate it simply by addition of an appropriate volume of pure water or suitable buffer solution.

At very low temperatures, a liquid may behave in one of two ways. Eutectic solutions undergo a sharp liquid–solid freezing transition over a very narrow temperature range, whereas amorphous liquids are characterized by a glass transition in which viscosity increases dramatically with cooling and the solution takes on the macroscopic properties of a solid, even though it has not crystallized. Below this glass transition temperature (T_g'), virtually no adverse chemical or biological reaction can take place. Above T_g' , however, the viscous, rubbery material is very prone to deterioration (6) (see Fig. 1). Understanding these low-temperature features of liquids is important for effective freeze-drying. Eutectic and glass transitions will greatly influence the freeze-drying protocol (the way the freeze-dryer is run), and the choice of substances used as preservatives or excipients in the product formulation subjected to lyophilization. Understanding of freeze-drying as a process has grown in recent years (6,8), and knowledge of the effects of additives has also grown (9,10). Together these advances have allowed lyophilization to be undertaken on a more rational and less strictly empirical basis. These matters are discussed in more detail in Section 3. Reference 11 contains a useful treatment that touches on the underlying theory.

Effective freeze-drying is time consuming and cannot be accomplished in a hurry. Process times of 72 h or greater are usual, depending on the nature of the product formulation and the properties of its constituents. The complex lyophilizer apparatus will have high capital and running costs (the latter owing to its power consumption and the time required). Freeze-drying is usually reserved, therefore, for high-value proteins, or is used in cases where alternative product presentations (such as ammonium sulfate precipitates) are unsuitable or give insufficient shelf lives.

A typical freeze-dryer comprises a set of shelves that can be cooled to about -40°C or heated above ambient temperature, a condenser usually cooled to -60°C , and a high-performance vacuum pump. The shelves allow rapid freezing of the protein solution and, later, gentle programmed heating for primary and secondary drying. The vacuum pump reduces the pressure within the system to allow sublimation of the bulk water

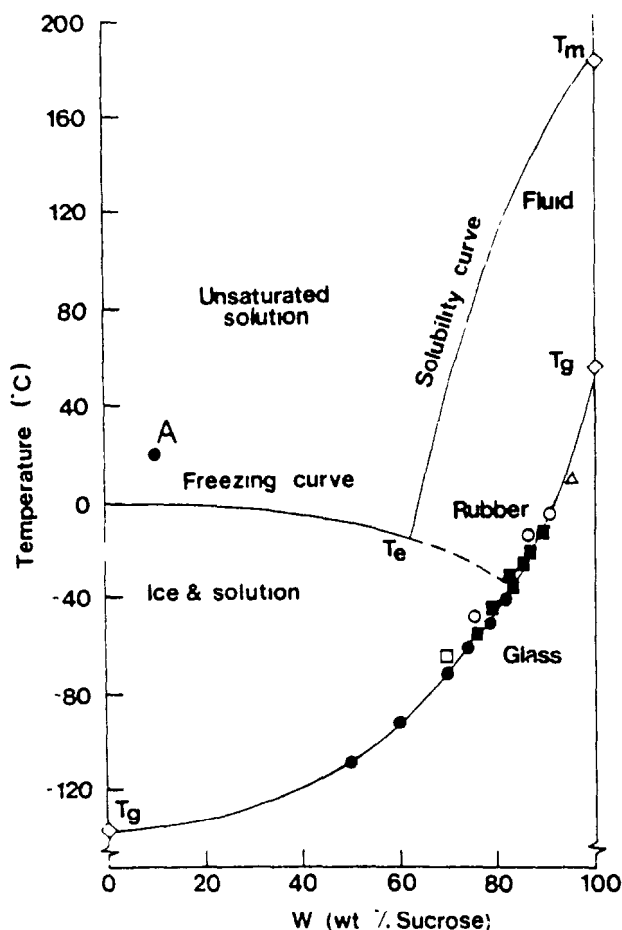
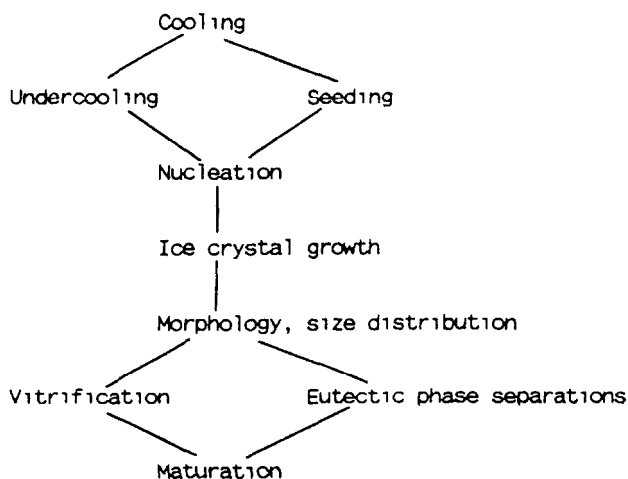


Fig. 1. Solid-liquid state diagram for the sucrose-water system, showing the glass transition T_g /concentration profile, the equilibrium solubility curve, running between the melting point and the eutectic temperature, T_e , and the region of metastability (supersaturation) beyond T_e (broken line). Symbols represent experimental T_g data taken from different sources. Point A represents a typical dilute solution that is to be dried to a stable solid product. From ref. 7

(i.e., primary drying), whereas the condenser acts as a lower temperature trap for the sublimed water that collects in the form of ice (i.e., it provides a temperature gradient within the apparatus). This explains why the condenser temperature must be considerably less than that of the shelves.

FREEZING



DRYING

Ice sublimation (Product Temperature, Chamber pressure)

Desorption (Temperature profile, Residual moisture)

STORAGE

Residual moisture/temperature control

Scheme 1. Physical principles of freeze-drying. From ref 7.

A complex interplay of chemical and physical phenomena takes place during freeze-drying (*see* Scheme 1). The product yield (that is, the percentage recovery of the initial active protein) depends on the formulation in which the protein is placed prior to lyophilization (6,10), whereas its ease of rehydration and its stability on long-term storage (or “shelf life”) are influenced by the processing regime (6). Section 3.1. describes, in broad terms, the operation of a typical freeze-dryer apparatus. Some critical factors concerning the operations of freezing, and primary and secondary drying are outlined in Sections 3.2.–3.4.

2. Materials

Freeze-dryers differ greatly in their specifications depending on the model or manufacturer. Regardless of the type, one must use it at all

times in accordance with the manufacturer's instructions. The method discussed in Section 3.1. assumes that the reader studies this chapter together with the user's handbook.

The most basic lyophilizer equipment will consist of a condenser/vacuum pump unit to which one may attach a centrifugal test tube holder or a manifold for the drying of multiple round-bottomed flasks. Freezing is accomplished in a separate cooling bath, usually filled with an alcohol. Such equipment can be used successfully for small-volume samples, but fine and reproducible control of the overall process may not be possible. Neither can one attain continuous recording of sample temperature. Higher grade equipment, with temperature-programmable shelves and a number of temperature probes, is to be preferred. Shelf-equipped freeze-dryers are especially suitable for use with rubber-capped pharmaceutical vials or with microtiter plates. Often an externally operated screw press will be present, allowing one to seal vials (partially closed with rubber stoppers so as not to restrict gaseous movement) under vacuum before releasing air into the chamber.

Obviously, one should thoroughly familiarize oneself with the machine to be used. The user should know the locations of the main components and indicators together with appropriate settings for the meters, controls, and valves.

The characteristics of the vials (usually glass) in which product is lyophilized will influence the process. The vial diameter, glass type, and bottom shape and thickness will all affect the rate of heat transfer from the shelves to product. The vials should withstand freezing and pressure changes, and should be uniform with respect to internal diameter and bottom thickness. The vial bottoms should be completely flat to make good contact with the shelves. The same model of vial should be used consistently, since a change of vial type will introduce a variable into an optimized process (8). Ensure that any other vessels used (e.g., round-bottomed flasks) are of sufficient quality to withstand the temperatures and pressures associated with freeze-drying.

3. Methods

3.1. Operation of Freeze-Dryer

These directions for freeze-drying are of necessity very general. Exact details will depend on the material and apparatus in question, and on the user's requirements. In pilot or manufacturing operations, Good Manu-

facturing Practice, ISO 9000, or regulatory authorities will likely impose many procedural disciplines that are beyond the scope of this chapter.

3.1.1. *Start-Up*

Ensure that the valve connecting the vacuum to the drying chamber is closed. Start the vacuum pump and allow it to evacuate. Observe the decrease in pressure on the vacuum indicator. It is important to start the pump first so as to reach a steady-state high vacuum long before evacuation of the main chamber. It is also important that the pump warms up thoroughly before the condenser or shelves are cooled: water in any form reaching the pump can cause damage. A warm-up time of 30 min will usually suffice. Close the condenser drain outlet, which should always be left open when the freeze-dryer is not in use. (If not, open it, allow any water to drain completely, and then close tightly once again.) Switch on the condenser, and allow to cool to -60°C . At this point, it may be convenient to cool the shelves slightly: this will aid the overall chilling process later. Do not cool the shelves more than a few degrees below ambient temperature, however, to prevent atmospheric condensation. Condensation will add to the water load, which must be removed during the lyophilization process.

3.1.2. *Filling and Loading*

Now loading of the shelves with the preparation to be lyophilized can begin (*see* Note 1). Fill only minimal amounts of material into each container to ensure a high ratio of surface area to volume: This will aid effective freeze-drying. The product matrix will tend to inhibit sublimation of water vapor from the surface of the ice crystal. This resistance depends on the depth of liquid and on the solids content of the product (8). Also, a good head space in the vial or ampule will allow easier and better gaseous movement. One can conveniently load filled vials into metal trays at the bench and then place the trays into the drying chamber. The trays should have level bottoms to make good contact with the freeze-dryer shelves. Many vials have narrow necks, which one closes with special rubber stoppers. The design of these stoppers allows one to cap the vials partially while maintaining contact between their contents and the atmosphere. (Sealing of the vials takes place later under vacuum.) If one wishes to use such closures, one should partially stopper the vials before loading them into the freeze-dryer. If using microtiter plates, place

them within special metal frames, which then come into contact with the shelves.

The freeze-dryer will likely have a number of flexible temperature probes; carefully clean each of these. Place some of the probes so as to monitor the temperatures of the shelves. Dip others directly into the material to be lyophilized. A temperature record of the actual solution (as distinct from the shelf underneath it) is well worth the loss of a small number of product vials. Arrange the temperature probes in different locations throughout the chamber: in the centers of shelves and also at the edges. Conditions will not be homogeneous across all vials, and monitoring should be as complete as equipment will allow. Insert the temperature probes right to the bottom of vials. Ensure that the probes remain in contact with the intended sample or shelf, and do not become displaced during product loading or manipulation or during lyophilization.

3.1.3. The Freeze-Drying Operation

Once loading is complete, close and secure the door of the drying chamber. Now the freezing process may begin. Bring the material well below its freezing temperature (or glass transition temperature) as quickly as possible. This is particularly important for a number of reasons (*see* Section 3.2.). Allow the shelf/product temperature to fall further to a steady -40°C before drawing the vacuum. Check that the drying chamber door is properly closed and sealed; then evacuate the chamber by opening the vacuum valve. The vacuum gage reading will indicate atmospheric pressure as the valve is opened and may take a few minutes to show a vacuum once again as the air within the drying chamber evacuates. Soon, however, the pressure within the drying chamber will decrease, and a steady-state high vacuum will result.

There now exists a high vacuum within the drying chamber together with a temperature gradient from the shelves/product (-40°C) to the condenser (-60°C). These conditions permit sublimation of the bulk water in the product over a period of hours as the shelves are gently heated. The sublimed water will collect as ice on the condenser. Sublimation, or primary drying, removes only the bulk water in the system; it is insufficient to remove the "bound" water closely associated with the protein molecules. Removal of this bound water, or secondary drying, requires heating to elevated temperatures. This is usually applied through the shelves on which the product rests (and has previously been

frozen). One can program the shelves to heat to a particular temperature at a defined rate appropriate for the product undergoing lyophilization. One must select the heating regime with particular care (*see* Sections 3.3. and 3.4.)

3.1.4. Termination of Run and Removal of Product

One may terminate the freeze-drying process when the cake has a good appearance and the product has reached a sufficiently low steady-state percentage of moisture (determined in a separate series of experiments) Seal vials, partially capped with rubber stoppers, before insertion into the chamber, by operating the special screw press. This operation will insert the stoppers fully before air is admitted into the drying chamber. Sealing of vials while still under vacuum has an important advantage over the use of an inert, moisture-free gas, such as nitrogen. Air will rush in when a vial, sealed under vacuum, is uncapped. The sound of the in-rushing air will be absent from a defective vial that has failed to seal or where the seal has broken down. In this way, the user will immediately know of the defect and be aware that the vial contents may have deteriorated (1).

Close the valve to the vacuum pump firmly. Slowly release the vacuum within the drying chamber by a minimal opening of the air inlet valve: air admission should be as gentle as possible to prevent undue sudden stresses on sealed vials, and also to prevent disturbance or upset of chamber contents by a strong jet of incoming air. The vacuum/pressure gage will show a rise in drying chamber pressure; this will eventually equalize to atmospheric, and one can then open the chamber door. Vials of freeze-dried product can be removed for storage (preferably at refrigerated temperatures). Store microtiter plates immediately in a desiccator or (better) vacuum-sealed within foil packets containing desiccant sachets.

3.1.5. Shutting Down

If the freeze-dryer is to be reused immediately, one must be certain that the condenser's ice capacity will withstand the accumulation of ice from two runs (*see* Note 1). If there is no more material for lyophilization, shut down the apparatus carefully. Remove any material spilled in the drying chamber and clean the shelves carefully according to the manufacturer's instructions. Leave the chamber door slightly ajar to allow circulation of air, and to prevent sticking and compression of the door seals. Switch off the condenser, and open its drain outlet. (The drain should remain open until the freeze-dryer is next used.) Over a period of hours,

the ice on the condenser surfaces will melt and drain away. Once the condenser temperature has returned to ambient and all the melted ice has drained away, allow the vacuum pump to run for a further 3 h before shutting it down. This is to prevent any damage to the pump owing to occurrence or accumulation of condensation. To maintain good pump performance, change the oil often according to the handbook's directions.

3.1.6. Using Simpler Freeze-Dryers

Use of simpler apparatus with manifold or centrifuge accessories is carried out in much the same stepwise fashion as above. In these cases, aliquots of the product are frozen in individual opennecked flasks or tubes. Switch on the vacuum pump and condenser, and allow to run as noted. Freeze flask contents by immersion in an alcohol cooling bath or in liquid nitrogen (follow the normal precautions). Swirl or rotate the flask during the freezing step to effect even distribution of the product over the widest possible surface area. This will minimize the depth of material through which water loss must occur (*see* Section 3.1.2. and ref. 8). Connect the frozen material directly to the manifold assembly (or load into the centrifugal tube dryer), and draw a vacuum in the chamber as quickly as possible so as to minimize back-melting of ice under atmospheric pressure. The rapid reduction of pressure by the vacuum pump will aid sublimation and help minimize melting to water. Although control of temperature and heating rates are more problematic with such accessories, visual inspection of the material and of the dried cake is much easier than in an enclosed chamber.

3.2. Freezing

In freezing, liquid water crystallizes to yield solid ice. Water, in common with other substances, including many solutes, is an eutectic material with a sharp transition between the liquid and solid states. Noncrystalline materials undergo a glass transition and become rigid at a certain low temperature (T_g). This is the end result of a notable increase in viscosity with decreasing temperature. The material becomes increasingly rubbery until rigidification occurs at T_g and liquid movement effectively ceases (*see* Note 2). Even for eutectics, ice formation usually does not occur at the thermodynamic freezing point: the actual temperature of freezing is generally 10–15°C lower than this, so that supercooling (or undercooling) is required (6,8). It is generally satisfactory to cool the product to just below 0°C initially, but not so low as to induce crystalli-

zation. When all the product has cooled to this set temperature, reduce the shelf temperature sharply to crystallize bulk water to ice. Typical temperatures lie in the range -20 to -30°C . Allow time for ice to form in all containers before proceeding to the next stage, which involves further shelf cooling below the lowest eutectic temperature (for crystalline materials) or below the glass transition temperature (T_g' , for amorphous substances) (*see* Note 3). A usual final temperature is -40°C . The product is reliably solidified only below these temperatures, and only now may primary drying begin. It is vital to maintain the freezing temperature below T_g' before and during primary drying. If the temperature rises above this value, the material ceases to be a solid and becomes a viscous rubber very prone to deleterious reactions, with resulting losses of activity. The rubber will also undergo a mechanical collapse, take on a different appearance, and be difficult to rehydrate (6). Collapsed product is always unacceptable even if reasonable biological activity remains.

Other notable changes occur in the product during freezing, many of which can lead to significant protein denaturation. As the bulk water freezes to ice, the amount of liquid water remaining naturally decreases. This leads to great increases in the concentrations of solutes, such as salts. Such freeze-concentration can have far-reaching effects. Concentration will increase the rates of unwanted chemical reactions, such as oxidations. Buffer components may crystallize differentially, leading to pronounced pH shifts; also, pK_a values are temperature-dependent. For phosphate buffers, one should use potassium or mixed salts in preference to sodium salts (6,12). For all of these reasons, one should accomplish the freezing steps (crystallization of bulk water and cooling below the eutectic or T_g' temperatures) in any lyophilization process as quickly as possible.

Suitable excipients or protective substances can lessen or overcome some of these damaging freezing effects: they can particularly influence melting or collapse temperatures (13). Excipients may be classed as bulking agents, tonicity modifiers, buffers, and cryoprotectants/lyoprotectants (13). One should use bulking agents where the product's solids content is low. They help to ensure the development of a plug of dried material, which will be of good appearance. Bulking agents can also prevent blow-out: where the material has a very low solids content of the order of 1%, disruption of cake structure by the issuing water vapor can result in loss of the dried product from the container along with the vapor (10). Typical bulking agents include polyols and certain sugars, mostly nonreducing.

Buffer substances must be chosen with great care. One must consider possible variations with temperature in pK_a and solubility as well as compatibility with the protein(s) in question and other constituents. Useful information can be found in ref. 14, and in monographs and articles dealing with the handling of proteins.

Many bulking agents will double as tonicity modifiers in preparations for administration *in vivo* to ensure that the preparation is isotonic with body fluids. Any agent used for this purpose must, of course, be pharmaceutically acceptable.

Cryoprotectants and lyoprotectants are substances that stabilize a protein against the deleterious effects of freezing and of lyophilization/storage, respectively. These protecting additives appear to be preferentially excluded from the protein surface. This results in a strengthening of the water "shell" surrounding the protein (15). Polyols and certain salts and sugars can be beneficial. In general, nonreducing sugars are preferable, since, unlike reducing sugars, they cannot participate in Maillard reactions. Xylitol's T_g' is -46.5°C in a freeze concentrate of 42.9 wt% water, whereas T_g' values for sorbitol, sucrose, and trehalose are -43.5 , -32 , and -29.5°C at wt% water values of 18.7, 35.9, and 16.7, respectively (6). Glycerol has a notably low T_g' value of -65°C at 46 wt% water (6). This means that glycerol-containing formulations must reach very low temperatures to become glassy. Glycerol, therefore, is not an ideal lyophilization excipient, since such temperatures may be difficult to achieve. Useful discussions and examples of cryoprotectants occur in refs. 9, 15, 16; see also Note 2 and Section 3.3. of Chapter 32 in this volume.

3.3. Primary Drying

Primary drying is the sublimation under vacuum of bulk ice from the product to the much colder condenser typically held at -60°C . Sublimation will be faster at higher shelf temperatures, so heat the shelves to a few degrees below T_g' to quicken the process (*see* Note 2). **Never** allow the shelf temperature to equal or exceed T_g' , however. If it does, collapse and product deterioration may occur. A usual safety margin is 2 – 5°C below the eutectic or collapse temperature (8). (The term "collapse temperature" is usually equivalent to T_g' for an amorphous substance.) The rate of sublimation depends on the vacuum and on the condenser temperature, and therefore is largely determined by the characteristics of the lyophilizer apparatus (6). Drying often becomes easier as the tempera-

ture approaches T_g' or the eutectic temperature: The sublimation rate can increase by about 13% for each 1°C rise in temperature (8). Resistance to drying also decreases with decreasing product thickness (i.e., filling height) and with increasing vial diameter (which influences the area of the drying surface) (8). It is important that the temperature of the actual product remains constant (and therefore is monitored) throughout primary drying. The sublimation rate will decrease as primary drying proceeds, and therefore, the degree of product cooling owing to sublimation will decrease also. The purpose of shelf warming is to counter this sublimation cooling. Accordingly, be sure to adjust the heat input to the shelves as the product dries to prevent a net rise in product temperature. An unchecked rise in product temperature could lead to collapse. One can use chamber pressure to control the length of the primary drying cycle (8). Even under uniform conditions, primary drying times may vary by up to 10% for a given process. Make sure to include a delay period (ascertained empirically and occupying a period of hours) at the end of the primary drying cycle to ensure that all ice has sublimed satisfactorily and to avoid collapse (8). See Note 4 for a reference to a quantitative model that has been used to predict drying times.

3.4. Secondary Drying

Secondary drying removes the remaining unfrozen, bound water from the lyophilizing material yielding a final product, low in residual moisture, which will be stable for an extended period without deterioration. The unfrozen water is removed by heating the shelves on which the primarily dried product rests. The rule of thumb for initiation of secondary drying is the equivalence of product and shelf temperatures. For safety, however, one should include a delay period at the end of primary drying in any freeze-drying protocol to prevent collapse of any vials that have not quite finished sublimation (8) (*see also* Section 3.3.). The partial pressure of water within the drying chamber drops at the end of primary drying as the last of the ice sublimes. If it can be monitored, this drop in water partial pressure can be a good indicator of the completion of primary drying. Even during secondary drying, with much of the original water gone, the preparation's temperature should never rise above T_g' , something that is seldom appreciated (6). T_g' , however, rises as the residual water content drops (8). One can, therefore, increase the temperature (within limits) during secondary drying. The vacuum need not be exhaus-

tive during secondary drying; indeed, one should use a pressure in the region of 0.2 torr for secondary drying (8). There is little evidence that one can harm a protein by overdrying it (8).

3.5. Quality Indices

Most freeze-dryers will have a built-in chart recorder to provide a profile of shelf temperatures (and perhaps other parameters, such as vacuum) during the lyophilization run. Study this carefully, and retain with other batch information. Note the cake shape and texture and any variations between vials, especially those located at different positions within the chamber. Choose a representative number of vials from different chamber locations for scrupulous moisture determination (*see* Note 5). Measure the yield or recovery (%) of the initial biological activity by appropriate assay following rehydration of a representative number of vials. While waiting to assay, note the time required for complete product rehydration. Also note whether any turbidity remains on rehydration or after what interval turbidity appears in a clear sample (6). Persistence of the rehydrated biological activity can be measured at suitable or convenient time intervals. Accelerated degradation methods can predict shelf lives of the lyophilized preparation at temperatures of interest for long-term storage. These methods involve extrapolation of an Arrhenius plot. Accelerated degradation protocols can be of value, provided the activity decay is first-order at each of the temperatures tested and all data are scrupulously accurate and precise. Reference 17 gives guidelines for the proper use of accelerated methods; *see also* Section 3.6. of Chapter 32 in this volume. Note that the formulation used greatly influences yield, whereas the process parameters affect ease of rehydration and shelf life (6).

4. Notes

1. The volumes of bulk liquid subjected to freeze-drying must never exceed the manufacturer's recommendations. If the condenser's ice capacity is reached or exceeded, the degree of product drying will be insufficient and many problems can result.
2. The glass transition temperature, T_g' , and the product water content are critical parameters in the freeze-drying process. It is essential to prevent or minimize damage to the protein of interest during freezing. Inclusion of excipients (additives) with high T_g' values in the protein formulation to be freeze-dried can be very useful. The mixture will form a glass at relatively low temperatures, minimizing freezing damage. A high T_g' will also allow

the use of higher temperatures during primary drying with less danger of product collapse. Any constituent that will lower the unbound water content of the freeze concentrate will help shorten the secondary drying operation (6), but uncrystallized salts will decrease T_g' since any salt will bring about a depression of freezing point. Thus, the salts content of the product formulation should be as low as is practicable (6). The optimum temperature for freezing and primary drying depends on the ratio protein:protectant additive:salt in the freeze concentrate rather than in the initial solution (6,10). The ratio protein:other solids in the freeze concentrate influences T_g' (6).

3. The glass transition temperature can be determined by differential scanning calorimetry, although a sensitive instrument is required (6). Electrical resistance measurements can sometimes be useful, but microscopic observation of freeze-drying over a range of temperatures is arguably the most direct, sensitive, and unambiguous method for determination of the collapse temperature (7).
4. Reference 18 contains a brief quantitative treatment of freeze-drying, discussing it in terms of heat and mass fluxes to derive an equation for the prediction of drying times. The model successfully predicted times for removal of 65–90% of the total initial water. However, the actual times required for removal of the remaining 10–35% of the water were greater than those predicted.
5. The residual moisture content of, and its distribution throughout, the lyophilized preparation will dictate its long-term stability. Uneven moisture distribution between vials often leads to biphasic activity loss profiles on extended storage (6). Each 1% of moisture can depress T_g' by more than 10°C (11).

Acknowledgments

The author thanks F. Franks for permission to reproduce figures from his work and P. K. Walsh for a critical reading of the manuscript.

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CHAPTER 32

Storage of Pure Proteins

Ciarán Ó Fágáin

1. Introduction

There is often a need to store an isolated or purified protein for varying periods of time. If the protein in question is to be studied, it will take some time to characterize the properties of interest. If the protein is an end product or is for use as a tool in some procedure, it will likely be used in small quantities over a period of time. It is vital, therefore, that the protein retains as much as possible of its original, postpurification, biological (or functional) activity over an extended period of storage. This storage period or "shelf life" may vary from a few days to more than 1 yr. Shelf life can depend on the nature of the protein and on the storage conditions. This chapter explores the means by which activity losses occur on storage, and discusses a range of measures to prevent or lessen the inactivating events. The chapter also describes the use of accelerated storage (accelerated degradation) testing for the prediction of shelf lives at particular temperatures.

Apart from extremes of temperature and pH (which will, naturally, be avoided as conditions for routine or long-term storage), a variety of factors can lead to loss or deterioration of a protein's biological activity. These include proteolysis, aggregation, and certain chemical reactions (1). Proteolysis may arise from microbial contamination or be an intrinsic hazard in the case of proteolytic enzymes. A protein may lose an essential cofactor, or the subunits of oligomeric proteins may dissociate from each other, resulting in each case in loss of activity (1). Adsorption to surfaces may also lead to inactivation (1,2).

From *Methods in Molecular Biology*, Vol. 59 *Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

The purely chemical reactions are few and well defined. Deamidation of glutamine and asparagine can occur at neutral to alkaline pH values, whereas peptide bonds involving aspartic acid undergo cleavage under acidic conditions. Cysteine is prone to oxidation, as are tryptophan and methionine. Alkaline conditions lead to reduction of disulfide bonds, and this is often followed by β -elimination or thiol-disulfide exchange reactions (1). Where reducing sugars are present with free protein amino groups (N-termini or lysine residues), there may be destructive glycation of the amino functions by the reactive aldehyde or keto groups of the sugar (the Maillard reaction) (3). Elevated temperatures favor all of these reactions, but it is important to note that aggregation and deleterious chemical reactions can occur at moderate temperatures also. Virtually complete aggregation of lyophilized bovine serum albumin occurred over 24 h at 37°C following addition of just 3 μ L of physiological saline. The degree of aggregation was less, but still significant, at lower temperatures. The underlying cause was formation of an intermolecular disulfide bond by a thiol-disulfide exchange. Ovalbumin, glucose oxidase, and β -lactoglobulin underwent similar aggregation under the same conditions, but by a somewhat different mechanism (4) (*see* Note 1). These findings underline how important it is to ascertain correct storage conditions for the protein of interest. Simple reliance on the laboratory refrigerator to minimize activity losses may not suffice over an extended period.

Other factors may work against the refrigerator. Certain proteins are more stable at room temperature than in the refrigerator and are said to be cold labile. This cold denaturation has been well characterized for myoglobin and a few other proteins (5,6). It is a property of the protein itself and is distinct from freezing inactivation (*see* Section 3.2. of Chapter 31). This phenomenon arises from the fact that it is thermodynamically possible for a protein to unfold at low as well as at high temperatures (*see* ref. 7 for a summary of the notable features of cold denaturation). Although cold denaturation is completely reversible (7), it is inconvenient and can waste valuable time as the protein refolds (as well as causing laboratory scientists some unpleasant shocks).

Storage concerns a protein's long-term or kinetic stability. Kinetic stability is distinct from (and need not correspond with) thermodynamic stability. Thermodynamic stability refers to a protein's conformational stability in terms of the change in heat capacity or the Gibbs (free) energy on reversible unfolding. (Unfolding is often reversible *in vitro*, but other

events occurring subsequently can lead to irreversible inactivation.) Kinetic stability measures the persistence of activity with time (or, to put it another way, the progressive loss of function). It can be represented by the scheme:



where N is the native, functional protein, I is an irreversibly inactivated form, and k_{in} is the rate constant for the inactivation process. The equation:

$$V_{in} = -d[\text{N}]/dt = k_{in}[\text{N}] \quad (2)$$

describes the process mathematically, where V_{in} is the experimentally observed rate of disappearance of the native form (1). Often the activity loss will be first-order, although more complex inactivation patterns are well documented (8). Note that an apparently unimolecular first-order time-course of inactivation may mask a more complex set of inactivating molecular events (8).

This chapter does not consider the special requirements of pharmaceutical regulatory authorities, good manufacturing practice (GMP), or ISO 9000 specifications. Readers needing to meet these directives and standards should consult materials produced by the appropriate authorities in order to ensure compliance. Finally, mention of suppliers names does not imply endorsement of any particular product(s).

2. Materials

1. All containers used for storage of pure proteins should be of good quality, and should tolerate temperatures as low as -20°C or even -70°C if freezer storage is desired or necessary. A number of manufacturers (such as Sarstedt, Nümbrecht, Germany, or Nunc, Roskilde, Denmark; there are many others) supply presterilized screw-cap plasticware with good mechanical and low-temperature properties. Clean glassware exhaustively, and sterilize it by dry heat. Screw caps or rubber stoppers that will not withstand dry heat can be autoclaved.
2. Membrane or cartridge filters, of pore size $0.22\ \mu\text{m}$ for sterile filtration, are available from such companies as Gelman (Ann Arbor, MI) or Sartorius (Göttingen, Germany), with or without a Luer lock attachment for extra secure attachment to a hand-held syringe.
3. Use an ordinary domestic refrigerator for storage at temperatures of a few degrees Celsius. Modern machines are often combined with a freezer unit that can easily maintain temperatures as low as -20°C . Storage at -70°C or below will require a specialized low-temperature freezer.

4. A number of standard constant-temperature laboratory incubators will be required if accelerated storage testing is to be performed.
5. Highly purified forms of the following chemicals will be useful:
 - a. Antimicrobials: sodium azide, thiomersal.
 - b. Protease inhibitors: phenylmethylsulfonyl fluoride (PMSF).
 - c. Chelators: ethylenediamine tetra-acetic acid (EDTA).
 - d. Salts: ammonium sulfate, other ammonium, citrate, sulfate, acetate, and phosphate salts.
 - e. Osmolytes: sucrose, glucose, trehalose, xylitol, glycerol, polyethylene glycol (PEG).
 - f. Reducing agents: dithiothreitol, 2-mercaptoethanol.

3. Methods

3.1. Prevention of Bacterial Contamination

3.1.1. Antimicrobials

Microbial contamination can lead to significant losses of a pure protein by proteolysis. Contamination can be detected by standard microbiological plating techniques, but the aim must always be to avoid it in the first place. Even if one can achieve successful elimination or removal of contaminating microorganisms, the protein of interest may already have lost at least some activity or may have deteriorated in ways difficult to detect. Addition of antimicrobial compounds, such as sodium azide or thiomersal (sodium merthiolate, a mercury-containing compound), can prevent microbial growth. (Both of these compounds are poisonous: handle them with care.) Add sodium azide to a final concentration of 0.1% (w/v) or thiomersal to a final concentration of 0.01% (w/v). Note that azide will inactivate oxygen-binding proteins, such as hemoglobin or peroxidase (*see also* Note 2).

3.1.2. Filtration

Where one wishes to ensure sterility or to avoid use of the antimicrobials discussed in Section 3.1.1., filtration offers a useful alternative. (One can use both strategies together, of course.) A filter of pore size 0.22 μm will exclude all bacteria; indeed, this method is used in industry to sterilize labile materials that cannot be autoclaved or irradiated. Disposable filter cartridges are widely available in a variety of configurations; most have very low protein-binding capacities. Typically, one draws the solution to be sterilized into a syringe, and then removes the needle or tube. Next, connect the filter to the syringe nozzle, ensuring it is firmly mounted. Uncap a suitable sterile storage container directly beneath

the filter outlet (using standard aseptic manipulations to avoid contamination of container or cap), and depress the syringe plunger to force the protein solution through the sterilizing filter into the container. Recap immediately. It will not be possible to flame plastic containers in a Bunsen burner as part of aseptic technique: it is much better practice to perform filtration operations of this sort in a Class 2 laminar flow microbiological safety cabinet, the design of which prevents contamination of the sample. Following the manufacturer's instructions closely, turn on the cabinet's fans, and allow to run for at least 10 min to allow adequate filtration of cabinet air. Open and remove the front door. Swab the cabinet's internal surfaces and the outer surfaces of storage containers brought inside the cabinet with 70% alcohol, and allow to evaporate. Carry out the filtration maneuvers, remove the storage containers, and dispose of waste materials appropriately. Swab the internal surfaces of the cabinet with alcohol once again, replace the front cover and allow to run for 10 min (or according to the user's handbook) before shutting down. It is not always possible to use a filter as fine as $0.22\ \mu\text{m}$ directly (*see* Note 3).

3.2. Avoidance of Proteolysis

It can be difficult to remove proteases completely during purification of a target protein. Unless the object protein is completely pure (homogeneous), even tiny amounts of contaminating proteolytic enzymes can cause serious losses of activity during extended storage periods. The molecular diversity of proteases complicates the situation: there are exopeptidases, which remove amino acid residues from the N- or C-termini, and endopeptidases, which cleave internal peptide bonds within protein chains. In addition, there are four types of proteases classified by their molecular reaction mechanisms: the serine, cysteine (or thiol), acid, and metalloproteases (9) Use EDTA in the concentration range 2–5 mM to complex the divalent metal ions essential for metalloprotease action. Pepstatin A is a potent, but reversible inhibitor of acid proteases. It is used at concentrations of around $0.1\ \mu\text{M}$, as are similar protease inhibitors. The compound PMSF reacts irreversibly with the essential serine in the active site of serine proteases, inactivating them. It can also act on some thiol proteases. Use it at a final concentration of 0.5–1 mM, having dissolved it first in a solvent, such as acetone (it is poorly soluble in water) (10). One must, of course, ensure before addition that none of these compounds will adversely affect the protein of interest (*see also* Note 4).

If the protein of interest is itself a proteolytic enzyme, use of protease inhibitors is not feasible. One may need to store such a protein in dried form (Section 3.5.) or as a freeze-dried preparation (Chapter 31). Alternatively, one can place it in a solution with a pH value far removed from the protease's optimum. Trypsin, for example, is most active at mildly alkaline pH values. Stock solutions of trypsin are often prepared in 1 mM HCl, where the very acid pH value renders the enzyme effectively incapable of catalysis. This helps prevent autolysis during the course of the experiment. The enzyme molecule does not inactivate under these conditions and is fully active on dilution into a suitable assay solution (11).

3.3. Use of Stabilizing Additives

3.3.1. Background

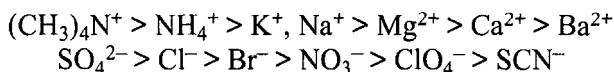
It has long been known that inclusion of low-mol-wt substances, such as glycerol or sucrose, in protein solutions can greatly stabilize the critical protein's biological activity. However, it took some time for the exact mechanism of this stabilization to be ascertained. Timasheff and Arakawa (12) showed that these types of substances are preferentially excluded from the vicinity of the protein molecules, since their binding to the protein is thermodynamically unfavorable. The protein molecule is preferentially hydrated by the solvent water. Loss of the protein's compact, properly folded structure (denaturation) will increase the protein-solvent interface. This, in turn, will tend to increase the degree of thermodynamically unfavorable interaction between the additive and the protein molecule. The result is that the protein molecule is stabilized by the additive (12,13). This preferential exclusion means that there is less of the solute (additive) immediately surrounding the protein than there is in the bulk solution; it does not necessarily mean that no solute molecules can penetrate the protein molecule's hydration shell (14).

It is important to note that the additives discussed in the following sections are generally applicable as stabilizing agents for proteins, but a given substance may not be effective for a particular protein. Both sucrose and PEG, for instance, are good stabilizers of invertase, but have denaturing effects on lysozyme; the same additive has contrary effects on the two enzymes (15).

3.3.2. Addition of Salts

Certain salts can significantly stabilize proteins in solution. The effect varies with the constituent ions' positions in the Hofmeister lyotropic series,

which relates to ionic effects on protein solubility (16,17) This series ranks both cations and anions in order of their stabilizing effects. In the series below, the most stabilizing ions are on the left, whereas those on the right are actually destabilizing:



The stabilizing ions force protein molecules to adopt a tightly packed, compact structure by salting out hydrophobic residues. This helps prevent the unfolding, which is the initial event in any protein deterioration process (*see* Section 1.). Most stabilizing ions seem to act via a surface tension effect (12,13). Ions can also stabilize proteins by shielding surface charges and can act as osmolytes by affecting the bulk properties of water (18). Note that ammonium sulfate, which is widely used as a stabilizing additive and as a noninactivating precipitant, comprises two of the most stabilizing ions from this list, the NH_4^+ cation and the SO_4^- anion. To stabilize proteins in solution while avoiding precipitation, add ammonium sulfate to a final concentration in the range 20–400 mM (18). One can do this by adding a minimal volume of a stock solution of ammonium sulfate of known molarity or by the careful addition of solid ammonium sulfate. Sprinkle the solid salt, a few grains at a time, into the protein solution. Ensure that each portion of ammonium sulfate added dissolves fully before addition of the next lot. This procedure will prevent accumulation of high local salt concentrations that are undesirable. In addition to ammonium sulfate, salts containing citrate, sulfate, acetate, phosphate, and quaternary ammonium ions are generally useful (18). Note, however, that the nature of the counterion will influence the overall effect of such compounds on protein stability (12,13).

This discussion assumes that ions used or added do not act as substrates, activators, or inhibitors of the enzyme or protein in question, and that added ions do not interfere with or precipitate possibly essential ions already in solution. Ammonium ion, for instance, is actually a substrate for the enzyme glutamate dehydrogenase.

3.3.3. Use of Osmolytes

Osmolytes are a diverse group of substances comprising such compounds as polyols, mono and polysaccharides, neutral polymers (e.g., PEG), and amino acids and their derivatives. They are not strongly charged and have little effect on enzyme activity below 1M concentra-

tion. In general, they affect water's bulk solution properties and do not interact directly with the protein (18).

Use polyols and sugars at high final concentrations: Typical figures range from 10–40% (w/v) (13,18). Sugars are reckoned to be the best stabilizers, but reducing sugars can react with protein amino groups leading to inactivation (3,13). This problem can be avoided by using nonreducing sugars or the corresponding sugar alcohols. Glycerol is a very widely used low-mol-wt polyol. Its advantages include its ease of removal by dialysis and its noninterference with ion-exchange chromatography (18). However, glycerol suffers from two significant disadvantages: It is a good bacterial substrate (18) and it greatly lowers the glass transition temperature (T_g) of materials to be preserved by lyophilization (see Section 3.2. of Chapter 31) or drying (Section 3.5., this chapter). The five-carbon sugar alcohol, xylitol, can often replace glycerol; it can be recycled from buffers and is not a convenient food source for bacteria (18).

Polymers, such as PEG, are generally added to a final concentration of 1–15% (w/v). They increase the viscosity of the single-phase solvent system and so help prevent aggregation. Note, however, that higher polymer concentrations will promote the development of a two-phase system. The protein of interest will concentrate in one of these phases, and this may actually lead to aggregation (18).

Amino acids with no net charge, notably glycine and alanine, can act as stabilizers if used in the range 20–500 mM (18). Amino acids and derivatives occur as osmolytes in nature (13). Some related compounds, such as γ -amino butyric acid (GABA) and trimethylamine *N*-oxide (TMAO), can be good stabilizers in the 20–500 mM range used for amino acids (13,18).

3.3.4. Substrates and Specific Ligands

Addition of specific substrates, cofactors, or competitive (reversible) inhibitors to purified proteins can often exert great stabilizing effects. (Indeed, their inclusion may be necessary where an essential metal ion or coenzyme is only loosely bound to the apoprotein.) Occupation of the target protein's binding or active site(s) by these substances leads to minor, but significant conformational changes in the polypeptide backbone. The protein adopts a more tightly folded conformation, reducing any tendency to unfold (19) and (sometimes) rendering it less prone to proteolytic degradation. Occlusion of the protein's active site(s) by a bound substrate molecule or reversible inhibitor will protect those amino acid side chains

that are critical for function. A starch-degrading amyloglucosidase enzyme (from an *Aspergillus* species) stored in the presence of 14% (w/v) partial starch hydrolysate was 80% more stable over a 24-wk period at ambient temperature than the corresponding enzyme preparation stored in the hydrolysate's absence (20)

Note that dialysis (or some other procedure for the removal of low-molecular-mass substances) may be necessary to avoid carryover effects of the substrate or inhibitor when the protein is removed from storage for use in a particular situation where maximal activity is desired.

3.3.5. Use of Reducing Agents and Prevention of Oxidation Reactions

The thiol group of cysteine is prone to destructive oxidative reactions. One can prevent or minimize these by using reducing agents, such as 2- (formerly β -) mercaptoethanol (a liquid with an unpleasant smell) or dithiothreitol (DTT or Cleland's reagent, a solid with little odor). Add 2-mercaptoethanol to reach a final concentration of 5–20 mM, and then keep the solution under anaerobic conditions. To achieve these anaerobic conditions, gently bubble an inert gas, such as nitrogen, through the solution and fill it to the brim of a screw-cap container to minimize headspace and the chances of gaseous exchange. DTT is effective at lower concentrations: usually 0.5–1 mM will suffice (10). Indeed, Schein advised that the DTT concentration should not exceed 1 mM. It can act as a denaturant at higher temperatures and is not very soluble in high salt (18). Note that these reducing agents are themselves prone to oxidation. (This is why solutions containing them must be stored under anaerobic conditions.) DTT oxidizes to form an internal disulfide that is no longer effective, but will not interfere with protein molecules (10). 2-Mercaptoethanol, on the other hand, participates in intermolecular reactions and can form disulfides with protein thiol groups (10). Such thiol–disulfide exchanges are highly undesirable and may actually lead to inactivation or aggregation. It is probably best to add reducing agents only in situations where they are known (or can be demonstrated) to be beneficial (16)

Much of the oxidation of thiol groups is mediated by divalent metal ions, which can activate molecular oxygen. Complexation of free metal ions (where they are not themselves essential for activity) can prevent destructive oxidation of thiol groups. See Section 3.2. regarding the use of EDTA to complex metal ions.

3.3.6. *Extremely Dilute Solutions*

Very dilute protein solutions are highly prone to inactivation. This is especially true of oligomeric proteins where dissociation of subunits can occur at low concentration. The individual polypeptide chains comprising the oligomer may lack activity alone and/or may denature with consequent loss of activity. Protein solutions of concentration $<1\text{--}2\text{ mg/mL}$ should be concentrated as rapidly as possible (10) by ultrafiltration or sucrose concentration (see Note 5).

Where rapid concentration is not possible, inactivation may be prevented by addition of an exogenous protein, such as bovine serum albumin (BSA), typically to a final concentration of 1 mg/mL . Alkaline phosphatase from *E. coli* is unstable at room temperature at concentrations $<10\text{ }\mu\text{g/mL}$, but can be stabilized by addition of BSA (21). Scopes discussed possible reasons for the undoubted benefits of BSA addition (10). It may seem foolish to add an exogenous, contaminating protein, such as BSA, deliberately to a pure protein preparation, but occasionally this may be the price to be paid in order to avoid inactivation.

3.4. *Low-Temperature Storage*

Refrigeration at $4\text{--}6^{\circ}\text{C}$ is often sufficient for the preservation of a protein's biological activity provided the hints in Sections 3.1.–3.3. of this chapter are followed judiciously. Many proteins are supplied commercially in 50% glycerol or as slurries in approx $3M$ ammonium sulfate. Freezing of such preparations is not necessary and should be avoided. Occasionally one may observe the phenomenon of cold denaturation; this has been discussed briefly in Section 1.

Some proteins can deteriorate at refrigerator temperatures and require storage at temperatures $<0^{\circ}\text{C}$. Usually, temperatures between -18 and -20°C , attainable by a domestic freezer, will allow for stable storage (see Note 6). Sometimes, however, it may be necessary to use temperatures below -20°C . In these cases, it is normal to use a low-temperature laboratory freezer designed to maintain temperatures in the range of -70 to -80°C (see Note 7).

Most protein solutions will undergo freezing to a solid at temperatures below 0°C . (Mixtures containing high concentrations of glycerol will remain liquid at -20°C ; see Section 3.3.3. in this chapter and Section 3.2. of Chapter 31.) The events occurring on freezing of a protein-containing mixture or biological system are much more complex than the simple macroscopic phase change would suggest. Differential freezing of particular

components of the mixture can lead to enormous concentration effects and to dramatic changes of pH at low temperatures. These chemical processes can lead, in turn, to a notable degree of protein inactivation. The subject of freezing damage and its avoidance is discussed in Section 3.2. of Chapter 31. Note that the problem can often be minimized by judicious choice of stabilizing additives as discussed in Section 3.3.; *see also* ref. 14.

Prevention of freezing will, of course, avoid freezing damage. It is possible to undercool liquids without freezing by preventing the nucleation of ice crystals. This means that proteins can be stored well below 0°C in the liquid phase. The preparation of protein-containing aqueous-organic emulsions, which can maintain complete biological activity in the liquid state over extended periods at -20°C, has been described (22), *see* Note 8. The method is very useful for small volumes of valuable proteins, avoids the need to use additives, and is more economical than freeze-drying. The same process is used for many different proteins, and one can remove portions of a sample without effect on the activity of the remainder. The actual storage temperature matters little, provided the upper temperature is <4°C and the lower temperature remains > -40°C, the nucleation temperature for ice crystal formation (23)

3.5. Drying for Stable Storage

The advantages of water removal as a protein storage/stabilization strategy have been set out in Section 1. of Chapter 31. Lyophilization can remove more than 95% of water from a protein preparation, but there is the risk of freezing damage (Section 3.2. of Chapter 31). It is possible to design protein-compatible formulations with T_g' values typically as high as 37°C (24). With these high T_g' values, controlled evaporative drying can be used in place of lyophilization to stabilize proteins in the solid state. Worthwhile evaporation rates will occur below these high T_g' values at reduced pressure. Evaporation is faster, less costly, and more easily controlled than freeze-drying (23,24). The high T_g' values also mean that one can sometimes store the resulting dried product at ambient temperature: If room temperature is less than T_g' the protein formulation will not undergo a glass/rubber transition during storage at room temperature. The glass-forming compounds are typically carbohydrates; maltose and maltohexose are particularly valuable, whereas sucrose can be useful if the moisture content can be reduced to 2% or less (24). Reconstitution of the solid protein preparation is accomplished simply by rehydration with added water or

buffer. The method has been patented and is described in ref. 25. An alternative formulation for vacuum-drying of proteins, involving the use of a cationic, soluble polymer (e.g., diethylaminoethyl dextran) and the sugar alcohol, lactitol, has also been patented (26) and published (27)

3.6. Stability Analysis and Accelerated Degradation Testing

3.6.1. Background

The distinction between conformational and kinetic stability has already been drawn in Section 1. Kinetic stability is usually measured at elevated temperatures (1), but the inactivating event(s) at high temperatures may not mirror that/those at the much lower temperatures used for storage. It is not feasible, however, to monitor stability in real time at the actual storage temperature; the experiment would take too long. Inaccuracy may result over shorter intervals, since only minimal losses, virtually indistinguishable from the starting activity, would be apparent.

Fortunately, there is a methodology that can in many cases overcome these difficulties, namely accelerated degradation (or accelerated storage) testing. This involves the periodic assay of samples incubated at different temperatures and use of the Arrhenius equation to predict shelf lives at temperatures of interest. The Arrhenius equation is conveniently expressed in logarithmic form as:

$$\ln k = -E_a/R \cdot T + \ln A \quad (3)$$

where k is the first-order rate constant of activity decay, E_a is the activation energy, R is the gas constant, and T is the temperature in Kelvin. This log form of the Arrhenius equation yields a straight-line plot of $\ln k$ against $1/T$ with slope $-E_a/R$ (28). Extrapolation of this plot can give the rate constant (and hence the useful life) at a particular temperature; see Note 9. Accelerated storage testing has been used as a practical means of quality assurance for biological standards (29) and has been employed in some scientific investigations (30,31)

3.6.2. Setting Up a Test

An accelerated storage test must use the minimum amount of the (often precious) test protein that is compatible with achievement of precise and accurate results. Since each experiment will occupy a period of weeks, it is important that every test yields a meaningful outcome. Accordingly, take great care in setting up an accelerated degradation test run.

It is essential to prevent microbial contamination of test samples (*see* Section 3.1.), especially since one assumes that no (or very little) sample degradation takes place at the reference temperature during the period of the experiment. Place test samples at a series of elevated temperatures (e.g., 48, 45, 42, 37, 33, 30°C) and at a suitable lower reference temperature, e.g., 4°C (*see* Note 6). Ensure that the reference temperature will not cause freezing of liquid samples. (The liquid–solid phase change will introduce a further variable and may also lead to freezing damage, as discussed in Section 3.2. of Chapter 31; *see also* Note 10.) Remove samples at intervals from the various incubation conditions, bring them to the same temperature, and assay for activity under standardized, optimal conditions. Ensure that stock solutions used in assays are carefully prepared and standardized: it is likely that different batches of assay solutions will be needed over the period of the accelerated storage test, so variations must be minimized. Be equally scrupulous with regard to procedural details of the assay and the performance of the instruments used.

Kirkwood made some practical recommendations for successful accelerated storage testing (32), that are summarized here. Use at least three elevated temperatures plus a low reference temperature. Set up 10 or more samples at each of these temperatures. The following schedule avoids waste of material owing to testing at inappropriate intervals. It also allows checking of the order of reaction. An acceptable result is very likely if enough material is placed on test at the beginning.

1. At intervals, test samples stored at the highest temperatures against the low-temperature reference samples. Ignore intermediate temperatures until a loss of 25% or greater (against the low-temperature reference sample) has occurred at the highest temperature. Now test a second sample to confirm the first result. If the results disagree, continue this stage 1 testing. If they agree, move on to stage 2.
2. Test the next two highest temperatures against a reference sample. Fit all available data to the Arrhenius equation. This procedure will not give a final precise result, but it will facilitate progress to stage 3.
3. Use the approximate result from stage 2 to ascertain further periods of storage at all temperatures such that measurable activity losses will have occurred at the intermediate temperatures. Assay samples from all available temperatures, and check all available data for their ability to fit the Arrhenius equation. This should result in a reasonable estimate of the low-temperature degradation rate, especially if multiple assays have been performed. If too little degradation has occurred to yield a precise result, one can repeat stage 3 after a longer time (32)

3.6.3. Analysis of Results

It is clear that raw experimental data must undergo a number of transformations for use in the Arrhenius plot, notably conversion to natural log or reciprocal forms. Error relationships are often significantly affected by these sorts of transformations. Use a computer for all statistical fitting to minimize such errors. Use of good-quality replicate results is very important.

The activity decay must be first-order at all of the temperatures used in the Arrhenius plot. Verify this by fitting the time course of activity loss at each test temperature to a first-order exponential decay. Assess goodness of fit by inspection of the graphic fit and of parameters, such as standard errors or chi-square values. First-order exponentials yield straight-line plots when transformed into semilogarithmic form, unlike higher order functions (i.e., a plot of $\ln[\text{Activity}]$ against time is linear for a first-order decay). Deviations from a first-order function are more likely to occur at higher temperatures (29).

Using the k -values determined at different temperatures, plot the Arrhenius graph ($\ln k$ is the ordinate, $1/T$ the abscissa). A linear plot of negative slope results. Extrapolate the line to a temperature of interest (e.g., 0, 4, or 25°C; these are respectively 273, 277, and 298 K) and estimate the value of k at this temperature. It is easy to estimate a true half life at this temperature by using the equation:

$$t_{1/2} = 0.693/k \quad (4)$$

Specially designed programs for the analysis of accelerated storage data are available (*see* Note 11).

4. Notes

1. Loss or decrease of the protein's biological or functional activity will be the main and most important index of deterioration. Often, however, the degree or time course of activity loss will not give any indication of the underlying molecular cause (although aggregation may be readily visible). Reference 1 provides a useful table of methods to identify the molecular changes leading to inactivation of the protein.
2. Thiomersal and azide are totally unacceptable in any product for administration. Do not discard azide compounds or azide-containing solutions down laboratory sinks. Not only is azide toxic, but it can accumulate in old lead piping, leading to the formation of potentially explosive compounds.
3. Some biological matrices, particularly sera, will not filter effectively through a 0.22- μm filter alone. One may need to prefilter the material in-

tially through a coarser 0.45- μm filter to which the desired 0.22- μm filter is connected in series. Alternatively, one can accomplish the finer filtration as a separate operation. One can best filter larger volumes (hundreds of milliliters or liters) using a stack of filters clamped in a special filtration unit. A filter as coarse as 1 μm may be used directly in contact with the solution of interest, the stack comprising progressively finer filters until the sterilizing 0.22- μm filter is encountered at the bottom of the stack. Technical representatives of filtration manufacturers can give specialist advice for individual cases.

4. Many suppliers offer specific inhibitors of proteases or classes of protease. These inhibitors are often peptides or proteins, e.g., aprotinin, soybean trypsin inhibitor. A cocktail of protease inhibitors is available in tablet form from Boehringer Mannheim (Mannheim, Germany) under the trade name Complete. The product is stated to give effective inhibition of serine, cysteine, and metalloproteases during protein extractions from a variety of tissues and sources.
5. A variety of vessels and membranes for laboratory-scale ultrafiltration, with a range of defined mol-wt cutoffs, are commercially available. These may comprise permanent stirred pressure cells with replaceable membranes (for volumes in the range of 10–500 mL) or disposable centrifugal concentrators (for volumes up to 10 mL) (18). The Amicon and Millipore corporations are prominent suppliers of such apparatus. Schein gave some useful observations on ultrafiltration and suggested some other means of achieving protein concentration (18).

Sucrose concentration is an effective and rapid means of concentrating a dilute protein solution. Place the solution of interest into a suitably treated, softened dialysis tube, and secure the ends. Tear off a piece of aluminum foil such that the dialysis tube will rest on the foil with roughly 5–6 cm to spare all around. Shake some solid sucrose onto the foil, and then rest the dialysis tube on top of the sucrose. Shake more sucrose on top of the dialysis tube, wrap the foil around the sucrose and dialysis tube to form a parcel, and place in the refrigerator. Water from the dilute protein solution will move by osmosis through the pores of the dialysis tube to the surrounding solid sucrose, leading to concentration of the protein. Examine the dialysis tubing every 15–20 min. The surrounding sucrose will gradually form a viscous liquid, which can be removed periodically and replaced with fresh solid material. Volume reduction can take place quite quickly. The method has the drawback that sucrose will enter the dialysis tube in amounts not readily calculable. (The sucrose will help to stabilize the protein, of course.) If the presence of sucrose is undesirable, gently pull the dialysis tube between finger and thumb to force its contents into one end. Knot or clamp the dialysis tube as close as possible to the concentrated solu-

tion, and then dialyze the shortened dialysis tube against a suitable buffer to remove the sucrose. Note that the dialysis tube will swell in dilute buffer as water moves by osmosis into the protein solution, which will have a high sucrose concentration. The dialysis tube must be clamped as short as possible to prevent undue redilution of the sucrose-concentrated protein solution.

- 6 It can be a good idea to place a maximum/minimum thermometer inside the refrigerator, freezer, or incubator(s) close to the containers of interest in order to record any significant variations of temperature that may occur over an extended period. (Ensure first that the thermometer will withstand the low or high temperatures!)
- 7 Low temperature freezers typically function at -70 to -80°C . These temperatures are extremely cold and can inflict a cold burn on exposed skin. Always wear insulating or autoclave gloves when handling low-temperature items: latex laboratory gloves are not sufficient.
8. Special fluids for the preparation of undercooling emulsions are available commercially under the trade name U-COOL.
9. The Arrhenius equation is empirical but a similar equation results from the applications of Eyring's transition-state theory: *see* ref. 28 for a fuller discussion of this point.
10. Amorphous solid preparations will follow Arrhenius kinetics provided they remain in the glassy state. However, if any of the elevated temperatures used exceeds the glass transition temperature (T_g), the product will become rubbery and will no longer obey the Arrhenius equation. (In fact, it will deteriorate much faster than predicted by the Arrhenius equation.) *See* ref. 24 for a description of this phenomenon and of an alternative kinetic scheme. Other situations in which deviations from Arrhenius kinetics may occur are outlined in ref. 33; *see also* Note 5 of Chapter 31.
- 11 There follows a brief description, for information only, of two programs for analysis of accelerated degradation data. Their underlying statistical methodology is beyond the scope of this chapter, and the author makes no endorsement or judgment of their merits or claims. Each user must determine the suitability of any package for his or her purposes.

The DEGTEST program carries out a three-stage analysis of data. First, the % activities at each temperature (relative to the low-temperature reference sample) are used to give estimates of relative degradation rates, assuming first-order decay. Second, these rates are fitted to the Arrhenius equation, and the goodness of fit is ascertained. Third, the decay rate at the proposed storage temperature is predicted from the Arrhenius fit, and the statistical precision of this value is calculated. The program can fit to the Arrhenius equation or to the more rigorous Eyring equation (34). In practice, however, little difference is observed between the two equations (35).

The POTENCYLOSS program is based on a linear model for stability prediction that uses a double logarithmic primary plot together with an altered secondary plot corresponding to the Arrhenius plot. The program can distinguish between zero- and first-order data and gives loss rate and shelf life parameters at a desired designated temperature. In addition, one can obtain estimates of activity values at any temperature or future time period (36)

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CHAPTER 33

Electroelution of Proteins from Polyacrylamide Gels

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1. Introduction

The ability of polyacrylamide gel electrophoresis (PAGE) techniques to resolve the individual components of complex mixtures of several thousand proteins has resulted in this group of methods being one of the most widely used laboratory techniques in biochemistry. Until relatively recently, these techniques were used almost exclusively as analytical tools to characterize proteins in terms of their size, charge, relative hydrophobicity, and abundance. However, the development of a battery of highly sensitive techniques of microchemical characterization, including N-terminal and internal protein sequencing, amino acid compositional analysis, peptide profiling, and mass spectrometry (1,2), has resulted in PAGE often being the method of choice for the micropreparative purification of proteins for subsequent chemical analysis.

A major obstacle to successful chemical characterization is efficient recovery of the separated proteins from the polyacrylamide gel, since most procedures are not compatible with the presence of the gel matrix. The most popular approach to this problem is currently the use of Western electroblotting techniques in which the proteins, after separation by PAGE, are transferred ("blotted") onto the surface of inert membranes (e.g., polyvinylidene [PVDF]), which are compatible with the reagents and solvents used in chemical characterization procedures (*see* Chapter 34).

An alternative approach that has been extensively used is electroelution. In this technique, protein zones are localized after electrophore-

From *Methods in Molecular Biology*, Vol. 59 *Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

sis by staining with Coomassie brilliant blue R-250. Protein-containing gel pieces are then excised and placed in an electroelution chamber, where the proteins are transferred in an electric field from the gel into solution and concentrated over a dialysis membrane with an appropriate mol-wt cutoff. Although this method can have a high efficiency (>90%) of protein recovery, it suffers from a number of disadvantages, including:

1. Relatively slow isolation of a small number of samples;
2. Contamination of the eluted protein with SDS, salts, and other impurities,
3. Peptide chain cleavage during staining or elution; and
4. Chemical modification during staining or elution, which can lead to N-terminal blockage (3).

In the simplest system for electroelution, gel slices containing the protein of interest are mixed with a small volume of polyacrylamide or agarose gel, and cast into a glass tube (such as that used for rod gel electrophoresis). A small sack made from dialysis tubing and containing electrophoretic buffer is then fastened with elastic bands over the end of the tube. The tube is then inserted into a rod gel electrophoresis apparatus and the protein eluted into the dialysis sack, from which it can subsequently be recovered. Electroelution is generally more efficient if 0.1% (w/v) SDS is added to the elution buffer, since it helps to maintain proteins in solution and provides them with a high negative charge density, thereby increasing their electrophoretic mobility.

More sophisticated variants of this approach have been described (3), and commercial equipment is now available that allows several different gel segments to be eluted into a series of elution chambers. In this chapter, the use of one such commercial device, the Biometra Elucon (Biometra Maidstone, Kent, UK), to recover proteins separated by SDS-PAGE is described.

2. Materials

Prepare all solutions freshly from analytical grade reagents and dissolve in deionized water.

1. Fixing and destaining solution: 450 mL methanol, 100 mL acetic acid, made up to 1 L in water.
2. Staining solution: 0.2 g Coomassie brilliant blue R-250 made up to 100 mL in water.

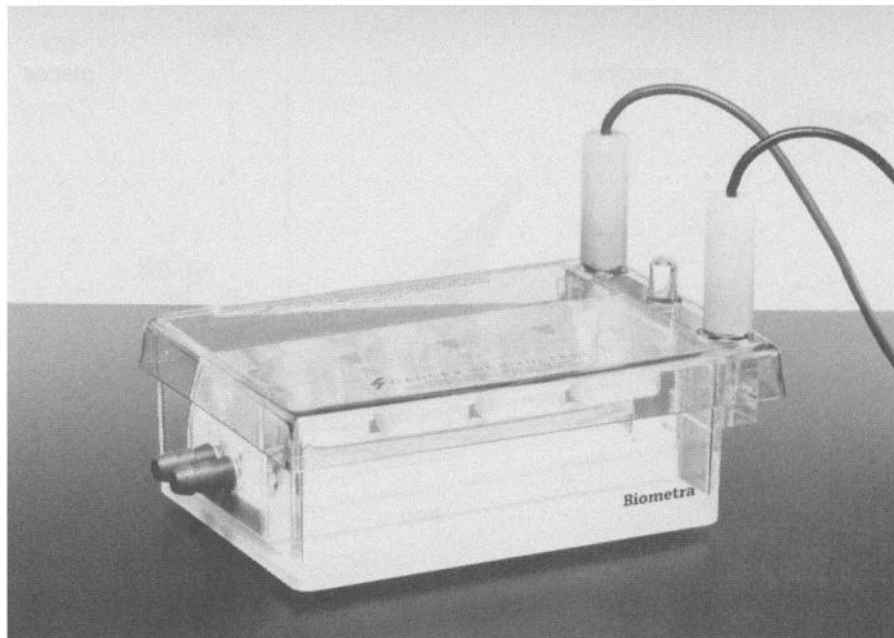


Fig. 1. Biometra Elucon apparatus for the simultaneous electroelution of proteins from up to four different polyacrylamide gel samples. Photograph courtesy of Biometra.

3. Electroelution apparatus: Biometra Elucon apparatus for the simultaneous elution of up to four samples (Fig. 1).
4. Dialysis membranes: Precut disks of dialysis membrane, exclusion limit 2 kDa (Biometra).
5. Elution buffer: For subsequent chemical characterization, it is preferable to use a volatile, basic buffer, such as 50 mM NH_4HCO_3 or 50 mM *N*-ethylmorpholine acetate, pH 8.5, containing 0.1% (w/v) SDS (*see* Note 1).

3. Methods

3.1. Gel Staining

1. Following separation of the proteins by SDS-PAGE, place the gel in fixing solution for at least 1 h at room temperature with gentle agitation.
2. Place the gel in staining solution for at 1 h at room temperature with gentle agitation.
3. Transfer the gel to destaining solution, and agitate gently at room temperature. After about 24 h and with several changes of destaining solution, the

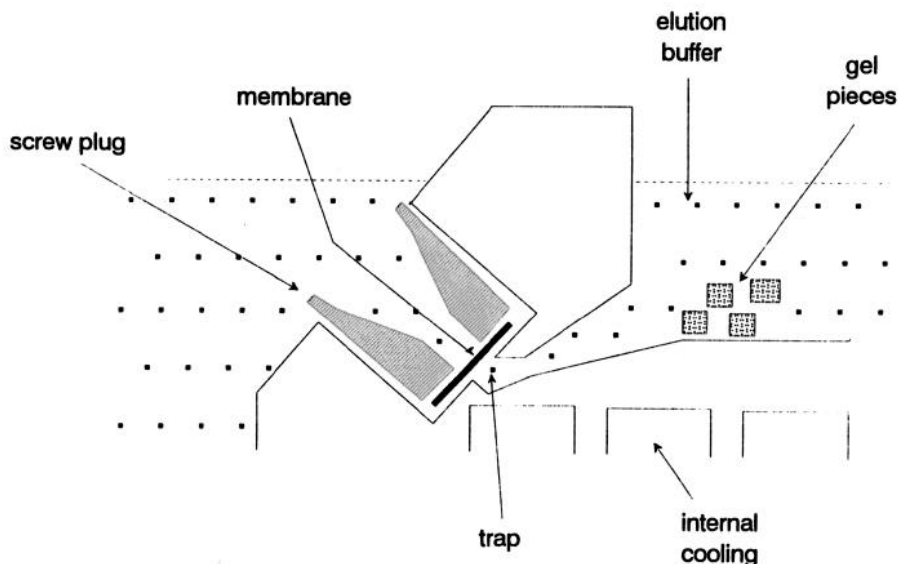


Fig. 2. Diagrammatic cross-section through an elution channel of the Elucon apparatus.

gel background will become colorless, leaving the separated proteins as blue-colored bands.

3.2. Electroelution

1. Connect the Elucon to a recirculating, cooling water bath set to 20°C.
2. Soak the dialysis membrane disks in elution buffer.
3. Fill the Elucon chamber up to 4 cm with elution buffer.
4. Place each prewetted dialysis membrane disk on top of a screw plug (*see* Fig. 2), and fix the screw with the aid of a spatula into the elution channel. Seal any elution channels not being used for electroelution with a rubber sealing disk in place of a dialysis membrane disk.
5. Fill each side of the elution channel with elution buffer, avoiding entrapment of air bubbles on either side of the fixed membrane.
6. Excise the stained protein bands of interest, and cut each band into small (3–5 mm) pieces.
7. Place the gel pieces in front of each elution channel (*see* Fig. 2).
8. Connect the Elucon to an electrophoresis power supply, and elute the proteins at 200 V constant for the appropriate time (*see* Note 2).
9. The protein is eluted out of the gel pieces and is concentrated in front of the dialysis membrane in the trap (*see* Fig. 2).

10. At the completion of electroelution, reverse the polarity of the Elucon for 1 min to release any protein that has adsorbed to the surface of the dialysis membrane.
11. Disconnect the Elucon from the power supply.
12. Insert a micropipet into the elution channel, and remove approx 100 μ L of protein-enriched solution from the trap in front of the dialysis membrane (see Note 3)

3.3. Protein Precipitation

1. Lyophilize the electroeluted protein sample, or dry using a rotary vacuum evaporator.
2. Add 50 μ L of water.
3. Add 450 μ L of ice-cold acetone acidified to a final concentration of 1 mM HCl and incubate for 3 h at -20°C .
4. Pellet the precipitated protein using a microcentrifuge.
5. Wash the pellet three times with 100 μ L ice-cold acetone.
6. Air-dry the pellet, which can then be redissolved in the appropriate solvent for the method of chemical characterization to be employed (see Note 4).

4. Notes

1. Electroelution is generally more efficient in the presence of 0.1% SDS, since it helps to maintain proteins in solution and provides them with a high negative charge density, thereby increasing their electrophoretic mobility. It can be reduced or omitted in the case of some highly soluble and highly charged proteins.
2. The time taken for electroelution is dependent on the molecular weight of the protein to be eluted, ranging from <1 h for proteins up to 50 kDa to 5 h for proteins of 150 kDa (see Fig. 3). Proteins larger than 200 kDa should be eluted overnight, but care must then be taken that the apparatus is effectively cooled (see Section 3.2., step 1) to avoid damage to the apparatus by overheating.
3. The efficiency of protein recovery by electroelution should be between 70 and 90%.
4. After electroelution, the protein-enriched solution also contains SDS, Coomassie Brilliant Blue stain, buffer salts and other contaminants. It has proven possible to apply electroeluted proteins directly onto sample targets for matrix-assisted laser desorption/ionization mass spectrometry (MALDI-MS) (4), where volatile buffer salts are removed by air-drying. However, for most high-sensitivity chemical characterization techniques, it is essential that the protein be separated from the impurities by precipitation before being redissolved for subsequent analysis.

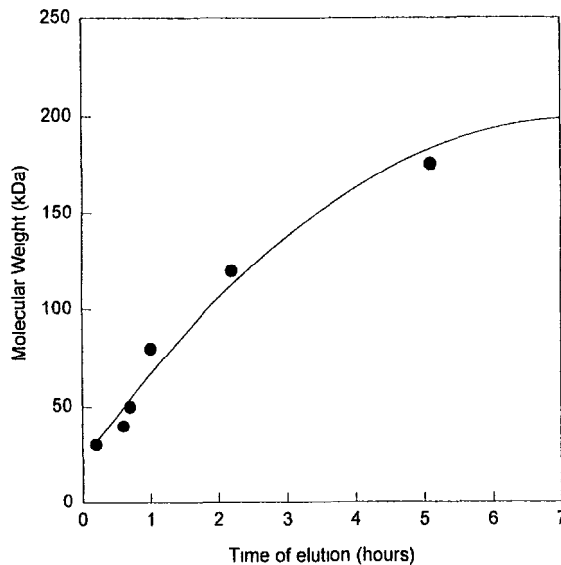


Fig. 3 Electroelution time of marker proteins in the Elucon chamber (data from Biometra) Buffer: 0.05M Tris-glycine, pH 8.6, 0.05% SDS. Electroelution conditions: 200 V, 20 mA at 20°C with cooling.

Acknowledgment

I am grateful to Peter Wolstenholme of Biometra for providing the photographic illustration of the Elucon apparatus.

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CHAPTER 34

Electroblotting of Proteins from Polyacrylamide Gels

Michael J. Dunn

1. Introduction

The ability of polyacrylamide gel electrophoresis (PAGE) techniques to resolve individual proteins of complex mixtures has resulted in this group of methods being indispensable to the protein biochemist. Although PAGE procedures can provide characterization of proteins in terms of their charge, size, relative hydrophobicity, and abundance, they provide no direct clues to the identity or function of the separated components.

A powerful approach to this problem was provided by the development of protein blotting techniques, based on those developed by Southern (1) for the analysis of electrophoretic separations of DNA. In these procedures the proteins, after separation by electrophoresis, are transferred ("blotted") onto the surface of an inert membrane, such as nitrocellulose. When immobilized on the surface of a membrane, the proteins are readily accessible to interaction with probes, such as antibodies or other ligands specific for the protein(s) being analyzed.

More recently, blotting has become the method of choice for the micropreparative purification of proteins for subsequent chemical characterization. In this approach, proteins separated by 1-D or 2-D PAGE are blotted onto an appropriate membrane support, the total protein pattern visualized using a total protein stain, and the protein band or spot of interest is excised. The protein, while still on the surface of the inert membrane support, can then be subjected to a battery of sensitive micro-

From *Methods in Molecular Biology, Vol. 59 Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

chemical characterization techniques, including N-terminal and internal amino acid sequencing, amino acid compositional analysis, peptide profiling, and mass spectrometry (2,3).

The most popular method for the transfer of electrophoretically separated proteins to membranes is the application of an electric field perpendicular to the plane of the gel. This technique of electrophoretic transfer, first described by Towbin et al. (4), is known as Western blotting. Two types of apparatus are in routine use for electroblotting. In the first approach (known as "tank" blotting), the sandwich assembly of gel and blotting membrane is placed vertically between two platinum wire electrode arrays contained in a tank filled with blotting buffer (5). The disadvantages of this technique are that:

1. A large volume of blotting buffer must be used;
2. Efficient cooling must be provided if high current settings are employed to facilitate rapid transfer; and
3. The field strength applied (V/cm) is limited by the relatively large inter-electrode distance.

In the second type of procedure (known as "semidry" blotting), the gel-blotting membrane assembly is sandwiched between two horizontal plate electrodes, typically made of graphite (6). The advantages of this method are that:

1. Relatively small volumes of transfer buffer are used;
2. Special cooling is not usually required, although the apparatus can be run in a cold room if necessary; and
3. A relatively high field strength (V/cm) is applied owing to the short interelectrode distance resulting in faster transfer times.

In the following sections, both tank and semidry electroblotting methods for recovering proteins separated by gel electrophoresis either for immunoprobng or for subsequent microchemical characterization are described. In addition, total protein staining procedures compatible with immunoprobng and microchemical techniques are discussed.

2. Materials

2.1. Electroblotting

Prepare all buffers from analytical-grade reagents and dissolve in deionized water. The solutions should be stored at 4°C and are stable for up to 3 mo.

1. Blotting buffers are selected empirically to give the best transfer of the protein(s) under investigation. The following compositions are commonly used:
 - a. For immunoprobings of proteins with pIs between pH 4.0 and 7.0: Dissolve 2.42 g Tris base and 11.26 g glycine, and make up to 1 L (*see Note 1*). The pH of the buffer is 8.3 and should not require adjustment.
 - b. For chemical characterization of proteins with pIs between pH 4.0 and 7.0 (*see Note 2*): Dissolve 6.06 g Tris base and 3.09 g boric acid, and make up to 1 L (*see Note 1*). Adjust the solution to pH 8.5 with 10*N* sodium hydroxide (7).
 - c. For proteins with pIs between pH 6.0 and 10.0: Dissolve 2.21 g 3-(cyclohexyl-amino)-1-propanesulfonic acid (CAPS) and make up to 1 L (*see Note 1*). Adjust the solution to pH 11.0 with 10*N* sodium hydroxide (8)
2. Filter paper: Whatman 3MM (Maidstone, UK) filter paper cut to the size of the gel to be blotted.
3. Transfer membrane:
 - a. For immunoprobings: Hybond-C Super (Amersham International, Aylesbury, UK) cut to the size of the gel to be blotted (*see Note 3*).
 - b. For chemical characterization: FluoroTrans (Pall, Havant, UK) cut to the size of the gel to be blotted (*see Note 4*).
4. Electroblotting equipment: A number of commercial companies produce electroblotting apparatus and associated power supplies. For tank electroblotting the TE 42 Transphor Unit from Hoefer Scientific Instruments (Newcastle-under-Lyme, UK) is used, whereas for semidry electroblotting, the NovaBlot apparatus from Pharmacia Biotechnology (St. Ablans, UK) is used.
5. Rocking platform.
6. Plastic boxes for gel incubations.

2.2. Protein Staining

2.2.1. Immunoprobings

1. Instaview Nitrocellulose staining kit (BDH, Poole, UK) (*see Note 5*). The kit comprises three components:
 - a. 50X stain concentrate.
 - b. 100X enhancer concentrate.
 - c. 10X destain concentrate.

2.2.2. Chemical Characterization

1. Destain: 450 mL methanol, 100 mL acetic acid made up to 1 L in deionized water.
2. Stain. 0.2 g Coomassie brilliant blue R-250 made up to 100 mL in destain

3. Methods

3.1. Electroblothing

3.1.1. Semidry Blotting

1. Following separation of the proteins by gel electrophoresis, place the gel in equilibration buffer, and gently agitate for 30 min at room temperature (*see* Note 6)
2. Wet the lower (anode) plate of the electroblothing apparatus with deionized water
3. Stack six sheets of filter paper wetted with blotting buffer on the anode plate, and roll with a glass tube to remove any air bubbles.
4. Place the prewetted transfer membrane (*see* Note 7) on top of the filter papers, and remove any air bubbles with the glass tube.
5. Place the equilibrated gel on top of the blotting membrane, and ensure that no air bubbles are trapped.
6. Apply a further six sheets of wetted filter paper on top of the gel, and roll with the glass tube
7. Wet the upper (cathode) plate with deionized water, and place on top of the blotting sandwich.
8. Connect the blotter to power supply, and transfer at 0.8 mA/cm^2 of gel area (*see* Note 8) for 1 h at room temperature (*see* Note 9).

3.1.2. Tank Blotting

1. Following separation of the proteins by gel electrophoresis, place the gel in equilibration buffer, and gently agitate for 30 min at room temperature (*see* Note 6)
2. Place the anode side of the blotting cassette in a dish of blotting buffer
3. Submerge a sponge pad taking care to displace any trapped air, and place on top of the anodic side of the blotting cassette.
4. Place two pieces of filter paper onto the sponge pad, and roll with a glass tube to ensure air bubbles are removed.
5. Place the prewetted transfer membrane (*see* Note 7) on top of the filter papers, and remove any air bubbles with the glass tube.
6. Place the equilibrated gel on top of the blotting membrane, and ensure that no air bubbles are trapped.
7. Place a sponge pad into the blotting buffer taking care to remove any trapped air bubbles, and then place on top of the gel.
8. Place the cathodic side of the blotting cassette on top of the sponge, and clip to the anode side of the cassette.
9. Remove the assembled cassette from the dish, and place into the blotting tank filled with transfer buffer.
10. Connect to the power supply, and transfer for 6 h (1 5-mm thick gels) at 500 mA at 10°C (*see* Note 9)

3.2. Protein Staining

3.2.1. Immunoprobings

1. Prepare a working staining solution by the addition of 1 mL of 100X enhancer concentrate and 2 mL 50X stain concentrate to 97 mL deionized water. Mix well.
2. Place the membrane to be stained in the staining solution, and agitate for 2–5 min.
3. While the membrane is staining, prepare a working solution of the enhancing solution by addition of 1 mL 100X enhancer concentrate to 99 mL deionized water, and mix well.
4. Pour off the staining solution from the membrane, and replace with the working enhancing solution. Agitate until a clear background is obtained (usually <5 min) (*see* Note 10).
5. At this stage, the membrane may be dried and stored, or destained (*see* step 6 and Note 11).
6. Prepare a working destaining solution by addition of 10 mL 10X destain concentrate to 90 mL deionized water.
7. Place the membrane in the destain solution, and agitate until all the color has been removed (typically 5–10 min).

3.2.2. Chemical Characterization

1. Remove the blotting membrane from the sandwich assembly.
2. Place the membrane into a dish containing the Coomassie blue staining solution for 2 min, and agitate gently on the rocking platform.
3. Place the membrane into destaining solution, and agitate for 10–15 min (or until the background is pale).
4. Wash the membrane with deionized water, place on filter paper, and allow to air-dry.
5. Place the membrane into a clean plastic bag, and seal until required for further analysis. The membrane can be stored in this state at room temperature for extended periods without any apparent adverse effects on subsequent chemical characterization.
6. An example of a membrane stained by this method is shown in Fig. 1.

4. Notes

1. Methanol (10–20% [v/v]) is often added to transfer buffers, since it removes SDS from protein–SDS complexes and increases the affinity of binding of proteins to nitrocellulose. However, methanol acts as a fixative and reduces the efficiency of protein elution, so that extended transfer times must be used. This effect is worse for high-mol-wt proteins, so that methanol is best avoided if proteins >100 kDa are to be transferred.

M 1 1 2 2 3 3 4 4

Fig. 1. PVDF Western electroblot transfer of human endothelial cell proteins separated by 10%T SDS-PAGE stained with Coomassie brilliant blue R-250. Different amounts of protein were applied to each lane: (1) 40 μ g, (2) 20 μ g, (3) 10 μ g, (4) 5 μ g, (M) mol-wt marker proteins.

2. The use of transfer buffers containing glycine or other amino acids must be avoided for proteins to be subjected to microchemical characterization.
3. Nitrocellulose is the most popular support for electroblotting, since it is compatible with most general protein stains, it is relatively inexpensive, and has a high protein binding capacity (249 μ g/cm² [9]). Hybond-C Super is a 0.45- μ m pore size supported nitrocellulose membrane, which is more robust than an unsupported matrix. Nitrocellulose membranes of smaller pore size are available (0.1 and 0.2 μ m) and can give better retention of small proteins (<1500 kDa). Polyvinylidene difluoride (PVDF) membranes, which have a high mechanical strength and a binding capacity similar to that of nitrocellulose (172 μ g/cm² [9]), are compatible with most immunoblotting protocols.
4. Nitrocellulose membranes are not compatible with the reagents and organic solvents used in automated protein sequencing. A range of PVDF-based (FluoroTrans, Pall; ProBlott, Applied Biosystems [Warrington, UK]; Immobilon-P and Immobilon-CD, Millipore [Watford, UK]; Westran, Schleicher and Schuell [Dassel, Germany]; Trans-Blot, Bio-Rad [Hercules,

CA)], glass fiber-based (Galssybond, Biometra [Maidstone, UK]; PCGM-1, Janssen Life Sciences [Geel, Belgium]), and polypropylene-based (Selex 20, Schleicher and Schuell) membranes have been developed to overcome this problem (2,10,11) PVDF-based membranes have a higher protein binding capacity, and result in better average repetitive and initial sequencing yields (7). Nitrocellulose can be used as a support in applications, such as internal amino acid sequence analysis and peptide profiling, where the protein band or spot is subjected to proteolytic digestion prior to characterization of the released peptides.

5. A variety of methods providing different levels of sensitivity have been described for the detection of total protein patterns on nitrocellulose and PVDF membranes following electroblotting (12). However, most of these methods (including Coomassie brilliant blue R-250, Amido Black 10B, India ink, colloidal gold) are incompatible with subsequent immunoprobng. Instaview Nitrocellulose is a rapid and sensitive (10 ng/band) method for visualization of protein patterns on nitrocellulose or PVDF membranes, and the membrane can be rapidly destained without loss of immunoreactivity. This allows a second probing of the membrane after initial protein detection, either with specific activity stains or antibody detection methods.
6. Gels are equilibrated in blotting buffer to remove excess SDS and other reagents that might interfere with subsequent analysis (e.g., glycine). This step also minimizes swelling effects during protein transfer. Equilibration may result in diffusion of zones and reduced transfer efficiencies of high-mol-wt proteins. It is important to optimize the equilibration time for the protein(s) of interest.
7. Nitrocellulose membranes can be wetted with blotting buffer, but PVDF-based membranes must first be wetted with methanol prior to wetting with the buffer.
8. The maximum mA/cm² of gel area quoted applies to the apparatus that was used. This should be established from the manual for the particular equipment available.
9. Blotting times need to be optimized for the particular proteins of interest and according to gel thickness. Larger proteins usually need a longer transfer time, whereas smaller proteins require less time. Proteins will also take longer to be transferred efficiently from thicker gels. The transfer time cannot be extended indefinitely (>3 h) using the semidry technique, since the small amount of buffer used will evaporate. If tank blotting is used, the transfer time can be extended almost indefinitely (>24 h) providing that the temperature is controlled.
10. With PVDF membranes, a blue background staining may still be observed.

11. The destaining step completely destains the protein bands and the background of the blot, allowing the subsequent use of specific methods of probing the membrane using normal protocols.

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CHAPTER 35

High-Performance Electrophoresis Chromatography

*Serge Desnoyers, Sylvie Bourassa,
and Guy G. Poirier*

1. Introduction

Protein purification is important because it gives essential data on an enzyme, like its catalytic activity, its interactions with other proteins or DNA, amino acid quantitation, N-terminal microsequencing, and other information. These analyses will provide important data for designing oligodeoxynucleotide probes for gene cloning and peptide synthesis for antibody production. Purified proteins can be obtained by several methods: gel-filtration chromatography, ion-exchange chromatography, immunoaffinity chromatography, immunoprecipitation, and salt precipitation (1).

Micropreparative electrophoresis is a relatively new protein purification method (2–4) that gives fast results (usually a few hours) and yields enough purified protein (≤ 200 μ g) to allow microsequencing either directly from the elution chamber (5) or by centrifugation on polyvinylidenedifluoride (PVDF) membrane (6). Micropreparative electrophoresis purification is based on the principle of relative mobility of proteins in sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE).

Originally described by Ornstein (7) and subsequently modified by Laemmli (8), SDS-PAGE is still the most reliable analytical system for proteins. However, it can also be used as a preparative system. This system is based on the effect of an electrical field causing charged molecules to move toward the electrode of opposite polarity. The mobility of each molecule decreases as it interacts with the surrounding matrix, in

From Methods in Molecular Biology, Vol 59 Protein Purification Protocols
Edited by S. Doonan Humana Press Inc., Totowa, NJ

the present case, polyacrylamide gel. The gel acts as a molecular sieve. The amphoteric nature of proteins implies that they will migrate toward the electrode opposite their net charge. In SDS-PAGE, the net charge of each protein is the same because the proteins are solubilized in an ionic detergent, SDS. The anion dodecyl sulfate gives each molecule of protein a negative charge. Proteins are thus separated according to their molecular weight, and small proteins move faster than larger ones. The criterion of purity is to obtain a single band on SDS-PAGE, on which is based the principle of protein purification by micropreparative electrophoresis. Instead of stopping the run when the front of migration reaches the bottom of the gel, electrophoresis is continued until each zone is eluted from the gel. If eluate is fractionated, each zone can be collected as a purified one.

Several devices have been described for micropreparative electrophoresis (2,9), and there are now commercialized apparatuses, such as the high-performance electrophoresis chromatography system (HPEC™) 230A from Applied Biosystems Inc. (Foster City, CA), ELFE from Genofit Inc. (Grand Lancy, Switzerland), Prep-Cell 491 from Bio-Rad (Hercules, CA), and 1100PG from BRL (Gaithersburg, MD). This chapter describes a method originally described for a slab gel apparatus (10), but applied on a commercialized micropreparative electrophoresis apparatus from Applied Biosystems, model 230A HPEC (11). The HPEC apparatus is provided with an on-line UV detector, an elution module, and a fraction collector. Proteins are chromatographed in a polyacrylamide gel matrix under the influence of an electrical field. The gel column stands between the cathodic module and the anodic module. Buffers are constantly renewed by a pressurized tubule system, eliminating heat build-up and ion depletion.

The protein sample is loaded onto the top of the gel at the cathodic end. Once current is applied, the proteins are electrophoresed under thermostated conditions. Since each separated zone is eluted from the bottom of the gel, proteins are swept through the detector by a flow of elution buffer and collected by a fraction collector. The proteins so purified can be analyzed further.

Most of the known buffer systems existing for slab gel can be applied for HPEC purification, with slight modifications (12). The inconvenience with the Laemmli buffer system Tris-glycine-SDS is that the sample, once purified, needs to be handled for desalting (6) to allow microsequencing of the purified protein, which implies a loss of material. We

describe here a micropreparative electrophoretic system buffer that is completely compatible with the microsequencing of protein permitting one-step protein purification and direct microsequencing. This buffer increases the resolution of the different proteins and is faster to recover the proteins because its use makes the blotting or the desalting step unnecessary. Virtually 100% of the proteins eluted from the HPEC is recovered.

2. Materials

It is recommended that reagents of the highest grade of purity possible or of electrophoresis grades be used.

2.1. Apparatus

Separation system HPEC 230A (Applied Biosystems):

1. Gas cylinder and regulator.
2. Gel tubes.
3. Column filters (Zytex[®] membrane).
4. Dialysis membrane.
5. Forceps.
6. Razor blade.
7. Pasteur pipet.
8. 0.5-mL Eppendorf tube with ventilation holes.

2.2. Stock Solutions and Buffers

1. Acrylamide/bis-acrylamide concentrate solution (30%T, 2.7%C): Dissolve 29.2 g of acrylamide and 0.8 g of bis-acrylamide in 80 mL of deionized (DI) water. Stir and once completely dissolved make up to 100 mL with DI water. Filter using a 0.22- μ m filter. Store at 4°C in a brown bottle for 1 mo (*see* Note 2).
2. Gel buffer (10X): 0.6M 2-dimethylamionethanol (DME) solution titrated with HCl at pH 8.3. Add 16 mL of DME (straight from the commercial bottle) to 70 mL of DI water. Adjust pH to 8.3 with concentrate HCl. Make up to 100 mL with DI water. Store at 4°C in a brown bottle for 6–8 wk.
3. 10% Ammonium persulfate (APS): Add 50 mg of APS to 0.5 mL DI water. Prepare fresh before each use.
4. Upper buffer (1X): 120 mM DME-HCl, 150 mM boric acid, 0.1% SDS (*see* Note 1). Add 24 mL of DME (straight from the commercial bottle) to 1.5 L of DI water. Dissolve 9.27 g of boric acid. Stir and once dissolved make up to 1.98 L with DI water, filter using a 0.22- μ m filter, and sparge with helium for 20 min. Add 20 mL of 10% SDS, stir gently. The pH should be close to 9.3.
5. Lower and elution buffers (1X): 150 mM boric acid-DME, pH 8.3. Dissolve 13.9 g of boric acid in 2.8 L of DI water and adjust pH to 8.3 with a few

Table 1
Gel Formulation for 10 mL of Final Mixture

Solutions	7.5%T	10%T	15%T
Acrylamide-bis, mL	2.44	3.33	4.88
10X gel buffer, mL	1	1	1
Water, mL	6.44	5.6	4
10% APS, μ L	70	70	70
TEMED, μ L	7	7	7

drops of DME (usually around 4 mL). Make up to 3 L with DI water, filter using a 0.22- μ m filter, and sparge with helium for 20 min.

6. Sample buffer (2X): 0.06M DME-HCl, 6% SDS, 20% glycerol, 0.4% 2-mercaptoethanol. Combine 1 mL 0.6M DME-HCl, 3 mL 20% SDS, 4 mL 50% glycerol, and 40 μ L 2-mercaptoethanol. Mix all reagents in water to a final volume of 10 mL.

3. Methods

3.1. Tube Gel Preparation

1. Gels are poured into glass tubes, which must have been thoroughly cleaned with detergent, such as Alconox[®] overnight
2. Rinse with DI water and dry.

3.2. Dialysis Membrane

This is an important component because it is placed between the elution block and the lower block of the HPEC230A, preventing proteins from going into the lower buffer tubule. Boil the membrane in DI water for 30 min. Handle the membrane with forceps. Store at 4°C in a 50:50 water:methanol solution.

3.3. Gel Preparation (see Table 1)

1. Put a plastic cap at one end of the tube, and a plastic collar at the other end. Tubes must be maintained in a vertical position.
2. Prepare the required gel volume, excluding APS and TEMED, according to the number of desired gel columns to be used (one gel column = 1 mL)
3. Deaerate the gel solution for 15 min under vacuum.
4. Add catalyst (APS and TEMED), and gently swirl the solution (see Table 1).
5. Fill tubes using a 9-in. Pasteur pipet.
6. Remove any air bubbles.
7. Add a layer of water-saturated butanol over the gel solution.
8. Allow to polymerize for 1 h (see Note 3)

9. Remove the collar, and cut the protruding portion of the gel with a razor blade while leaving 2 mm of gel protruding for storage.
10. Seal the end with a plastic cap and Parafilm™. Put in a small container (Falcon 50-mL conical tube with 1 or 2 mL of water). Gels can be conserved for 2 wk in such conditions.

3.4. Buffer Installation and Circulation

1. Use buffers described in Section 2.2.
2. Fill appropriate bottles with upper, lower, and elution buffers.
3. Adjust pressure to 1–2 psi in the tubule system to allow buffers to circulate (*see* Note 4).

3.5. Gel Installation

1. Carefully remove cap and collar from the gel tube by cutting the plastic with a razor blade. Be careful not to touch or move the gel and thus introduce an air pocket.
2. With razor blade, cut the protruding gel at the top and bottom of the gel tube, which must be as even as possible. Cut the gel under water to avoid introduction of an air pocket.
3. Wet the bottom and top ends of the gel tube with a drop of upper and lower buffer, respectively. Place a Zytex® membrane on each end (they will stay in place because of the surface tension).
4. Place the gel tube between the upper and lower electrode assembly. Be sure there is no leak of running buffers.
5. Purge the tubule system with buffer to make sure there is no air bubble in the system, especially in the elution line, which goes directly through the UV detector.
6. Elution buffer flow rate should be 15–20 $\mu\text{L}/\text{min}$ for good analysis and collection of the sample protein.

3.6. Sample Preparation

The sample must be dissolved in the 2X sample buffer and in the smallest volume possible (10 μL) even if the apparatus allows a sample volume of 90 and 200 μL (13) (*see* Note 5).

1. If the sample is solid, dissolve the proteins in 5–10 μL of DI water and mix an equal volume of 2X sample buffer. Heat at 100°C for 2 min. Place on ice for 5 min, and spin at 13,000g for 2 min (*see* Note 6).
2. For sample of <25 μL , mix an equal volume of 2X sample buffer. Heat at 100°C for 2 min. Place on ice for 5 min and spin at 13,000g for 2 min.
3. For a sample of >25 μL , concentrate the sample before solubilization as a liquid (*see* Note 7)

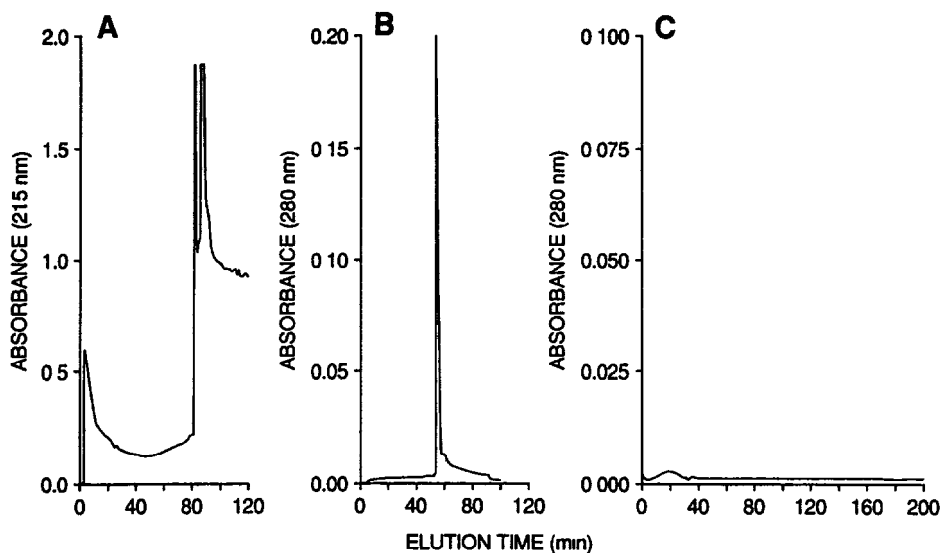


Fig. 1. Prerunning in Tris-glycine-SDS compared with DME buffers. The Tris-glycine-SDS buffer shows a typical shift in the optical density when the glycine ions replace the chloride ions (A). There is no shift in the optical density with the DME buffer system, but removal of UV-absorbing matter is still visible (B). The baseline in such a system is very stable (C).

3.7. Prerunning

Each new gel used for micropreparative electrophoresis must be prerun to remove UV-absorbing contaminant. This is done by applying a current of 1 mA for 1–2 h. As depicted in Fig. 1, it is possible to see the peaks of absorption of these contaminants in both of the systems, i.e., Tris-glycine-SDS and DME-borate. The shift in the baseline level is caused by glycinate ions for the Tris-glycine-SDS system (Fig. 1A), but in DME-borate system, there is no shift because of the presence of DME in both the lower and elution buffers (Fig. 1B; *see* Note 8).

3.8. Electrophoresis

Electrophoresis is conducted at 1 mA/gel after loading the sample at the top of the gel column. Eluted fractions are collected in a fraction collector in 0.5-mL conical tubes with ventilation holes (Fig. 2). Sample proteins can be further analyzed by microsequencing (Fig. 3; *see* Note 9).

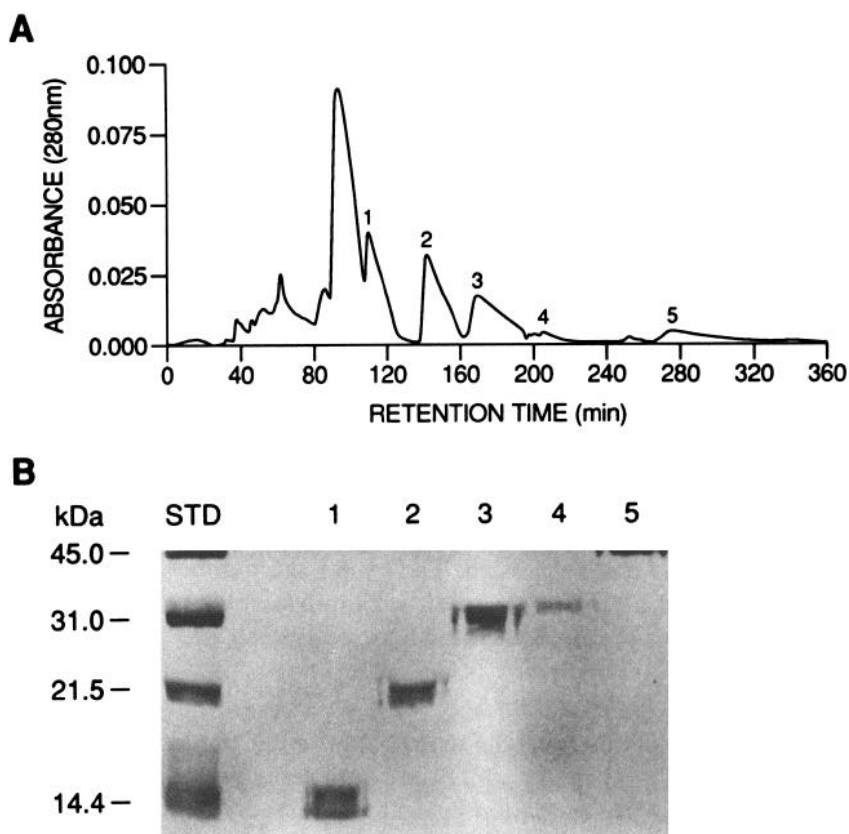


Fig. 2. Separation of four different proteins using a 2.5×50 -mm polyacrylamide column (15%T) and DME buffer. Twenty micrograms of protein were loaded onto the column, and electrophoresis was performed for 360 min at 1 mA (A). An aliquot of each peak was analyzed on a mini-SDS-PAGE and silver-stained (B). 1, lysozyme; 2, soybean trypsin inhibitor; 3 and 4, carbonic anhydrase; 5, ovalbumin.

4. Notes

1. The quality of SDS is important because an impure C_{12} SDS source can cause an electrophoretic pattern to change (13). Bio-Rad SDS has been proven to be of the highest quality.
2. Acrylamide is a neurotoxic agent. Wear protective gloves and mask when handling the powder and the liquid. To dispose of old liquid acrylamide solution, add an excess of catalyst (APS and TEMED), and let polymerize. Discard the solid product. Handle bis-acrylamide with the same precaution as acrylamide, even though there are no data on its toxicity.

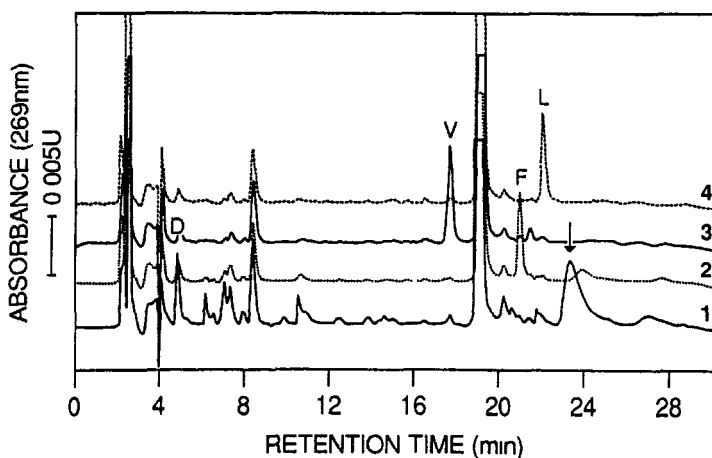


Fig 3 Microsequencing of HPEC-DME buffer-eluted soybean trypsin inhibitor. The protein, after elution from the HPEC, was directly applied to a polybrene membrane and sequenced. The cycles are numbered in the figure. The first cycle shows a contaminant peak at 23 min (arrow), which is out of the range of separation of the amino acids. Microsequencing was performed on an ABI Model 473A.

3. Polymerization of polyacrylamide is an exothermic process, and control of heat production might be important in a high percentage gel because of the possibility of bubble formation in the gel. Polymerization should then be conducted at 4°C
4. Gas flows from the gas tank to pressurize the buffer bottles. Upper buffer flows across the cathodic electrode and out to the waste bottle. Lower buffer flows across the anodic electrode and out to waste. Elution buffer flows between the anodic end of the gel and the lower buffer.
5. Samples must be loaded in small volumes because there is no stacking gel in the HPEC system. Glycine can be eliminated and a single buffer (DME) used in place of Tris/glycine. DME contains no primary amino groups and is therefore compatible with microsequencing, unlike Tris.
6. Centrifugation is important to remove particulate insoluble material from the sample. If particulate material is loaded on the gel, it may dissolve during the run and cause a "ghost" peak to appear.
7. Protein samples can be concentrated by ethanol or trichloroacetic acid precipitation or lyophilization. Precipitated proteins are usually difficult to resolubilize and require longer heat treatment. If a sample cannot be concentrated sufficiently, it can be dialyzed against 1X sample buffer without 2-mercaptoethanol, which has to be added just prior to heating. Another pos-

sibility is to achieve the composition of 1X sample buffer within the sample using concentrated gel buffer, SDS, 2-mercaptoethanol, and glycerol.

A minimum of protein must be loaded on the gel column to obtain maximum recovery. Recovery of more than 95% is achieved with 50 μ g or more of protein. Loading of less protein could yield a recovery of as low as 25% after an HPEC run (14)

8. Prerunning the gel column in the HPEC apparatus is necessary to establish a continuous system of ions in the gel and is a useful diagnostic tool. In the standard Tris-glycine-SDS system, the prerun showed (Fig. 1A) two peaks eluted between 80 and 100 min of the run. These two peaks are gel contaminants and must be removed before the run. As also shown, the baseline value increased compared with the initial value at the beginning of the run. This increase in baseline value was caused by glycine ions replacing chloride ions in the gel. Thus, the prerun is necessary to establish a continuous system of ions. Glycine and SDS are not recommended for Model 230A. Glycine generated high background absorbance at low wavelength and interferes with microsequencing. The SDS caused foaming to occur and disturbed the baseline. The DME buffer system is better used at 280 nm for detection, because it has a strong absorption at 215 nm, and baseline stability is thus affected.
9. To demonstrate the compatibility of the elution buffer with the microsequencing, soybean trypsin inhibitor collected after micropreparative electrophoresis was directly applied to a polybrene membrane and sequenced. The first four amino acid cycles are shown in Fig. 3: Asp (D), Phe (F), Val (V), and Leu (L). There is a single additional peak in the first cycle (arrowed), which presumably represented a contaminant. This peak did not interfere with the separation and identification of any authentic amino acids in the cycle. Since DME contains no primary amine that would be derivatized, there is little probability that these contaminant peaks come from DME. With a repetitive yield of 95.7% and an initial yield of 95.6 pmol, which is 10% of that applied on HPEC and almost 100% of that recovered from HPEC, this yield compared very well with sequencing results from soluble proteins. Similar results were obtained by sequencing lysozyme separated under the same conditions (data not shown). Although sample recovery on the HPEC is better in the Tris system (25%, data not shown), the separation is far better in DME-borate (Fig. 2 and ref. 14) because the soybean trypsin inhibitor is completely separated from lysozyme, which is not the case in the Tris system.

Acknowledgments

This work was supported by a grant from the Medical Research Council of Canada (grant no. 12344). The authors thank Michael H. P. West

for his contribution to the method described in this chapter, and Van Luu The for making the HPEC available for our research program. Figures are reprinted with permission from ref. 5, *Anal. Biochem.*, Academic, San Diego, CA.

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CHAPTER 36

Practical Column Chromatography

Shawn Doonan

1. Introduction

Many of the most powerful methods of protein fractionation involve the use of column chromatography. Earlier chapters in this volume describe specific chromatographic techniques, but the focus is mainly on the particular application rather than on the generalities of how to do column chromatography in practice. This final chapter is intended to fill in some of the practical gaps in terms of what materials and equipment are needed and how they should be used.

There is no doubt that for convenience, and to some extent for quality of the results obtained, it is ideal to set up a laboratory with a completely automated set of fractionation equipment and to use commercial chromatography columns and ancillary equipment. This is not always realistic. For example, if the intention is to do a one-off purification for a particular purpose, then the expense involved in setting up a dedicated laboratory will not be justified. It is, therefore, important to realize that much can be achieved with a minimum of equipment and that lack of more sophisticated facilities should not deter one from attempting protein purification by chromatographic methods; in principle, all that is needed is a column and some plastic tubing! In what follows, these low-cost options are dealt with in addition to a survey of available commercial equipment. Detailed descriptions of how to set up a particular piece of equipment are always provided by the supplier and are not given here.

There are several suppliers of fractionation equipment, but reference will be made mainly to Pharmacia Biotech (Milton Keynes, UK), since this

From *Methods in Molecular Biology, Vol 59 Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

is the equipment with which the author is familiar; catalogs from other suppliers (Bio-Rad [Hercules, CA], Watson-Marlow [Falmouth, UK], Whatman [Maidstone, UK], Amicon [Beverly, MA], Pierce [Rockford, IL]) should be consulted for alternatives. Approximate prices for equipment are given to provide an idea of the cost of setting up a laboratory. No reference is made to specialist equipment for FPLC or for HPLC, since these topics are covered in Chapters 26 and 27, respectively. Similarly, Fig. 2 in Chapter 25 gives a diagram of the basic arrangement for column chromatography, and this is not repeated here.

2. Equipment and Materials

2.1. Columns

For most applications columns should have a length-to-diameter ratio in the range 5/1–20/1 (*see* Note 1). Pharmacia Biotech markets two main series of columns for general laboratory-scale work. The XK series is suitable for all applications, except for use with organic solvents (where the SR series is required). Internal diameters of 16, 26, and 50 mm are available in each case with four tube lengths in the range 20–100 cm. Each column is supplied with a variable-flow adaptor that allows complete enclosure of the matrix bed, so that sample and buffer application is directly onto the bed with no dead space. The adaptors are continuously adjustable up to 18 cm, so that a 40-cm tube can have an enclosed matrix bed anywhere in the range 22–40 cm. Flow adaptors can be used at both ends of the column, giving greater flexibility in bed height, although normally a fixed-end piece is used at the bottom of the column. By appropriate choice of column dimensions and with the use of adaptors, any bed volume up to 1880 mL can be obtained. The columns come with thermostatic jackets for temperature control by circulation of coolant from an external source. Prices are in the range of \$165 for the XK 16/20 to \$330 for the XK 50/100.

The more basic C range of columns come with diameters of 10, 16, and 26 mm with lengths in the range 10–100 cm (40-cm maximum for the 10-mm diameter column). They are supplied without flow adaptors and without thermostatic jackets, although these are available as optional extras, and in this basic format are about 50% of the cost of the corresponding XK column. If purchased without the thermostatic jacket, then the column must be run in a cold room if temperature control is required. Packing reservoirs, which effectively extend the length of the column, are available for both the XK and the C series.

A cheap alternative is provided by standard laboratory chromatography columns. These are essentially glass tubes drawn out to a taper at one end and with a sintered glass disk (usually of porosity P160; pore size 100–160 μm) sealed in above the taper. For example, BDH (Poole, UK) markets these with diameters of 10, 20, and 30 mm, and 30-cm long at a cost of about \$15 each. Larger sizes can be made by purchasing Buchner filter funnels with diameters in the range 30–95 mm (porosity P160), and then getting a professional glass blower to join a piece of glass tube of the same diameter and of the desired length onto the funnel. A 70 mm $\times$ 50 cm column (bed volume 2 L) made in this way is very useful for large-scale work and easier to handle than commercial columns of a similar size. The major problem with them is the dead space under the sinter in which substantial mixing can occur, but for initial relatively crude separations, this is not important. At the other end of the range for very small-scale work, a Pasteur pipet or small syringe plugged with glass wool makes a perfectly adequate column.

For most purification schedules, a range of column sizes will be required. A good selection from the Pharmacia Biotech range would be one each of C 10/20, C 16/40, and C 26/40 with a flow adaptor in each case (or the corresponding XK columns if a cold room is not available) plus one or two large homemade columns. The Pharmacia Biotech columns can, however, be substituted by cheap glass columns if funds are limited (*see* Note 2).

2.2. Pumps

Columns can be operated under gravity flow with control exercised by the height of the solvent reservoir above the column and/or by a screw clip on the tubing at the column outlet. Ideally, however, a peristaltic pump should be used. Pharmacia Biotech markets a basic single-channel pump (P-1) at about \$1300. This gives a continuous range of flow rates from 0.6–500 mL/h, which is adequate for most purposes (*see* Note 3). This range is achieved by using tubing (silicone rubber is standard) of three different internal diameters (1.0, 2.1, and 3.1 mm) with two continuous ranges of roller speeds differing by a factor of 10. Pumping can be in either direction. The pump can be controlled from Pharmacia Biotech fraction collectors or from a chromatography controller. It can also be safely used in a cold room, which is essential for many applications. More expensive piston pumps that give greater flow rates and can exert moderately high pressures are available (e.g., the HiLoad P-50 from

Pharmacia Biotech, maximum rate 50 mL/min, cost \$3300), but there are few situations in which their use is essential.

2.3. Fraction Collectors

Although it is obviously possible, it is very tedious to collect fractions from a column by hand, so a simple fraction collector is an important item of equipment (*see* Note 4). The basic RediFrac from Pharmacia Biotech has a standard rack for 95 test tubes of 10–18-mm diameter; accessory racks for 45 tubes of 28-mm diameter or for 175 tubes of 12-mm diameter are also available. The cost is about \$1500 with extra racks at about \$80 each. Collection can be on the basis of time or of drop count, and there is a facility for marking tube changes on an attached recorder. Shut-down can be programmed and linked to switching off the peristaltic pump, so that flow through the column is terminated. The fraction collector can be used in a cold room. The GradiFrac collector has the same basic features, but with additional capabilities for control of ancillary equipment (pumps, monitors, switch valves) and for analysis of chromatograms; it costs about \$3000. At the top of the range (\$4300) is the SuperFrac collector, which has very sophisticated control functions, including a peak slope detection system, which allows for accurate collection of poorly resolved peaks. It is also easily adaptable for the collection of large-volume fractions. These features might be in demand in a laboratory that is dedicated to protein purification, but they are luxuries that can be dispensed with, and for most applications the basic RediFrac is perfectly adequate. Certainly for an active laboratory, two cheaper fraction collectors would be much better use of resources than one top-of-the-range model.

2.4. Monitors

Fractions from a column can be analyzed for protein content by manual absorption measurements at an appropriate wavelength (generally 280 nm; *see* Note 5), but it is more convenient to use a monitor attached to a chart recorder for continuous display of the elution profile. The UV-1 monitor uses a mercury discharge lamp with filters to select wavelengths of 254, 280, or 405 nm. It has a sensitivity range of 0.01–2 absorbance units for full-scale deflection and provides a 0–10 mV signal for connection to standard chart recorders. The cost of the basic instrument with flow cell and two filters is about \$5000. One problem with this monitor is that it

should not be used in a cold room. This restriction does not apply to the Uvicord S II. The latter also provides a greater range of wavelengths (including 206 and 226 nm, which are useful for detecting proteins at very low levels) and has the capability of connection to a peak integrator. The cost is higher at about \$6300 (with three filters), but the extra expense is justified by the extra performance. Either machine requires, of course, a chart recorder for display of the monitor signal. Pharmacia Biotech offers single-channel (REC 101, about \$2300) and dual-channel (REC 102, about \$2800) machines, both of which are fully compatible with their other equipment. The latter is to be preferred if it is intended to monitor more than one property of the column effluent.

In addition to absorption monitors, Pharmacia Biotech also produces flow-through conductivity and pH monitors (both cost about \$2200). They are useful for monitoring gradients (particularly the former, since salt gradients are much more commonly used than are pH gradients), but neither can be considered as essential pieces of equipment.

2.5. Gradient Makers

Many chromatographic procedures require elution of adsorbed protein with a gradient usually of increasing salt concentration. Pharmacia Biotech markets a gradient mixer (GM-1, cost about \$800) that produces linear gradients of between 40 and 600 mL. These are produced by having two reservoirs each of 300-mL capacity and of equal diameter connected at the base, such that as solvent is withdrawn from the reservoir originally containing start buffer, the limit buffer flows in to maintain equal levels in the reservoirs (*see* Note 6). Mixing is provided by a motor-driven paddle.

Such devices can also be easily made by a competent glass blower. All that is required is two flat-bottomed glass cylinders, one (for the limit buffer) with a single outlet near the base, and the other (for the start buffer) with two such outlets. The two reservoirs are connected by a sort piece of flexible tubing that can be closed off with a screw clip. Buffer is drawn off from the reservoir initially containing start buffer through plastic tubing connected to its second outlet. Mixing in this reservoir is provided by a magnetic stirrer and pellet. It is useful to have a range of reservoirs available with volumes in the range 100 mL to 2 L.

It is even easier to construct a system with two measuring cylinders connected by a syphon of narrow-bore plastic tube buffer being with-

drawn from the cylinder originally containing start buffer through a second plastic tube. With this arrangement, it is essential to ensure that the plastic tubings do not slip out during gradient operation. This can be done by passing the plastic tubings through lengths of glass tube and fixing the latter in place with tape to the tops of the measuring cylinders. Beakers can be used instead of measuring cylinders, but they should be tall and narrow. Otherwise small differences in level of the beakers will cause a large movement of buffer from one to the other.

Yet another possibility is to use a peristaltic pump to deliver limit buffer into a mixing vessel from which solvent is withdrawn onto the column by a second pump. If the flow rate of the pump delivering limit buffer is one-half of the rate at which solvent is removed onto the column, then the gradient will be linear as with the above devices. The problem with this method, of course, is that it ties up a second peristaltic pump.

2.6. Valves

It is convenient, but by no means essential, to include three- or four-way valves in the solvent stream before and after a column. These can be used for sample application and for diversion of the effluent stream to waste, respectively. Pharmacia Biotech markets manual valves (LV-3 and LV-4) at about \$170. If the chromatography system is to be automatically controlled, then the valves must be solenoid-driven (e.g., the PSV-50 at about \$330). The latter can be controlled by the fraction collector in an automated system.

2.7. Tubing

It is important to have available a supply of flexible plastic tubings for transporting solvents from one part of the chromatographic system to another. These are marketed by most suppliers of laboratory materials, but again a convenient range is available from Pharmacia Biotech. Standard polyethylene tubing with an internal diameter of 1 mm and external diameter of 1.8 mm is generally useful. Polyvinyl chloride tubings with larger internal diameters (1.6–4.0 mm) are useful when faster flow rates are required. A supply of flexible Microperpex (silicone rubber) tubings with internal diameters of 1.3, 2.7, and 4.0 mm is very useful for making joins between lengths of capillary tubings or joining these to glass tubing. For example, a 1-cm length of the 1.3-mm Microperpex tubing makes a good air-tight connection between lengths of the standard 1.8-mm polyethylene tubing.

3. Methods

3.1. Packing Columns

1. Suspend the appropriate amount of matrix (*see* Note 7) in equilibration buffer. Check the pH, and if it is far removed from the target value, then adjust it by careful addition of the acidic or basic component of the buffer.
2. Resuspend the matrix by gentle stirring, and allow to settle until a firm bed has formed. If there is fine material in suspension, then remove this by aspiration with a water vacuum pump or by syphoning. Repeat this step until no more fines remain (*see* Note 8).
3. Choose a column of such a size that the matrix bed will fill it sufficiently for the adaptor to reach the top of the bed (if using a commercial column with adaptor) or to about 80% of its height. Clamp the column vertically in a position where it is protected from drafts or sources of radiant heat (convection currents owing to these will cause uneven column packing). Ideally, the column should be packed at the temperature at which it is to be run.
4. Seal off the outlet tube of the column, and pour in some equilibration buffer. Open the outlet, and allow buffer to flow through to remove air from the bottom net or sinter and from the outlet tubing. Clamp off the outlet again.
5. Suspend the settled matrix in about an equal volume of buffer, and pour the slurry into the chromatography column being careful not to trap any air bubbles (pouring it down a glass rod touching the side of the column can be helpful). If using a column packing extension or reservoir, the remainder of the slurry can also be poured in.
6. Open the bottom tube of the column, and allow buffer to flow through at a rate about 20% greater than that at which it is intended to run the column (*see* Note 9); flow rate can be controlled either with a peristaltic pump or by adjustment to the pressure on a screw clip attached to the outlet tubing.
7. As packing proceeds, a discontinuity will be seen between the packed bed, and the suspension of matrix above it. If using a packing extension then allow packing to continue until this discontinuity has reached the desired height, and remove any remaining suspension from the column. If not using a packing extension, then as clear liquid forms above the suspension, remove it by aspiration and replace it with fresh suspension until the bed has reached the desired height (*see* Note 10). Stop flow through the column.
8. If using a column with a flow adaptor, then layer buffer on top of the matrix bed, being careful not to disturb it, and fill the column to the top. Slide the upper adaptor across the top of the column so that no air is trapped between the net, and the liquid meniscus, push the adaptor just into the column, and tighten the compression nut so that the sealing O-ring makes contact with the sides of the column. Push the adaptor slowly down the col-

umn until the net is in contact with the matrix bed, during this process, excess buffer will exit through the tubing of the flow adaptor, the end of which should be submerged in the buffer reservoir. When the adaptor is in place, then tighten the compression nut so that the O-ring makes a firm contact with the sides of the column. If using a column without an adaptor, then carefully layer buffer on the top of the matrix (the depth depending on the size of the column), and place a tightly fitting rubber bung with a glass tube passing through it in the top; the latter should have a length of capillary tubing passing through, such that the tubing inside the column ends somewhat above the matrix bed (and ideally, touches the wall of the column) and the other end is immersed in a buffer reservoir; the whole arrangement must be air-tight (*see* Note 11)

9. Flow equilibration buffer through the column either under the control of a peristaltic pump (*see* Note 12) or under gravity; in the latter case, flow rate can be controlled by a combination of adjusting the height of the buffer reservoir above the top of the column and adjustment of a screw clip on the outflow tubing. Generally two or three column volumes of buffer are sufficient for equilibration, but this should be confirmed by measuring the pH of the effluent and, if possible, the conductivity

3.2. Sample Application

1. The sample should have been equilibrated in application buffer either by dialysis or by gel filtration (*see* Chapter 11). Except when using gel filtration (*see* Chapter 25), there is usually no limit on the volume of sample that can be applied (*see* Note 13); protein loading will, of course, depend on the size of the column (*see* Note 7).
2. If using a flow adaptor and peristaltic pump, stop flow through the column by switching off the pump (*see* Note 14). Transfer the inlet tubing to the sample container, and restart the flow when sample will be sucked onto the column. Be careful not to introduce air bubbles; if there is a bubble at the end of the inlet tubing when it is transferred to the sample container, then briefly reversing the direction of the peristaltic pump with the tubing immersed in the sample will get rid of it. Stop the pump, transfer the inlet tube to the buffer reservoir, and restart the flow.
3. If not using a flow adaptor, then remove the rubber bung from the top of the column, and continue buffer flow until the meniscus is flush with the top of the bed. Stop the flow. Layer sample carefully on top of the bed using a pipet or a syringe, taking great care not to disturb the surface; allowing the sample to flow down the wall of the tube helps. Restart the flow. If the volume of sample is large, then replace the bung in the column, and place the inlet tube in the sample container; sample will be sucked onto the column by vacuum. Run the sample completely on to the matrix bed (*see*

Note 15), stop the flow through the column, and wash the sample in with a few milliliters of buffer. Layer buffer over the matrix bed, replace the bung, and restart the flow with the inlet tube in the buffer reservoir.

3.3. Column Development

1. Wash the column with starting buffer until no more protein is eluted (*see* Note 16). The flow-through peak can often be collected as a single fraction (*see* Note 17).
2. Elute remaining protein by application of one or more step changes in eluent or by application of a gradient (*see* Note 18). If using a column with flow adaptor, then be careful not to introduce air bubbles into the line when switching from the equilibration buffer reservoir to the gradient maker (*see* Section 3.2., step 2). If using a standard laboratory column, then the gradient can be applied by simply transferring the column inlet tube to the outlet of the gradient maker; some mixing between the gradient, and the liquid layer over the matrix bed is unavoidable. Make sure that the mixing paddle or the magnetic stirrer is switched on.
3. Collect appropriate size fractions (*see* Note 19).
4. Combine fractions using appropriate criteria (*see* Note 20).
5. Regenerate and store the column as necessary (*see* Note 21)

4. Notes

1. This is the optimum range for applications such as ion-exchange and other types of adsorption chromatography. Short, fat columns give better flow rates and are less prone to development of high back-pressures. They are, however, less easy to pack evenly. Poor packing can lead to distortion of bands of material passing through the column, which in turn can give poor resolution if two bands are eluting close together. Long, thin columns (up to 100/1 length/diameter) are appropriate for true partition chromatography applications, such as gel filtration.
2. The great advantage of commercial columns is in the design of the end pieces, i.e., the parts in contact with the matrix bed. These have very low liquid volumes, so the possibility of mixing (and of dilution) of eluted materials is minimized. This is important in the most demanding applications, e.g., in gel filtration or in adsorption chromatography where poorly resolved peaks are eluted by application of a gradient. With standard laboratory columns, it is impossible to avoid some dead volume and consequent mixing in the space under the sintered support. In addition, there is always a liquid layer over the matrix bed, which interferes with application of gradients. That being said, there are few occasions where the refinements offered by commercial columns are essential (gel filtration being one such), and it is perfectly feasible to carry out protein purification pro-

cedures using unsophisticated homemade columns. (Note that if money is no object, Pharmacia Biotech markets many of its chromatography matrices as prepacked columns. These offer consistency of behavior as well as convenience, but of course, at a price. This company's catalog should be consulted for a full list of columns available.)

3. Very large columns (e.g., 1-L bed volume) of ion-exchange materials can be run at flow rates of up to 1–2 L/h. It is perfectly feasible to run these under gravity.
4. When using large columns at high flow rates, particularly at the beginning of a purification procedure where the separation required may be relatively crude, the fractions can be quite large and may conveniently be taken by hand. For example, with a 7×50 -cm column of CM cellulose (vol 2 L) developed with a 4-L salt gradient at a flow rate of 2 L/h, it might be adequate to collect 20 fractions of 200 mL each over the 2-h period of the chromatographic run, this is more easily done manually than by adapting a laboratory-scale fraction collector to cope with the large-volume fractions.
5. Nearly all proteins absorb light at 280 nm owing to their content of the aromatic amino acids tryptophan and tyrosine. A very useful rule of thumb is that in most cases, an absorption of 1 in a 1-cm cell corresponds roughly to a protein concentration of 1 mg/mL (or $A_{280\text{nm}}^{1\%} = 10$), although for proteins devoid of tryptophan, this may be wrong by a factor of 4. However, a plot of $A_{280\text{nm}}$ against volume provides a useful elution profile for a column. The absorbance at 254 nm is generally about one-half of that at 280 nm, and hence gives lower sensitivity; it is used only because there is a strong line in the mercury emission spectrum at this wavelength. For very low protein concentrations, absorbance at 226 or 206 nm can be used, since these wavelengths are in the region of absorption by the peptide bond, and the absorbance is very high. The problem is that many buffers absorb at these wavelengths (particularly at 206 nm), thus limiting the usefulness of the detectors. Even with detection at 280 nm, the possibility of buffer absorption should be considered if the buffering species is aromatic or contains conjugated double bonds. A low-level and constant buffer absorption can usually be blanked out, but problems will arise if a gradient of increasing buffer concentration is used.
6. The shape of the gradient obtained depends of the cross-sectional areas of the two reservoirs. It can be shown (1) that:

$$C = C_1 - (C_1 - C_i) (1 - v)^{A_1/A_2} \quad (1)$$

where C is the concentration after a fraction v of the total gradient has been withdrawn, C_1 and C_i are the limit and initial concentrations, respectively, and A_1 and A_2 are the cross-sectional areas of the mixing vessel and of the

reservoir for the higher concentration solution, respectively. In the simplest case where $C_1 = 0$, and the cross-sectional areas are equal, this reduces to.

$$C = C_1 \quad v \quad (2)$$

i.e., a linear gradient. Concave or convex gradients can be obtained by appropriate choice of the ratio A_1/A_2 , but this is rarely done. For example, if the protein of interest elutes early in a particular gradient, then better resolution could in principle be obtained either by using a concave gradient or, more simply, a linear gradient with a lower final concentration. If the protein of interest elutes late in the gradient, then improved results could be obtained with a convex gradient or, more simply, by using an initial concentration (C_1 in Eq. 1) > 0 . It should be noted that the method of making gradients with two interconnected vessels works by maintaining the heights of the solutions in the vessels equal, and hence, with vessels of the same cross-sectional areas, this means starting with equal weights of the two solutions. To a first approximation, this can be taken as equality of volumes, but if the densities are markedly different (e.g., 0 and 1M NaCl solutions), then some of the strong solution will move into the mixing vessel as soon as connection is made; this is only likely to be important if the protein of interest elutes very early in the gradient.

7. The amount required will depend on the type of chromatography to be done, on the capacity of the matrix (*see* previous chapters on specific techniques), and on the state of purity of the sample. To some extent, amounts have to be determined by trial, but there are some principles and practices that can be useful in deciding the approach to take.

In ion-exchange chromatography, for example, the protein binding capacities of the matrices vary in the range 10–30 mg/mL of column bed; more precise values are given in suppliers' literature. If the chosen protocol is for the protein of interest to be retained on the column and then eluted by a single step change in the elution buffer, then the whole of the binding capacity of the column may be used since no further separation can be achieved once the protein is displaced from the matrix. If the intention is to do gradient elution, however, then only a fraction (~20–30%) of the binding capacity of the column should be used so that the remaining matrix is available for ion exchange as the displaced material passes down the column. Similarly, if conditions are chosen so that the protein of interest passes through the column without retention, then the whole capacity of the matrix can be used to bind impurities.

An estimate of the amount of matrix required can be obtained in a trial using batch adsorption. For example, take 10 aliquots of the equilibrated protein solution, and add increasing quantities of equilibrated matrix to

each Mix gently for 10 min, and then sediment the matrix by centrifugation. Determine the amount of the protein of interest remaining in the supernatant in each case. This will allow an estimate of the minimum amount of matrix required to bind all of the protein of interest, and hence, the amount to used in practice can be chosen on the basis of the outlined considerations. If the objective is to bind impurities, but not the target protein, then a similar experiment can be used to determine the amount of matrix that gives maximum removal of protein from solution without loss of target protein. It is important to bear in mind that the ability of a matrix to bind a particular protein will depend on the state of purity of the sample. Hence if conditions for adsorption, and the binding capacity of a matrix for a particular protein, have been determined using a partially purified sample, then the capacity may be considerably less if a cruder preparation is used. In these circumstances, it may well happen that more strongly binding impurities saturate the column and displace the protein of interest, so that it emerges without adsorption to the column contrary to expectations. Conversely, if a protein is partially purified using a particular ion-exchange material followed by rechromatography on the same material under the same conditions, then a smaller amount of matrix will be required for the second step, since much of the impurities will already have been removed.

There is a temptation to "play it safe" and use much larger columns than are actually required; this should be resisted. The consequences will be reduced yield owing to nonspecific adsorption of protein on the matrix and unnecessary dilution of the active fraction. There will also be a cost penalty, which can be severe with the more expensive matrices.

8. Care should always be taken to remove fines. If this is not done, then the fine particles will pack into the interstices between matrix particles and block flow of solvent through the column. If this happens, then the only recourse is to repack the column.
9. If the flow rate of the column is too great, then equilibration of protein between the matrix and the solvent may not be achieved; in addition, compaction of the matrix bed may occur. On the other hand, low flow rates will lead to peak spreading by diffusion with consequent dilution and loss of resolution, and possibly to loss of activity because of the extended time of the procedure. Flow rates are generally in the range 0.1–0.5 cm/min (i.e., 0.1–0.5 cm³/cm² of cross-sectional area/min).
10. It is important not to allow all the matrix to settle, and, then replace the supernatant liquid with more matrix suspension, and so on. Because the coarser particles settle faster, this would result in a series of discontinuities between bands of fine and coarse material in the packed bed with the consequence of inferior chromatographic performance, and the possibility of poor flow rates.

11. With this arrangement, as liquid is removed from above the matrix bed, the partial vacuum produced will cause buffer to be sucked out of the reservoir and into the column; there needs to be sufficient buffer above the matrix bed before flow is started to ensure that the syphoning starts before the bed goes dry. The tube carrying the buffer should extend to close to the top of the bed, so that incoming buffer does not "bomb" the bed and disrupt the surface. The arrangement must, of course, be air-tight. This can be arranged by pushing one end of a piece of flexible plastic tubing onto the glass tube, and then inserting one or more short lengths of tightly fitting plastic tubings into the other end until the bore of the innermost insert is somewhat less than the external diameter of the capillary tubing to be used to connect to the reservoir. The capillary can then be pushed through this last insert by a sufficient distance to end just above the matrix bed. If very small columns are being used with rubber bungs that are too small to bore easily, then a hypodermic needle can be substituted for the glass tube.
12. If a flow adaptor is being used, then the peristaltic pump can be placed before or after the column (except during packing, of course, when it must be after). One disadvantage of placing it after the column is that some mixing of the eluent will occur owing to the volume of the pump tubing and to the pumping action; this can cause loss of resolution. Another problem arises if it is attempted to pump liquid through the column at a rate greater than its natural flow rate when air bubbles will form in the outlet tubing. Nevertheless, it is common practice to leave the peristaltic pump positioned after the column after packing is complete and when the column is run.
13. Large-volumes of dilute protein solution do not usually pose a problem, particularly if the protein of interest binds to the column. Indeed, adsorption of the protein onto a column from a large-volume of solution followed by stepwise or gradient elution provides concentration as well as purification.
14. Having a three-way valve in the buffer line (before the peristaltic pump), with the tubing from the third port dipped into the sample reservoir, is useful here, since simply turning the valve will direct sample rather than buffer onto the column. If the sample tubing contains air, then reversing the direction of flow of the pump temporarily will displace the air by buffer from the column, reversing it again will then load the sample onto the column.
15. Some problems can arise at this stage. Particularly if the protein solution is concentrated, and the column diameter is large, imperfections in packing sometimes result in the protein solution channeling or slipping between the matrix bed and the walls of the column; this is easy to see if the solu-

tion is colored. The only remedy is to stop flow through the column, stir up the top section of the bed (to below the point of visible channeling), and allow it to settle again before applying the rest of the sample. Another problem can arise with crude protein samples that either have not been properly clarified before application to the column or from which protein precipitates (owing to the particular pH, and so forth) during application. The particulate matter will form a layer on top of the matrix bed; this will slow down or stop buffer flow, or may lead to channeling. The situation may be recoverable by stirring the top of the bed to disperse the layer of particulate matter, but subsequent running of the column is unlikely to be ideal. This emphasizes the desirability of using properly clarified protein solutions for chromatography. When flow adaptors are being used, particulate matter can block the applicator net, the only solution if this occurs is to dismantle the adaptor and clean or replace the net.

- 16 This can be judged by the trace on a recorder returning to base line or by manual measurements of $A_{280\text{nm}}$. This should occur with a volume somewhat larger than the liquid volume of the column, unless material is partially adsorbed, in which case the volume may be considerably greater; conditions should be chosen, if possible, to avoid the latter situation
- 17 This is obviously the case if the protein of interest is retained on the matrix (although it is always advisable to retain the unadsorbed fraction until it is confirmed that the target protein has indeed bound—failure to equilibrate the matrix or the sample properly or overloading the column can lead to unexpected behavior, and it is easier to rechromatograph the fraction than to start again from the beginning) Note that diversion of the flow-through fraction to a collecting reservoir is one of the functions that can be programmed into more sophisticated fraction collectors equipped with solenoid valves

If conditions have been chosen such that the protein of interest is not adsorbed onto the matrix, then the breakthrough material can usually still be collected in a single fraction, since it should have a uniform composition. One circumstance where this is not so is if either the target protein or some of the impurities are retarded, but not retained by the column. The target protein will then be enriched in either the back part or the front part of the breakthrough peak, and individual fractions will need to be taken for optimal purification. Similarly, if the column is overloaded so that the protein of interest, which was intended to be retained, is displaced by more tightly binding contaminants (*see* Note 7), then the target protein will be concentrated in the back part of the breakthrough peak, and potential purification will be lost if individual fractions are not collected. This situation should be suspected if the breakthrough peak is markedly asymmetric with increasing protein content in the back half.

18. The total volume of the gradient will generally be in the range of four to eight times the volume of the matrix bed. The most commonly used salt for gradients is NaCl. In the absence of previous information on the salt concentration required to elute a particular protein, a gradient of 0–1.0*M* should be tried. From this trial, the salt concentration at the maximum of the eluted protein peak can be calculated from the equation in Note 6. This value plus about 20% can then be used as the limit of the gradient for subsequent experiments. If the target protein elutes at relatively high salt concentration, then the gradient can be started at a value >0. For example, if the protein elutes at 0.6*M* NaCl, then a gradient from 0.4–0.75*M* would probably give good results. It is important not to have the protein eluting at the limit of the gradient because tailing will then occur.

To set up the gradient, pour equal volumes (*see* Note 6) of the start buffer, and the start buffer containing salt, into the mixing chamber and into the second reservoir, respectively, making sure that the connection between the vessels is closed off and that the vessels are at the same height. If it is intended to use a nonlinear gradient with vessels of unequal cross-section (*see* Note 6), then the vessels must be filled to the same level. At the time when the gradient maker is connected to the column, open the connection between the vessels, and switch on the stirring motor. If the gradient device consists of two cylinders or beakers that need to be connected by a syphon tube, then fill them to the desired level, insert the tubing into one of the vessels, suck liquid up it with a syringe, squeeze the tube so that liquid cannot escape, and place the end below the surface of the liquid in the second vessel.

19. It is not usually worthwhile collecting a large number of small fractions—this simply increases the amount of analysis to be done. Fraction volumes should be between 1/50th and 1/20th of the total elution volume of the column, depending on the degree of resolution expected. Take larger rather than smaller volumes, unless there is a good reason to do otherwise.
20. Chromatography at early stages of a purification is unlikely to result in the protein of interest being obtained in a symmetric, well-resolved protein peak. Combination of fractions is therefore likely to be a payoff between yield and purification. The first requirement is to construct a protein profile (if not using an automatic monitor) by plotting $A_{280\text{ nm}}$ against elution volume or fraction number. Next, the fractions containing the target protein must be established using a quantitative assay, and the activity profile superimposed on the protein profile. From this, it will be possible to see how to combine the fractions in such a way as to optimize both purification and yield. For example, it is usually worth rejecting fractions at the beginning and/or end of the peak of active material if their contents of

contaminating protein are very high. It is also very important to determine the total yield of the target protein, since a particular chromatographic procedure may result in substantial purification, but may still be unacceptable if it results in a low overall yield. The importance of having a quantitative assay for the protein of interest is difficult to overestimate, particularly for the analysis of fractions from chromatographic procedures. In addition, the assay should be as rapid as possible, since large numbers of fractions may have to be analyzed. Time devoted to developing a rapid quantitative assay will be well spent if a substantial amount of purification work is to be undertaken. A good example of what can be done with a little ingenuity is given in ref. 2.

21. With most matrices, regeneration can be done by washing with one or two column volumes of a solution of high salt concentration ($\sim 1M$), followed by a similar volume of equilibration buffer; manufacturers' instructions should be followed if different from the above. The column should then be stored in equilibration buffer, containing an antibacterial agent, such as sodium azide (0.2% [w/v]). Matrices used to fractionate crude protein mixtures may retain some protein and pigmented material after such a treatment. This can usually be removed by washing with 0.1M NaOH (if the matrix is stable under these conditions) before re-equilibration. As a last resort, the discolored matrix can be removed from the top of the column and replaced. Some matrices (particularly those based on Sephadex) shrink considerably when subjected to solutions of high ionic strength, and the bed may not return to its original dimensions on re-equilibration. If this occurs, then the column should be repacked.

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